

Full lattice geometry in the Connes–Consani programme

Divisors, arithmetic sites, absolute curves and spectral receivers

Programme derivations from the cited human sources

23 September 2026

The controlling object and its two exact coordinates are

$$G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup (R \times \{1_L\}), \quad (q, \sigma)(r, \lambda) = (r, \lambda).$$

Throughout, L is the entire bounded distributive lattice. Every label, both distinguished zeros, each join and meet, and the original nonzero-amplitude constraint are retained. The finite-lattice prime classification states its finite hypothesis separately; the coefficient and sheaf constructions apply to arbitrary nontrivial L .

The human sources are Alain Connes and Caterina Consani’s *Absolute algebra and Segal’s Γ -sets*, *Riemann–Roch for the ring $\mathbb{Z}$* , *Geometry of the Arithmetic Site*, *On the Absolute Geometry of $\text{Spec } \mathbb{Z}$ and the Fargues–Fontaine curve*, and *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* . The exact original TeX versions are retained with archive receipts. Every application below includes its proof locators. The full lattice foundation is The Clankers, *Split Support Geometry*, v11, Definition `def:lattice-split`. The user attributes the earlier scalar supported-zero prime chain to Gemini 2.5 Pro; the present text does not claim its rediscovery.

This cumulative component contains the complete derivations currently proved: coefficient functors and cancellation, bounded sections, divisor multiplication and morphisms, tropical base change and primes, absolute-curve characters and sheaf comparisons, and the original spectral receiver with its tensor and nilpotent sectors. It is not a claim to have completed the entire corpus synthesis, a Deligne purity comparison, or the Riemann hypothesis.

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The whole lattice map and its prime fibres

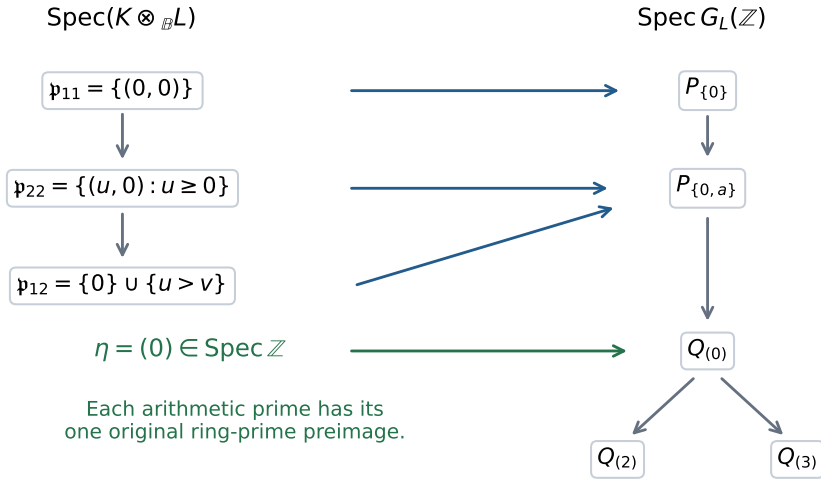
$L = \{0 < a < 1\}$, with join and meet; $K = ([0, \infty), \max, \cdot)$.

$$G_L(\mathbb{Z}) \longrightarrow \mathbb{Z} \times (K \otimes_{\mathbb{B}} L)$$

$$(r, \lambda) \longmapsto (r, \mathbf{1}_{\{j \leq \lambda\}})$$

$\tau = z_0$	$\longrightarrow$	$(0; (0, 0))$
z_a	$\longrightarrow$	$(0; (1, 0))$
$e = z_1$	$\longrightarrow$	$(0; (1, 1))$
$\hat{n} \ (n \neq 0)$	$\longrightarrow$	$(n; (1, 1))$

Prime contraction: the complete fibres



Grey arrows: strict prime inclusion. Blue/green arrows: contraction.
Only primes 2 and 3 are drawn; every rational prime remains.

The additional prime is retained

$$(2, 1) \in \mathfrak{p}_{12}, \quad (2, 1) \notin \mathfrak{p}_{11} \cup \mathfrak{p}_{22}.$$

$$\text{Fr}_0^{-1}(\mathfrak{p}_{12}) = \mathfrak{p}_{22}, \quad \text{Fr}_t^{-1}(\mathfrak{p}_{ij}) = \mathfrak{p}_{ij} \ (t > 0).$$

Proofs: ASL18–22, ASL31; LSP3–27. Tropical coefficients: Connes–Consani,
Geometry of the Arithmetic Site, arXiv:1502.05580v1, Definition defnpt.
Full lattice base: The Clankers, Split Support Geometry v11, def:lattice-split.

Figure 1: The full lattice map in the smallest example with an intermediate label. The three tropical primes map to two support primes; the arithmetic primes are supplied by the ring factor. Both coordinates of the original carrier remain. All displayed maps, fibres and Frobenius actions are proved in ASL18–22 and LSP3–27.

1 The full support lattice and the arithmetic site's coefficient extension

The human source used here is Alain Connes and Caterina Consani, *Geometry of the Arithmetic Site*, original author source `arithmeticsite_Adv_final1.tex`, 2074 lines, SHA-256 `bd9da2b9e3c8846e306af18dbf745d7cf603450d6630a00e239da4235a7320b4`.

The source mathematics actually read for this calculation is lines 331–407, 440–503, 598–704, and 850–1010. These ranges include the definition of the characteristic-one semirings and Frobenius, the rank-one description of points, the arithmetic-site definition, the complete stalk proof, global sections, the definition of valued points, and the stated pullback to the ordinary arithmetic spectrum. This is bounded reading, not a claim to have read the whole paper.

The known programme input is the full, arbitrary bounded distributive lattice L , not a Boolean lattice in its place. Its carrier, complete prime classification, and supported-zero localization are already proved in 031, Theorem `thm:rebuilt-full-support`, and Theorem `thm:e-stalk`, equation `eq:e-element-localization`. The latter states $G_L(R)[e^{-1}] \simeq L$. It is an existing result, not a new assertion of the present calculation. We prove its precise universal consequence for characteristic-one targets, and construct the resulting full-lattice coefficient extension of the arithmetic site, including its points, stalks, and Frobenius.

1.1 The characteristic-one universal property

Let R be a nonzero commutative unital ring and let L be a nontrivial bounded distributive lattice, finite or infinite. Retain

$$\begin{aligned} S &= G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\}, \\ (r, \lambda) + (s, \mu) &= (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu), \\ z_\lambda &= (0, \lambda), \quad \tau = z_{0_L}, \quad e = z_{1_L}, \quad 1_S = (1_R, 1_L), \quad \sigma : S \longrightarrow L, \quad \sigma(r, \lambda) = \lambda. \end{aligned} \tag{ASL1}$$

The operations on the semiring L are join and meet, with zero 0_L and unit 1_L . A semiring homomorphism below preserves both zero and unit.

For every commutative semiring K whose addition is idempotent, the map σ induces a natural bijection

$$\mathrm{Hom}_{\mathrm{sr}}(L, K) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{sr}}(S, K), \quad g \longmapsto g\sigma. \tag{ASL2}$$

To prove this, let $f : S \rightarrow K$. In S , write $v = (-1_R, 1_L)$, so $e = 1_S + v$ and $e + 1_S = 1_S$. Idempotence in K gives

$$f(e) + 1_K = (1_K + f(v)) + 1_K = 1_K + f(v) = f(e),$$

whereas the second source relation makes this equal to 1_K . Thus $f(e) = 1_K$. Since $e(r, 1_L) = e$, every supported amplitude satisfies $f(r, 1_L) = 1_K$. Define $g(\lambda) = f(z_\lambda)$. Addition and multiplication of the z_λ show that g preserves join and meet, and $g(0_L) = 0_K$, $g(1_L) = 1_K$. It follows that $f = g\sigma$. Conversely every such composite is a homomorphism. Surjectivity of σ gives uniqueness. This proves (ASL2) directly, with exactly the same localization map as the established result.

There is a more precise description for an arbitrary idempotent K . Use its natural partial order $x \leq y \iff x + y = y$ and let

$$\mathrm{Id}_{\leq 1}(K) = \{a \in K : a^2 = a, a + 1_K = 1_K\}. \tag{ASL3}$$

This is a bounded distributive lattice under addition and multiplication. For a, b in this set, $ab \leq a, b$, while any other c in the set with $c \leq a, b$ satisfies $c = c^2 \leq ab$. Thus multiplication is their meet. Also $(a + b)^2 = a + b + ab = a + b$, since $ab \leq a + b$, so addition gives the join in the same set. Distributivity is inherited from K . Every unital homomorphism from L has its image here, because $\lambda^2 = \lambda$ and $\lambda \vee 1_L = 1_L$. Consequently (ASL2) is also the exact bijection

$$\mathrm{Hom}_{\mathrm{sr}}(S, K) \simeq \mathrm{Hom}_{\mathrm{BLat}}(L, \mathrm{Id}_{\leq 1}(K)). \tag{ASL4}$$

1.2 All maps to the actual tropical coefficient semirings

Connes–Consani define their structure semiring as $\mathbb{Z}_{\max}$. Its stalk at the ordered group H is

$$H_{\max} = (H \cup \{-\infty\}, \max, +).$$

See their Definition `site`, lines 618–622, and Theorem `structure2`, lines 626–659. Write $\mathbf{0} = -\infty$ and $\mathbf{1} = 0_H$ in this additive presentation, so that the multiplicative unit is never confused with the absorbing zero. For any totally ordered abelian group H , the only multiplicative idempotents of $H_{\max}$ are $\mathbf{0}$ and $\mathbf{1}$: for $h \in H$, $h + h = h$ forces $h = 0_H$. The same assertion holds for their nonnegative min-version $(H_{\geq 0} \cup \{\infty\}, \min, +)$ (source lines 662–665), and for the multiplicative tropical real semifield $\mathbb{R}_+^{\max} = ([0, \infty), \max, \cdot)$.

Let $X = \text{SpecLat } L$ be the set of proper lattice prime ideals. For $P \in X$ set

$$\chi_P(\lambda) = \begin{cases} 0, & \lambda \in P, \\ 1, & \lambda \notin P. \end{cases} \quad \chi_P : L \longrightarrow \mathbb{B} = \{0, 1\}. \quad (\text{ASL5})$$

A lattice ideal is closed under finite joins and downward closed; primality means $\lambda \wedge \mu \in P$ forces one factor into P . These properties prove exactly that (ASL5) preserves zero, unit, join, and meet. Conversely the zero fibre of a bounded lattice homomorphism to $\mathbb{B}$ is such a prime ideal, and determines the homomorphism. Combining this fact with (ASL4) gives, for each of the tropical targets K just specified,

$$\text{Hom}_{\text{sr}}(S, K) \simeq X, \quad f_P = \iota_{\mathbb{B}, K} \chi_P \sigma. \quad (\text{ASL6})$$

Here the canonical inclusion sends the two Boolean elements to $\mathbf{0}_K, \mathbf{1}_K$. The complete fibres are

$$\begin{aligned} f_P^{-1}(\mathbf{0}_K) &= \{z_\lambda : \lambda \in P\}, \\ f_P^{-1}(\mathbf{1}_K) &= \{z_\lambda : \lambda \notin P\} \cup \{(r, 1_L) : r \in R \setminus \{0_R\}\}, \\ f_P^{-1}(a) &= \emptyset \quad (a \notin \{\mathbf{0}_K, \mathbf{1}_K\}). \end{aligned} \quad (\text{ASL7})$$

In particular all ordinary nonzero amplitudes and the supported zero e have value $\mathbf{1}_K$. This is forced by the universal property; it is not an assumption that the added prime has no effect.

The source’s Frobenius maps are $x \mapsto x^n$, $n \geq 1$, and on $\mathbb{R}_+^{\max}$ also $x \mapsto x^t$, $t > 0$ (source lines 395–406). They fix both Boolean elements. Thus

$$\text{Fr}_n^K f_P = f_P, \quad f_P(x^n) = f_P(x), \quad \text{Fr}_t^{\mathbb{R}_+^{\max}} f_P = f_P. \quad (\text{ASL8})$$

The middle identity holds for the actual powers in S , by multiplicativity. It does not assert that $x \mapsto x^n$ is an additive endomorphism of S ; that fails, for example, for $R = \mathbb{Z}$ and $n = 2$. On the reflected lattice, $\lambda^n = \lambda$, so its Frobenius is exactly the identity. Equations (ASL8) are the precise compatibility statements, with no additional action imposed on R .

1.3 The full lattice is recovered by its whole family of characters

Define the basic opens $D(\lambda) = \{P \in X : \lambda \notin P\}$. The map

$$L \longrightarrow \mathbb{B}^X, \quad \lambda \longmapsto (\chi_P(\lambda))_{P \in X} = \mathbf{1}_{D(\lambda)} \quad (\text{ASL9})$$

is an injective lattice homomorphism. Here and below we use the usual prime-ideal theorem for bounded distributive lattices, for which the needed separation proof is as follows. If $\lambda \not\leq \mu$, the ideal $\downarrow \mu$ and filter $\uparrow \lambda$ are disjoint. By Zorn’s lemma choose an ideal maximal among those containing the first and disjoint from the second. If a, b are outside it but $a \wedge b$ lies in it, adjoining a and adjoining b each produces an intersection with the filter. Thus filter elements are bounded above by $i \vee a$ and $j \vee b$ for some ideal elements i, j . Their meet is bounded above by $i \vee j \vee (a \wedge b)$,

by distributivity, and hence is in the ideal, a contradiction. The maximal ideal is therefore prime, contains μ , and avoids λ . This separates the two elements in (ASL9).

The same proof shows $D(\lambda)$ is compact. If it is covered by $D(\mu_i)$ but no finite subfamily covers it, then λ is not in the ideal generated by all the μ_i . The preceding ideal/filter argument gives a prime containing all μ_i and avoiding λ , contradicting the cover. Thus a finite subfamily covers it. Since

$$D(\lambda \vee \mu) = D(\lambda) \cup D(\mu), \quad D(\lambda \wedge \mu) = D(\lambda) \cap D(\mu), \quad (\text{ASL10})$$

every compact open of X equals a unique $D(\lambda)$. Consequently (ASL9) identifies L with the lattice of compact opens, not with the entire power set of X .

For finite L this is especially explicit. Every nonzero join-irreducible j is join-prime: if $j \leq a \vee b$, then $j = (j \wedge a) \vee (j \wedge b)$ forces $j \leq a$ or $j \leq b$. Its character is $\chi_j(\lambda) = 1$ exactly when $j \leq \lambda$. Conversely the complement of any prime ideal is a finite filter; its least element j is nonzero and join-prime, so every character arises this way. Thus

$$X \simeq J(L), \quad L \simeq \{\text{downward-closed subsets of } J(L)\}, \quad \lambda \mapsto \{j : j \leq \lambda\}. \quad (\text{ASL11})$$

For surjectivity in the last map, a downward-closed subset A has $\lambda = \bigvee A$; join-primality puts every irreducible below λ in A . No Boolean support assumption occurs. For the three-element chain, the character vectors are $(0,0), (1,0), (1,1)$; the vector $(0,1)$ is absent. Replacing this lattice by two independent Boolean channels would therefore change the object.

1.4 Prime centres, localization, and the support structure sheaf

The complete prime data of the original semiring, already proved in 031, is

$$\begin{aligned} P^{\text{supp}} &= \{z_\lambda : \lambda \in P\} \quad (P \in X), \\ Q_{\mathfrak{p}} &= \{z_\lambda : \lambda \in L\} \cup \{(r, 1_L) : r \in \mathfrak{p}\} \quad (\mathfrak{p} \in \text{Spec } R), \\ \text{Spec } S &= X \blacktriangleright \text{Spec } R, \quad D_S(e) = X, \quad V_S(e) = \text{Spec } R. \end{aligned} \quad (\text{ASL12})$$

For clarity, exhaustion follows directly from the source operations. An ideal containing a supported element contains e , hence every z_λ ; its supported amplitudes form a ring ideal. An ideal containing no supported element consists of the z_λ from a proper lattice ideal. The two primality tests are respectively ring primality and meet-primality. Every support prime lies strictly below every arithmetic prime, and the basic opens are $D_S(z_\lambda) = D(\lambda)$ and $D_S((r, 1_L)) = X \cup D_R(r)$. This proves the stated topology.

A tropical semifield has only its zero ideal as a proper ideal, because every nonzero element is a unit. The spectral map induced by f_P therefore has the single image P^{supp} , by (ASL7). More generally, every proper prime of any target in (ASL6) contracts to that same prime, since it contains $\mathbf{0}$ and excludes $\mathbf{1}$. In particular these maps do not centre on an arithmetic prime $Q_{\mathfrak{p}}$ containing e .

For a multiplicative subset $T \subset S$, a character map f_P extends to $T^{-1}S$ exactly when

$$\sigma(T) \cap P = \emptyset; \quad f_P(s/t) = f_P(s) \quad (t \in T). \quad (\text{ASL13})$$

Indeed all target values of denominators are either zero or unit one. For the universal property with arbitrary idempotent targets, localization commutes with the reflection:

$$(T^{-1}S)_{\text{idemp}} \simeq \sigma(T)^{-1}L. \quad (\text{ASL14})$$

Both sides represent maps from S to idempotent semirings in which all elements of T become units, using (ASL2), so the displayed isomorphism follows from the two inverse universal maps.

The full support localizations are explicit:

$$L[\lambda^{-1}] \simeq \downarrow \lambda, \quad \mu \mapsto \mu \wedge \lambda, \quad 1_{\downarrow \lambda} = \lambda. \quad (\text{ASL15})$$

The equivalence relation is $\mu \sim \nu \iff \mu \wedge \lambda = \nu \wedge \lambda$; all positive powers of the denominator are λ itself. For a prime P , the stalk has the description

$$L_P = L/(\mu \sim_P \nu), \quad \mu \sim_P \nu \iff \exists t \notin P : \mu \wedge t = \nu \wedge t, \quad S_{P^{\text{supp}}} \simeq L_P. \quad (\text{ASL16})$$

Every denominator in $L \setminus P$ is idempotent, so after being inverted it becomes the unit; this gives the stated equivalence and exhausts fractions. In $S_{P^{\text{supp}}}$, e is inverted first, giving L , and the remaining denominators are precisely those represented by $L \setminus P$. This proves the last isomorphism.

These semirings of sections form a sheaf on the support spectrum. On compact basic opens set

$$\mathcal{L}(D(\lambda)) = \downarrow \lambda, \quad \text{res}_{D(\mu)}(a) = a \wedge \mu \quad (\mu \leq \lambda). \quad (\text{ASL17})$$

For a finite cover with $\lambda = \bigvee_i \lambda_i$, compatible sections $a_i \leq \lambda_i$ obey $a_i \wedge \lambda_j = a_j \wedge \lambda_i$. Their unique gluing is $a = \bigvee_i a_i$: distributivity proves $a \wedge \lambda_i = a_i$, and if a second section has these restrictions then intersecting it with $\bigvee_i \lambda_i$ proves equality. Compactness gives a finite subcover for any cover of a basic open, and the same compatibility checks every remaining member. Thus this is a sheaf on the basis and has stalk (ASL16). It is the restriction of the localization structure sheaf of S to $D_S(e)$.

The stalk L_P generally retains more than a Boolean value. For the chain $0 < a < 1$ at the prime $P = \{0, a\}$, only 1 is inverted, so $L_P = L$ still has three elements. Its local residue map to $\mathbb{B}$ sends $0, a$ to zero and 1 to one. In general the extension of χ_P sends a class to one precisely when that class is the unit: $\mu \notin P$ becomes invertible and hence equals one; if $\mu \in P$ could equal one, then some $t \notin P$ would satisfy $t \leq \mu$, contradicting downward closure. The Bourne quotient by its zero fibre is exactly $\mathbb{B}$, because it collapses all nonunits to zero while leaving the unit distinct. Thus the residue character is related to the full support stalk by an exact local map; it is not a replacement for that stalk.

1.5 The universal tropical coefficient extension retains every support

Let K be a tropical semifield as above and form the commutative semiring tensor product

$$K_L = K \otimes_{\mathbb{B}} L. \quad (\text{ASL18})$$

There is no unital map $\mathbb{B} \rightarrow S$: it would require $1_S + 1_S = 1_S$, which in the nonzero ring amplitude would say $2 \cdot 1_R = 1_R$. The correct direct base change from S uses the initial semiring $\mathbb{N}_0$, and (ASL2) gives the canonical isomorphism

$$S \otimes_{\mathbb{N}_0} K \simeq L \otimes_{\mathbb{B}} K. \quad (\text{ASL19})$$

Indeed any semiring receiving a unital map from K is additively idempotent: $1 + 1 = 1$ implies $x + x = x$ for every element. Its map from S therefore factors uniquely through L . Maps from L and K agree on $\mathbb{B}$, and these are exactly the universal maps on the right side of (ASL19). This proves the isomorphism without pretending that S was already a Boolean algebra or a $\mathbb{B}$ -algebra.

The simultaneous character evaluations give a faithful model

$$K_L \longrightarrow K^X, \quad w = \sum_{i=1}^r a_i \otimes \lambda_i \longmapsto \left(P \longmapsto \max_{i: \lambda_i \notin P} a_i \right), \quad (\text{ASL20})$$

with the empty maximum equal to $\mathbf{0}_K$. Both operations are pointwise on the displayed functions. The map exists by the tensor universal property. Every tensor expression is a finite sum of pure tensors, because products of pure tensors are pure tensors. To prove injectivity, discard zero-coefficient terms and list the finitely many distinct remaining coefficients as $a_1 < \dots < a_r$ in

the natural total order of K . Let μ_i be the join of the supports whose coefficients are at least a_i . The original tensor expression equals

$$\sum_{i=1}^r a_i \otimes \mu_i, \quad \mu_1 \geq \cdots \geq \mu_r. \quad (\text{ASL21})$$

For verification, distributing each μ_i merely adds terms $a_i \otimes \lambda$ already bounded above by a term $a_j \otimes \lambda$ with $j \geq i$; their sum is the latter because $a_i + a_j = a_j$. Thus all original data is related by actual tensor relations. In the image function $w(P)$, the set where $w(P) \geq a_i$ is exactly $D(\mu_i)$. For two expressions, use the union of their finite coefficient lists in (ASL21). Equality of functions gives equality of every $D(\mu_i)$ and therefore of every μ_i by (ASL9). Hence the two tensors are equal, proving faithfulness.

Equivalently, (ASL20) identifies K_L with the finite-valued functions $w : X \rightarrow K$ whose sets $\{P : w(P) \geq a\}$ are compact open for every nonzero attained value a . Conversely, the finitely many nonzero values of such a function and their compact opens supply (ASL21), by (ASL10). In particular $1 \otimes L$ embeds faithfully as its compact-open characteristic functions. The fibre of evaluation at a specified P is completely described by the displayed maximum: it is zero exactly when all nonzero-coefficient supports belong to P ; a nonzero value a occurs exactly when a is the largest active coefficient. In level-set language this is $P \in D(\mu_{\geq a}) \setminus D(\mu_{>a})$.

The coefficient extension can also accompany the original ring amplitudes without losing them. The map

$$S \longrightarrow R \times K_L, \quad (r, \lambda) \longmapsto (r, 1 \otimes \lambda) \quad (\text{ASL22})$$

is an injective semiring homomorphism: its first coordinate retains the amplitude and (ASL20) makes its second coordinate faithful on the entire lattice. Its image consists exactly of pairs with a compact-open characteristic function in the second coordinate and with that function identically one whenever $r \neq 0$. This is an actual correspondence retaining both coordinates, not a claim that the ring amplitude itself embeds into a characteristic-one semiring.

1.6 The arithmetic site with full lattice coefficients

Let $\mathcal{A} = \widehat{\mathbb{N}}^\times$ be the source topos and $\mathcal{O} = \mathbb{Z}_{\max}$ with its Frobenius action, exactly as in Connes–Consani’s Definition **site**. Give the constant semiring object $\underline{L}$ the trivial action and set

$$\mathcal{O}_L = \mathcal{O} \otimes_{\mathbb{B}} \underline{L}, \quad (\mathcal{A}, \mathcal{O}_L) \longrightarrow (\mathcal{A}, \mathcal{O}). \quad (\text{ASL23})$$

The underlying geometric morphism is the identity, and its structure map is $\mathcal{O} \rightarrow \mathcal{O}_L$, $a \mapsto a \otimes 1$. By (ASL19) it is also the universal coefficient object receiving $\mathcal{O}$ and the constant full semiring $\underline{S}$ over $\underline{\mathbb{N}}_0$. The latter is expressly a constant object; no unproved additive Frobenius on the ring R is asserted.

The Frobenius action and joint character maps are

$$\text{Fr}_n \left(\sum_i a_i \otimes \lambda_i \right) = \sum_i a_i^n \otimes \lambda_i, \quad \text{ev}_P : \mathcal{O}_L \rightarrow \mathcal{O}. \quad (\text{ASL24})$$

Evaluation is (ASL20), so these are equivariant semiring maps and their family is jointly injective. In the additive notation of $\mathbb{Z}_{\max}$ the coefficient operation a_i^n means na_i ; the absorbing zero stays absorbing zero. For each P the exact coefficient specialization is

$$\mathcal{O}_L \otimes_{\underline{L}, \chi_P} \mathbb{B} \simeq \mathcal{O}. \quad (\text{ASL25})$$

This follows from the two tensor universal properties, or directly by substituting $\lambda \mapsto \chi_P(\lambda)$ in every pure tensor. All supports are present before this specified specialization.

At the point corresponding to a nonzero ordered subgroup $H \subset \mathbb{Q}$, the stalk is

$$(\mathcal{O}_L)_H \simeq H_{\max} \otimes_{\mathbb{B}} L. \quad (\text{ASL26})$$

To justify this precisely, the inverse-image functor of the point preserves finite products and arbitrary colimits. A semiring tensor product is constructed from finite products, coproducts and the coequalizers imposing its addition and multiplication relations. It is therefore preserved by that inverse-image functor. A constant object with trivial action has the same stalk: in the source's description $(X \times H_{>0})/\sim$ at lines 631–649, any two positive parameters have a common divisor in $H_{>0}$, and the trivial action identifies them while retaining the value in X . Finally the source's Theorem **structure2** identifies the original structure stalk as $H_{\max}$. These facts prove (ASL26). Its elements have exactly the full-lattice function description (ASL20)–(ASL21), now with coefficients in $H_{\max}$.

The global sections of the extended site are exactly

$$\Gamma(\mathcal{A}, \mathcal{O}_L) \simeq L. \quad (\text{ASL27})$$

For a one-object action topos, a global section is a simultaneous fixed element; this is the source's argument at lines 668–673. Under the faithful function representation (ASL20), invariance under Fr_2 forces each value a to satisfy $a^2 = a$. In $\mathbb{Z}_{\max}$ these are precisely **0**, **1**. Such a finite-valued function is a section of $\mathcal{O}_L$ exactly when its nonzero set is $D(\lambda)$, and then it is $1 \otimes \lambda$. Conversely these sections are fixed by every Frobenius. This proves (ASL27); the source's unextended case is $\Gamma(\mathcal{A}, \mathcal{O}) = \mathbb{B}$.

1.7 All tropical-valued points of the extended site

By the definition of a valued point in the source, lines 685–690, one supplies a topos point H and a semiring map from its structure stalk to $\mathbb{R}_+^{\max}$. The tensor universal property and (ASL6) therefore give the exact classification

$$\begin{aligned} \text{Hom}_{\text{sr}}(H_{\max} \otimes_{\mathbb{B}} L, \mathbb{R}_+^{\max}) &\simeq \text{Hom}_{\text{sr}}(H_{\max}, \mathbb{R}_+^{\max}) \times X, \\ (\beta, P) : \sum_i a_i \otimes \lambda_i &\longmapsto \max_{i: \lambda_i \notin P} \beta(a_i). \end{aligned} \quad (\text{ASL28})$$

There is no missing compatibility condition: every unital semiring map agrees on the two distinguished elements of $\mathbb{B}$. Thus the set of isomorphism classes of valued points of the extended site is the product of the original valued-point set with X . An isomorphism of underlying topos points can change H and its map β as in the source's equivalence relation, but it acts identically on the constant object L , hence cannot identify different P 's.

For a fixed embedding $H \subset \mathbb{Q}$, the first factor in (ASL28) can itself be given exactly. Every map has the form

$$\beta_c(-\infty) = 0, \quad \beta_c(h) = e^{ch} \ (h \in H), \quad c \geq 0. \quad (\text{ASL29})$$

Indeed group units must map to positive real units. Taking their real logarithms gives an additive order-preserving map $\ell : H \rightarrow \mathbb{R}$. Fix $h_0 > 0$. Since every h/h_0 is rational, additivity forces $\ell(h) = h\ell(h_0)/h_0$; order preservation makes $c = \ell(h_0)/h_0 \geq 0$. Conversely every formula (ASL29) preserves both multiplication and maximum. The value $c = 0$ is the degenerate map through $\mathbb{B}$; $c > 0$ is injective on the ordered group.

Under the source's real Frobenius the exact point action is

$$\text{Fr}_t : (H, \beta_c, P) \longmapsto (H, \beta_{tc}, P), \quad t > 0. \quad (\text{ASL30})$$

Thus the arithmetic site's original nontrivial coefficient dynamics is retained, and the full lattice contributes its whole space of prime characters as a fixed parameter. A direct map from S to a single tropical stalk sees only its one character, by (ASL7), whereas the extended coefficient sheaf, all its stalks, and their joint evaluations retain every element and every relation of L . Equations (ASL19), (ASL23), (ASL26), and (ASL28) are the exact connecting maps. No purity, cohomology comparison, or Riemann-hypothesis conclusion is inferred from them.

1.8 A precise distinction between prime ideals and valued points

The preceding statements classify all support primes of S , all lattice primes, and all tropical-valued points of the coefficient extension. They do not identify the ordinary semiring prime spectrum of K_L with X . There can be additional primes, with an explicit example already for the three-element chain $L = \{0 < a < 1\}$. For $K = \mathbb{R}_+^{\max}$, (ASL20) gives

$$K_L \simeq \{(u, v) \in [0, \infty)^2 : u \geq v\}, \quad \mathfrak{q} = \{(0, 0)\} \cup \{(u, v) : u > v\}. \quad (\text{ASL31})$$

The operations are coordinatewise maximum and multiplication. The set $\mathfrak{q}$ is a proper prime ideal. It is closed under maximum: whichever of two first coordinates is larger also exceeds both second coordinates. Multiplying (u, v) with $u > v$ by a nonzero (r, s) with $r \geq s$ gives $ur > vs$; multiplication by zero gives zero. Its complement is the set of positive diagonal pairs, which is multiplicatively closed. These observations prove the ideal and prime assertions. But $\mathfrak{q}$ is neither evaluation kernel: the first evaluation has only the zero kernel, the second kernel is $\{(u, 0) : u \geq 0\}$, and $(2, 1)$ belongs to $\mathfrak{q}$ but to neither of those. Thus the full support and valued-point comparison retains its precise stated scope even in this smallest non-Boolean case. It does not erase other prime geometry introduced by the tropical coefficient extension.

1.9 The exact pullback over the ordinary arithmetic spectrum

Connes–Consani’s geometric morphism over $\text{Spec } \mathbb{Z}$ uses the monoid $\mathbb{N}_0^\times$ that includes zero, not just $\mathbb{N}^\times$. This distinction is explicit in their Remark `caseno`, source line 660, and Theorem `thmspz`, lines 951–961. Put $\mathcal{A}_0 = \widehat{\mathbb{N}_0^\times}$ and let $\mathcal{O}^0$ be their extended structure semiring. In additive exponent notation its additional action is

$$\text{Fr}_0(-\infty) = -\infty, \quad \text{Fr}_0(h) = 0_H \quad (h \in H). \quad (\text{ASL32})$$

This preserves both operations: a maximum is absorbing zero exactly when both inputs are absorbing zero, and a product is absorbing zero exactly when one factor is. The extension $\mathcal{O}_L^0 = \mathcal{O}^0 \otimes_{\mathbb{B}} \underline{L}$ has

$$\text{Fr}_0\left(\sum_i a_i \otimes \lambda_i\right) = 1 \otimes \rho\left(\sum_i a_i \otimes \lambda_i\right), \quad \rho\left(\sum_i a_i \otimes \lambda_i\right) = \bigvee_{a_i \neq 0} \lambda_i. \quad (\text{ASL33})$$

The map ρ is well defined by the tensor universal property, using the semiring map $H_{\max} \rightarrow \mathbb{B}$ that sends every nonzero element to one. Equivalently, in (ASL20), $D(\rho(w))$ is exactly the set where w is nonzero. Thus (ASL33) retains a specific element of L , rather than only whether any support exists. It obeys $\text{Fr}_0 \text{Fr}_n = \text{Fr}_n \text{Fr}_0 = \text{Fr}_0$.

The original source constructs $\Theta : \text{Sh}(\text{Spec } \mathbb{Z}) \rightarrow \mathcal{A}_0$. At a rational prime p , its point is $H(p) = \mathbb{Z}[1/p] \subset \mathbb{Q}$, as defined at source lines 910–914; the generic point η goes to the trivial group. Applying its inverse-image functor to the full coefficient extension gives the exact sheaf and stalk formulas

$$\begin{aligned} \mathcal{T}_L &:= \Theta^* \mathcal{O}_L^0 \simeq \Theta^* \mathcal{O}^0 \otimes_{\mathbb{B}} \underline{L}, \\ (\mathcal{T}_L)_p &\simeq \mathbb{Z}[1/p]_{\max} \otimes_{\mathbb{B}} L, \quad (\mathcal{T}_L)_\eta \simeq L. \end{aligned} \quad (\text{ASL34})$$

The preservation argument proving (ASL26) applies to Θ^* as well. Connes–Consani’s Proposition `sheavesonspz0`, source lines 965–975, gives the original stalks $H(p)_{\max}$ and $\mathbb{B}$, so tensoring proves (ASL34). This is a pullback of semiring coefficients on the underlying topos of the arithmetic spectrum. It does not identify these coefficients with the ordinary ring structure sheaf $\mathcal{O}_{\text{Spec } \mathbb{Z}}$.

All its sections can also be described without suppressing the prime or support labels. For a nonempty open U of $\text{Spec } \mathbb{Z}$, a section of $\mathcal{T}_L$ is exactly the data

$$\begin{aligned} \lambda \in L, \quad w_p \in \mathbb{Z}[1/p]_{\max} \otimes_{\mathbb{B}} L \quad (p \in U \text{ a rational prime}), \\ \rho(w_p) = \lambda \quad \text{for every } p, \quad w_p = 1 \otimes \lambda \quad \text{for all but finitely many } p. \end{aligned} \quad (\text{ASL35})$$

The value at η is λ . Addition and multiplication are pointwise, with join and meet at η ; restriction discards the coordinates at the removed primes. The integer Frobenius applies $h \mapsto nh$ to every coefficient at every retained prime and acts identically on λ . Its zero action is (ASL33).

Here is a complete gluing proof. The source's Proposition **sheavesonspz0(ii)** says that an original section is either absorbing zero everywhere or a family of nonzero coefficients equal to the unit outside a finite set of primes. A local finite tensor expression therefore has a single generic support $\lambda = \bigvee_{a_i \neq 0} \lambda_i$; all its prime values have that same support, and outside a finite set every value is $1 \otimes \lambda$. Every sheaf tensor section has such expressions locally, by the construction of the tensor as the sheafification of the presheaf tensor. Every nonempty open U here is quasi-compact: a nonempty member of any open cover misses only finitely many points of U , which finitely many other members can cover. Thus finitely many local expressions suffice. Their generic values agree since every nonempty intersection contains η . Their finite exceptional sets have finite union. This proves that every section yields (ASL35).

Conversely take (ASL35), with finite exceptional set E . For each $p \in E$ write w_p as a finite tensor sum using nonzero coefficients $a_i \in \mathbb{Z}[1/p]_{\max}$. Its support labels have join λ . On the open $U_p = U \setminus (E \setminus \{p\})$ extend a_i to the original coefficient section with value a_i at p and the unit at every other prime; the source's section description makes this an actual section. Tensor these sections with their labels and add. The result has prescribed value w_p at p and value $1 \otimes \lambda$ elsewhere on U_p . The case $\lambda = 0_L$ has only zero tensors and is obtained from the zero section. Add the constant section $1 \otimes \lambda$ on $U \setminus E$. These local sections agree on their intersections, so the sheaf gluing axiom constructs the required global section. Uniqueness follows because sections of a sheaf on a topological space are determined by all their germs. This proves (ASL35) and its operation and restriction formulas.

2 The complete finite-lattice prime spectrum after tropical base change

The coefficient semiring throughout this calculation is exactly $K = \mathbb{R}_+^{\max} = ([0, \infty), \max, \cdot)$, with absorbing zero 0 and unit 1. This is the tropical coefficient semifield of Alain Connes and Caterina Consani, *Geometry of the Arithmetic Site*. The exact locators are lines 395–406 and Definition **defnpt**, lines 685–690 of the original author source:

`arithmeticsite_Adv_final1.tex.`

The complete support base $G_L(R)$ and its original spectrum are the existing programme construction cited and reproduced in (ASL1), (ASL12), and 031, Theorem **thm:rebuilt-full-support**. The ordinary-prime classification proved below is a derivation for the coefficient extension; it is not attributed to Connes–Consani.

Let L be any nontrivial finite bounded distributive lattice. Its original elements are retained through the exact isomorphism

$$L \longrightarrow \text{Down}(J), \quad \lambda \longmapsto \{j \in J : j \leq \lambda\}, \quad J = J(L), \quad (\text{LSP1})$$

whose inverse takes the join of a downset. The proof with all maps is (ASL11): join-irreducibles in a distributive lattice are join-prime, and every element is the join of those below it. In particular this is not a replacement by independent Boolean coordinates. By the faithful tensor calculation (ASL18)–(ASL21), the full coefficient extension is exactly

$$A_J := K \otimes_{\mathbb{B}} L \simeq \{f : J \rightarrow [0, \infty) : x \leq y \implies f(x) \geq f(y)\}, \quad (\text{LSP2})$$

with pointwise maximum and multiplication. For completeness the finite version follows directly: the nonzero superlevel sets of an antitone function are downsets, and its finitely many values express it as a sum of those values tensored with those downsets. Sorting the coefficient values and

using their cumulative downsets proves uniqueness by character evaluation, exactly as (ASL21). For every downset $D \subset J$, write e_D for its characteristic function. These are precisely the multiplicative idempotents of A_J ; each coordinate of an idempotent is zero or one.

An ideal here means an ordinary semiring ideal, closed under addition and multiplication by every element. It is not required to be subtractive. A prime ideal is proper and its multiplicative complement is closed. Retaining that distinction is necessary for the complete answer.

2.1 Exhaustive classification, including the primes with several labels

For $j \in J$, put $J_{\leq j} = \{x : x \leq j\}$. Let U be an upward-closed subset of $J_{\leq j}$ containing j . Define

$$\mathfrak{p}_{j,U} = \{f \in A_J : f(j) = 0 \text{ or } \exists i \in U : f(i) > f(j)\}. \quad (\text{LSP3})$$

Then every ordinary prime ideal of A_J is exactly one of (LSP3), and the pair (j, U) is unique.

First prove existence. If f, g belong to (LSP3), put $h = \max(f, g)$. When $h(j) = 0$, it belongs to the ideal. Otherwise one of f, g , say f , attains $h(j) > 0$ at j . Since f is in the ideal, there is $i \in U$ with $f(i) > f(j) = h(j)$, whence $h(i) > h(j)$. This proves additive closure. For a product fg , a zero at j persists. If $f(i) > f(j) > 0$ and $g(j) > 0$, then $g(i) \geq g(j) > 0$, so $f(i)g(i) > f(j)g(j)$. If $g(j) = 0$, the product has zero at j . This proves absorption. The constant unit is outside the ideal. Finally its complement is exactly

$$f(j) > 0, \quad f(i) = f(j) \quad (i \in U), \quad (\text{LSP4})$$

and is multiplicatively closed. Thus (LSP3) is prime.

For exhaustion, let $\mathfrak{p}$ be any prime of A_J . Its intersection with the e_D is a prime of the lattice of downsets. The complement of that lattice prime is a finite filter. Its least member D_0 is nonempty and join-prime: if $D_0 \subset B \cup C$ then $D_0 \subset B$ or $D_0 \subset C$, by primality of the complementary ideal. Since D_0 is the union of its principal downsets, $D_0 = J_{\leq j}$ for a unique j . It follows that

$$e_D \notin \mathfrak{p} \iff j \in D. \quad (\text{LSP5})$$

In particular $e_{J_{\leq j}} \notin \mathfrak{p}$, and primality gives

$$f \in \mathfrak{p} \iff e_{J_{\leq j}} f \in \mathfrak{p}.$$

We can therefore work on the full subposet $B = J_{\leq j}$, with unit e_B . This is the exact truncation map $f \mapsto f|_B$, followed by extension by zero outside B when viewed in A_J . The latter extension is a homomorphism with unit e_B in its image.

Every function on B with value zero at j is in the prime: it is multiplied identically by $e_{B \setminus \{j\}}$, which belongs to the prime by (LSP5). All remaining functions are positive at every coordinate, because j is the maximum of B . The positive constant functions are units, so multiplying by such a function does not change ideal membership. For a positive function f define the exact coordinates

$$a_x = \log(f(x)/f(j)), \quad C_j = \{a : B \rightarrow [0, \infty) : a_j = 0, a_x \geq a_y \text{ if } x \leq y\}. \quad (\text{LSP6})$$

This retains f through $f(x) = f(j) \exp(a_x)$ and a positive unit factor $f(j)$. Write $h_a(x) = \exp(a_x)$ and set

$$F = \{a \in C_j : h_a \notin \mathfrak{p}\}.$$

The set F contains zero and is closed under addition, by $h_{a+b} = h_a h_b$ and the multiplicative closure of the complement. It also has the stronger property

$$b \in F, \quad a \in C_j, \quad a_x \leq b_x \quad (x \in B) \implies a \in F. \quad (\text{LSP7})$$

Indeed let $g(x) = \exp(b_x)$ for $x \neq j$ and $g(j) = 0$. It is antitone, is supported on $B \setminus \{j\}$, and belongs to the prime. If h_a also belonged to it, then $\max(h_a, g) = h_b$ would belong, a contradiction. This proves (LSP7).

Let D be the union of the sets $\{x : a_x > 0\}$ over $a \in F$. It is a downset of B and excludes j . Since B is finite, there is a single $b \in F$ whose positive-coordinate set is exactly D : choose one member positive at each coordinate of D and add them. When D is empty take $b = 0$. For any $a \in C_j$ vanishing outside D , some positive integer N satisfies $a_x \leq Nb_x$ at every coordinate; take an integer at least every nonzero ratio a_x/b_x . Then $Nb \in F$ and (LSP7) gives $a \in F$. Conversely every member of F vanishes outside D by the definition of D . Hence

$$F = \{a \in C_j : a_x = 0 \text{ for every } x \in U\}, \quad U = B \setminus D. \quad (\text{LSP8})$$

The set U is upward closed and contains j . Returning through (LSP6) and retaining the functions with zero at j proves $\mathfrak{p} = \mathfrak{p}_{j,U}$.

Uniqueness of j follows from (LSP5). If $U \neq V$ have the same j , take $i \in U \setminus V$, interchanging them if necessary. Then $i \neq j$ and the principal downset $J_{\leq i}$ is disjoint from V . The function equal to $\exp(1)$ on $J_{\leq i}$ and to 1 elsewhere belongs to $\mathfrak{p}_{j,U}$ and not to $\mathfrak{p}_{j,V}$. This proves uniqueness completely.

For a three-point poset with incomparable a, b both below j , the upset $U = \{a, b, j\}$ gives the additional prime

$$\{f : f(j) = 0 \text{ or } f(a) > f(j) \text{ or } f(b) > f(j)\}. \quad (\text{LSP9})$$

It is distinct from both comparable-pair primes with $U = \{a, j\}$ or $U = \{b, j\}$, by the uniqueness proof. Thus comparable pairs alone do not exhaust the spectrum for a general finite lattice.

2.2 Specialization and every basic open

The complete specialization criterion is

$$\mathfrak{p}_{j,U} \subseteq \mathfrak{p}_{k,V} \iff j \leq k \text{ and } U \setminus \{j\} \subseteq V. \quad (\text{LSP10})$$

Necessity of $j \leq k$ follows by contracting to the idempotents: the lattice prime of j is contained in that of k exactly when $j \leq k$. Explicitly, if $j \not\leq k$, the downset $J_{\leq k}$ contains k and excludes j , contradicting the required inclusion. If $i \in U \setminus \{j\}$ but $i \notin V$, the function equal to $\exp(1)$ on $J_{\leq i}$ and 1 elsewhere has value 1 at j, k and all of V , but a strict excess at i . It contradicts prime inclusion. This proves the other necessity.

For sufficiency, take $f \notin \mathfrak{p}_{k,V}$. Then $f(k) > 0$ and $f(v) = f(k)$ for all $v \in V$. If $U = \{j\}$, the inequality $f(j) \geq f(k) > 0$ already puts f outside $\mathfrak{p}_{j,U}$. Otherwise every $i \in U \setminus \{j\}$ lies in V , and upward closure with $i \leq j \leq k$ also puts j in V . Thus $f(i) = f(j) = f(k)$ for every $i \in U$, proving the implication. The basic open of any original function is exactly

$$D(f) = \{(j, U) : f(j) > 0, f(i) = f(j) \text{ for all } i \in U\}. \quad (\text{LSP11})$$

Equations (LSP10)–(LSP11) determine the whole finite topological spectrum, with every original support label still present.

For the chain $J = \{1 < \dots < n\}$, every possible upset in $J_{\leq j}$ is $\{i, i+1, \dots, j\}$, with $1 \leq i \leq j \leq n$. The list becomes

$$\mathfrak{p}_{ij} = \{f : f_i = 0 \text{ or } f_i > f_j\}, \quad 1 \leq i \leq j \leq n, \quad |\text{Spec } A_J| = \frac{n(n+1)}{2}. \quad (\text{LSP12})$$

This formula equals (LSP3): when $f_j = 0$, either $f_i = 0$ or $f_i > f_j$, and when $f_j > 0$ the largest value in the indicated upset is f_i . The inclusion criterion is

$$\mathfrak{p}_{ij} \subseteq \mathfrak{p}_{k\ell} \iff j \leq \ell \text{ and } (i = j \text{ or } k \leq i). \quad (\text{LSP13})$$

In particular, for $n = 2$ the entire spectrum is the three-element chain $\{0\} = \mathfrak{p}_{11} \subsetneq \mathfrak{p}_{22} \subsetneq \mathfrak{p}_{12}$, including (ASL31). For general n its Krull dimension is exactly $2n - 2$. The integer $2j - i - 1$ strictly increases under each strict inclusion in (LSP13) and ranges from 0 to $2n - 2$, giving the upper bound. The chain

$$\mathfrak{p}_{11} \subsetneq \mathfrak{p}_{22} \subsetneq \cdots \subsetneq \mathfrak{p}_{nn} \subsetneq \mathfrak{p}_{n-1,n} \subsetneq \cdots \subsetneq \mathfrak{p}_{1n} \quad (\text{LSP14})$$

has exactly $2n - 2$ strict inclusions, proving equality, including the single-point case $n = 1$.

2.3 All localized coordinates and the exact maps between them

Fix (j, U) and put $B = J_{\leq j}$. The localization at its prime has the following concrete realization, with pointwise operations:

$$(A_J)_{\mathfrak{p}_{j,U}} \simeq \left\{ h : B \rightarrow [0, \infty) : \begin{array}{l} \{x : h(x) > 0\} \text{ is a downset of } B, \\ h|_U \text{ is antitone} \end{array} \right\}. \quad (\text{LSP15})$$

The map sends a fraction f/g , $g \notin \mathfrak{p}_{j,U}$, to $h(x) = f(x)/g(x)$ for $x \in B$. By (LSP4), every such denominator is positive throughout B and constant on U . Thus the indicated support and order properties hold. Equality of fractions implies coordinatewise equality there. Conversely coordinatewise equality implies equality after multiplication by e_B , which is outside the prime; hence this map is injective. This also shows explicitly why coordinates outside B disappear at this specified localization.

To prove surjectivity without guessing denominators, take any h on the right side. Set $g(x) = 1$ for $x \in U$, and construct its other positive values in reverse order, processing all strict successors of x first. Choose

$$g(x) = \max(\{1\} \cup \{g(y) : y > x\} \cup \{h(y)g(y)/h(x) : y > x\}) \quad (h(x) > 0), \quad (\text{LSP16})$$

and omit the last set when $h(x) = 0$. Every set here is finite. For $x \in U$, its successors are also in U and the antitone condition on $h|_U$ already verifies the required inequalities with $g(x) = 1$. For every other x , formula (LSP16) makes both g and hg antitone. In the zero case every successor also has $h(y) = 0$ because the nonzero support is a downset, so no inequality was lost by omitting the ratios. Thus g is positive, antitone and constant on U , while $f = hg$ is antitone. Extend both by zero outside B . They are elements of A_J , g is outside the prime, and $f/g = h$. This proves (LSP15) with an explicit inverse construction.

The units and maximal ideal in this local semiring are therefore

$$\begin{aligned} h \text{ is a unit} &\iff h(x) > 0 \ (x \in B), \quad h(i) = h(j) \ (i \in U), \\ \mathfrak{m}_{j,U} &= \{h : h(j) = 0 \text{ or } \exists i \in U : h(i) > h(j)\}. \end{aligned} \quad (\text{LSP17})$$

Indeed an inverse must be the coordinatewise reciprocal, so support must be all of B . Both $h|_U$ and its reciprocal are antitone exactly when values agree along every comparable pair. Every element of U is below j , forcing the displayed constant value. Conversely these conditions make the reciprocal belong to (LSP15). The complement of the units is the ideal displayed in (LSP17), so it is the unique maximal ideal.

If (LSP10) holds, the canonical map between the localizations is

$$(A_J)_{\mathfrak{p}_{k,V}} \longrightarrow (A_J)_{\mathfrak{p}_{j,U}}, \quad h \longmapsto h|_{J_{\leq j}}. \quad (\text{LSP18})$$

It has this value on every original fraction, hence is the required localization map. Its image consists exactly of the functions in (LSP15) whose restriction to $V \cap J_{\leq j}$ is antitone. Necessity follows by restriction. For sufficiency extend such a function by zero on $J_{\leq k} \setminus J_{\leq j}$. Its nonzero support remains a downset. Its restriction to V is antitone: the only cross-boundary comparisons

start in $J_{\leq j}$ and end outside, and then the latter value is zero. It is therefore an element of the source of (LSP18). The fibres of the map are exactly the admissible functions agreeing on $J_{\leq j}$. No additional surjectivity assertion is needed or made.

For completeness, primes of the localized semiring correspond to the pairs (k, V) whose prime is contained in $\mathfrak{p}_{j,U}$. Extension takes such a prime to the fractions with numerator in it. This is well defined: if $f/g = f'/g'$ then some inverted t satisfies $tf g' = tf' g$; all inverted factors are outside the smaller prime, so primality makes f belong exactly when f' does. The complement is multiplicatively closed, and contraction recovers the same prime. Conversely a prime of the localization cannot contain an inverted element; its contraction has this property and the fraction rule recovers it. Together with (LSP10), this proves the entire localized prime geometry, not just the local coordinate formulas.

2.4 Subtractive primes, quotient maps, and actual tropical residue maps

An ideal I is subtractive when $a + b \in I$ and $a \in I$ imply $b \in I$. Precisely the primes

$$\mathfrak{p}_{j,\{j\}} = \{f : f(j) = 0\} \quad (\text{LSP19})$$

are subtractive. They are kernels of evaluation at j , and the subtractive property follows by evaluating the defining implication in K . For any $U \neq \{j\}$ choose $i \in U \setminus \{j\}$. The function b equal to $\exp(1)$ on $J_{\leq i}$ and 1 elsewhere satisfies

$$b \in \mathfrak{p}_{j,U}, \quad \max(1, b) = b, \quad 1 \notin \mathfrak{p}_{j,U}. \quad (\text{LSP20})$$

This violates subtractivity. It also proves that the Bourne quotient by this prime is the zero semiring: the relation generated by $b = 0$ and $1 + b = b$ forces $1 = 0$. This is a property of that precise quotient construction; the original prime and localization in (LSP15) are still proper and nonzero.

For the subtractive prime the complete global Bourne quotient is

$$A_J / \mathfrak{p}_{j,\{j\}} \simeq A_{J_{\geq j}}, \quad f \mapsto f|_{J_{\geq j}}. \quad (\text{LSP21})$$

Here the quotient identifies f, g when $f + a = g + b$ for some a, b in the ideal. Such functions a, b vanish on $J_{\geq j}$, so equivalence forces equality on that whole upset. Conversely, if the restrictions agree, choose a real M at least all the finitely many values of f, g . The function $Me_{J \setminus J_{\geq j}}$ is in the ideal, and adding it to both functions makes them equal everywhere. This proves the kernel-congruence assertion. Every antitone function on $J_{\geq j}$ extends by a sufficiently large constant on its complementary downset, proving surjectivity. Thus (LSP21) is an actual quotient isomorphism; it need not equal K before localization.

After localization at this prime the residue quotient is exactly

$$(A_J)_{\mathfrak{p}_{j,\{j\}}} / \mathfrak{m}_{j,\{j\}} \simeq K, \quad h \mapsto h(j). \quad (\text{LSP22})$$

Evaluation is surjective on constants. If $h(j) = h'(j)$, choose a constant M dominating both functions on $B \setminus \{j\}$. The function equal to M there and zero at j belongs to the local maximal ideal and to (LSP15); adding it makes h, h' equal. Conversely any such equivalence preserves the value at j . This proves (LSP22). For $U \neq \{j\}$ the local Bourne quotient is also zero, by the image of (LSP20).

Any unital homomorphism to a tropical semifield has a subtractive zero kernel: if the image of $a + b$ and of a is zero, the image of b is zero. Consequently none of the additional primes can be the zero kernel of such a map. The exact tropical-valued-point classification (ASL28) and the larger ordinary-prime classification (LSP3) therefore agree through the specified kernel map; neither geometry has been substituted for the other.

2.5 The full original prime spectrum is recovered from the companion map

For a nonzero commutative ring R , retain the original, injective map (ASL22), with all amplitude and support coordinates:

$$G_L(R) \longrightarrow R \times A_J, \quad (r, \lambda) \longmapsto (r, e_{\{j \in J : j \leq \lambda\}}). \quad (\text{LSP23})$$

Injectivity follows because both displayed coordinates are retained and (LSP1) is injective; preservation of operations follows from join being union and meet being intersection of these downsets. Every prime of $R \times A_J$ comes from exactly one factor. Indeed the two idempotents $(1, 0), (0, 1)$ multiply to zero and sum to the unit; a proper prime therefore contains exactly one of them. It then contains the whole corresponding factor and is the inverse image of a prime of the other factor. Hence

$$\text{Spec}(R \times A_J) = \text{Spec } R \sqcup \text{Spec } A_J. \quad (\text{LSP24})$$

The contractions under (LSP23) are exactly

$$\begin{aligned} \mathfrak{r} \in \text{Spec } R &\longmapsto \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in \mathfrak{r}\}, \\ (j, U) &\longmapsto \{(0, \lambda) : j \not\leq \lambda\}. \end{aligned} \quad (\text{LSP25})$$

To check the second formula, an idempotent e_D belongs to $\mathfrak{p}_{j,U}$ exactly when $j \notin D$, by (LSP5). Every nonzero supported amplitude, and the supported zero at top support, maps to the unit function. The first formula follows directly from the amplitude coordinate. The original prime classification (ASL12) now proves that (LSP24) maps surjectively onto the whole original spectrum of $G_L(R)$. The fibre over an arithmetic prime is its one R -prime; the fibre over a support prime labelled j is exactly all upsets $U \subseteq J_{\leq j}$ containing j . This is the precise simultaneous relation between the ring amplitudes, every support prime, and the additional tropical prime geometry. It makes no cohomological or purity identification.

2.6 The complete Frobenius action on these primes

For every real $t > 0$, coefficient Frobenius is the automorphism $\text{Fr}_t(f)(x) = f(x)^t$. It preserves every zero, equality, and strict inequality in (LSP3), so

$$\text{Fr}_t^{-1}(\mathfrak{p}_{j,U}) = \mathfrak{p}_{j,U} \quad (t > 0). \quad (\text{LSP26})$$

Thus its action on this ordinary prime spectrum is the identity, although its action on coefficient functions and on the local semirings remains the indicated power operation. At the additional zero element of $\mathbb{N}_0^\times$, use the actual convention (ASL32), which gives $\text{Fr}_0(f) = e_{\text{supp } f}$. By (LSP5), the spectral action is precisely

$$\text{Fr}_0^{-1}(\mathfrak{p}_{j,U}) = \mathfrak{p}_{j,\{j\}}, \quad (j, U) \longmapsto (j, \{j\}). \quad (\text{LSP27})$$

More explicitly, the semiring maps $L \xrightarrow{D \mapsto e_D} A_J \xrightarrow{f \mapsto \text{supp } f} L$ compose to the identity on L , while their reverse composite is Fr_0 . The corresponding spectral maps are the projection $(j, U) \mapsto j$ and its section $j \mapsto (j, \{j\})$. This proves an actual retraction onto the evaluation primes. It does not identify the additional primes with them: their distinct specializations and localized functions remain (LSP10)–(LSP18).

2.7 Exact naturality for every finite bounded-lattice morphism

Let $\phi : L \rightarrow L'$ preserve zero, unit, join, and meet, with both lattices finite, nontrivial, and distributive. Put $J = J(L)$ and $J' = J(L')$. The dual monotone map $g : J' \rightarrow J$ is uniquely specified by

$$j' \leq \phi(\lambda) \iff g(j') \leq \lambda \quad (\lambda \in L). \quad (\text{LSP28})$$

To prove existence, evaluate ϕ at the character of j' . The resulting bounded lattice character of L is the character of a unique $j \in J$, by the proof of (LSP5), and this j defines $g(j')$. For monotonicity, if $j' \leq k'$ then $k' \leq \phi(\lambda)$ implies $j' \leq \phi(\lambda)$ for every λ . Taking $\lambda = g(k')$ in (LSP28) gives $g(j') \leq g(k')$. Under (LSP1) the entire map is therefore $\phi(D) = g^{-1}(D)$. Conversely inverse image under any monotone g preserves downsets, empty and full sets, unions, and intersections, and hence gives exactly such a ϕ .

The induced coefficient map has the full coordinate formula

$$\Phi = \text{id}_K \otimes \phi : A_J \longrightarrow A_{J'}, \quad \Phi(f) = f \circ g. \quad (\text{LSP29})$$

On a pure tensor $a \otimes D$ it sends its characteristic support to $g^{-1}(D)$ and keeps the coefficient a , so (LSP29) follows on all finite sums. Pointwise operations also prove directly that this is a unital semiring map. It commutes with every positive real Frobenius and with Fr_0 , since both power and the nonzero-support test commute with precomposition.

Fix an arbitrary target prime $\mathfrak{p}_{j',U'}$. Its exact contraction is

$$\Phi^{-1}(\mathfrak{p}_{j',U'}) = \mathfrak{p}_{j,U}, \quad j = g(j'), \quad U = \{x \leq j : \exists u' \in U', g(u') \leq x\}. \quad (\text{LSP30})$$

The set U is the upward closure of $g(U')$ inside $J_{\leq j}$, is upward closed there, and contains j . Membership of $\Phi(f)$ in the target prime says either $f(g(j')) = 0$ or $f(g(u')) > f(g(j'))$ for some $u' \in U'$. Every $g(u')$ belongs to U , so this implies membership on the right of (LSP30). Conversely, if $f(x) > f(j)$ for $x \in U$, choose $u' \in U'$ with $g(u') \leq x$. Antitonicity gives $f(g(u')) \geq f(x) > f(j)$. The zero alternative is identical on both sides. This proves equality of the two ideals, with no loss of the upward-closure labels.

The associated local map is just as explicit:

$$(A_J)_{\mathfrak{p}_{j,U}} \longrightarrow (A_{J'})_{\mathfrak{p}_{j',U'}}, \quad h \longmapsto h \circ (g|_{J'_{\leq j'}}). \quad (\text{LSP31})$$

Its domain is correct because $g(J'_{\leq j'}) \subseteq J_{\leq j}$. An allowed nonzero-support downset in (LSP15) pulls back to a downset. On U' , the arguments of h lie in $g(U') \subseteq U$, where h is antitone, so the image satisfies the target order condition. Formula (LSP30) makes every denominator outside the source prime map outside the target prime. Thus (LSP31) is the canonical localization map, as can also be checked on its fractions $\Phi(f)/\Phi(q) = (f/q) \circ g$. It is local: a value function h is a unit precisely when $h(j) > 0$ and it is constant on U , by (LSP17). Its image is a unit precisely when $h(g(j')) > 0$ and it is constant on $g(U')$. These conditions are equivalent, since each $x \in U$ lies between some $g(u')$ and j , all in U ; the antitone inequalities force $h(x) = h(j)$ when both endpoint values equal $h(j)$. This proves the inverse image of the target maximal ideal is the source maximal ideal.

All maps compose with their full labels. If a second bounded-lattice map $L' \rightarrow L''$ has dual $r : J'' \rightarrow J'$, the dual of its composite with ϕ is $g \circ r$ by (LSP28), and the coefficient maps compose by the ordinary identity $(f \circ g) \circ r = f \circ (g \circ r)$. The required equality for prime labels is

$$\uparrow_{\leq g(r(j''))} g \left(\uparrow_{\leq r(j'')} r(U'') \right) = \uparrow_{\leq g(r(j''))} (g \circ r)(U''). \quad (\text{LSP32})$$

Indeed an intermediate element y' obeys $r(u'') \leq y' \leq r(j'')$ for some $u'' \in U''$, hence $g(r(u'')) \leq g(y') \leq g(r(j''))$, proving one inclusion. The other follows because each $r(u'')$ is itself in the intermediate upward closure. This proves (LSP32) and the composition of (LSP30). Localized precomposition in (LSP31) then composes identically on every retained coordinate.

The maps also fit the original split-support base, not only its tropical coefficient extension. For any fixed nonzero commutative ring R , define

$$G_\phi : G_L(R) \longrightarrow G_{L'}(R), \quad (r, \lambda) \longmapsto (r, \phi(\lambda)). \quad (\text{LSP33})$$

This is defined on the original carriers: nonzero amplitudes have top support, which ϕ preserves, while zero amplitudes admit every image support. The operations in (ASL1) prove that it preserves addition, multiplication, zero, and unit. It also sends the global zero τ to τ and the supported zero e to e . The companion maps (LSP23) obey the exact equality

$$(\text{id}_R \times \Phi) \circ \iota_L = \iota_{L'} \circ G_\phi. \quad (\text{LSP34})$$

For verification at (r, λ) , both sides have amplitude r and characteristic support $g^{-1}(\{j : j \leq \lambda\})$, which equals $\{j' : j' \leq \phi(\lambda)\}$ by (LSP28).

Write $\mathfrak{s}_j = \{(0, \lambda) : j \not\leq \lambda\}$ for an original support prime and $Q_{\mathfrak{r}}$ for its original arithmetic prime at $\mathfrak{r} \in \text{Spec } R$, as in (ASL12). Direct contraction gives

$$G_\phi^{-1}(\mathfrak{s}_{j'}) = \mathfrak{s}_{g(j')}, \quad G_\phi^{-1}(Q_{\mathfrak{r}}) = Q_{\mathfrak{r}}. \quad (\text{LSP35})$$

The first follows from (LSP28), with no nonzero supported amplitude entering a support prime. The second follows because amplitude is unchanged and all zero-supported elements are in every arithmetic prime. Consequently the spectral contraction square associated to (LSP34) commutes explicitly: on tropical primes it is $(j', U') \mapsto (g(j'), U) \mapsto g(j')$, which also results from $(j', U') \mapsto j' \mapsto g(j')$; on arithmetic primes both paths retain $\mathfrak{r}$. Each entire fibre of upsets in (LSP25) therefore maps by the exact upward-closure formula (LSP30), not merely by the map on its base point.

Finally the full original support stalks participate in the same local square. In finite-lattice coordinates, (ASL15)–(ASL16) give

$$(G_L(R))_{\mathfrak{s}_j} \simeq \text{Down}(J_{\leq j}), \quad (r, \lambda) \mapsto \{x \leq j : x \leq \lambda\}. \quad (\text{LSP36})$$

Indeed among the lattice denominators containing j , the least is $J_{\leq j}$, and localizing that idempotent restricts every downset to it. The map of these stalks induced by G_ϕ is

$$D \mapsto \{x' \leq j' : g(x') \in D\}. \quad (\text{LSP37})$$

Their embeddings into the two local tropical semirings send D to e_D , which satisfies every support and order condition in (LSP15). Applying (LSP31) to such an idempotent yields exactly the characteristic function of (LSP37). This proves commutation of the local coefficient square, retaining each original stalk and each chosen prime label U, U' in its proper place.

3 The absolute arithmetic curve over the full support spectrum

3.1 Sources, conventions, and the objects retained

Alain Connes and Caterina Consani’s original author source `FF.tex`, *On the Absolute Geometry of Spec Z and the Fargues–Fontaine curve*, arXiv:2606.06604, was read in full, lines 1–1764. Its SHA-256 is

14c17a95040be7bfc57bb5f87e58907a05
cacd0bcb1b6967b2892cf3d9bc1229.

The precise inputs are Section `sect9.1`, Proposition `prop12`, and the following source locators:

```
prop:monoid_structure_morphisms
defnloc, propflat
prop:archimedean_points_C
prop:nonarchimedean_points_C.
```

The notation of that source is preserved:

$$\theta(\eta) = [\mathbb{Z}], \quad \Theta(\eta) = [\{0\}], \quad \Theta(p) = [\mathbb{Z}[1/p]], \quad \kappa\Theta = \theta. \quad (\text{ACF1})$$

The first equality concerns an arithmetic divisor class; the second concerns a point of the topos. In particular the source sheaf is

$$\begin{aligned} \mathcal{F} &= \Theta^{-1}\mathcal{O}_{\mathbf{F}_1}, & \mathcal{O}_{\mathbf{F}_1} &= \mathbf{F}_1[T], \\ \mathcal{F}_\eta &= \mathbf{F}_1, & \mathcal{F}_p &= \mathbf{F}_1[T^{H_p^+}], & H_p &= \mathbb{Z}[1/p], & H_p^+ &= H_p \cap [0, \infty). \end{aligned} \quad (\text{ACF2})$$

The original sheaf on the topos and its generic stalk are different objects in this formula.

For the coefficient foundation the input is *Split Support Geometry: Universal Zero Fibres, Arithmetic Curves, and Frobenius-Perfect Quantization*, source byline *The Clankers*, June 2026, pre-publication Version 11. The complete original proof file is pinned at commit a2851f9e. The locators used here are Definition `def:lattice-split`, Theorems `thm:lattice-ideal-classification`, `thm:lattice-prime-classification`, `thm:spectral-ordinal-sum`, Proposition `prop:classical-collapse`, and Definition `def:split-absolute-point-map`. Their relevant source ranges 1221–1311, 1606–1749, and 4591–4788 were read. All extensions proved below retain every lattice label and reproduce the necessary arguments.

Let L be a bounded distributive lattice, including the possible one-element lattice. For a nonzero commutative ring R set

$$\begin{aligned} S_R &= G_L(R) = \{z_\lambda = (0, \lambda) : \lambda \in L\} \cup \{\widehat{a} = (a, 1_L) : a \in R\}, \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), & (a, \lambda)(b, \mu) &= (ab, \lambda \wedge \mu), \\ \tau &= z_{0_L}, & e &= z_{1_L} = \widehat{0}, & 1 &= \widehat{1}, & q(a, \lambda) &= a. \end{aligned} \quad (\text{ACF3})$$

These operations close on the displayed carrier: an amplitude different from zero forces support 1_L , while a factor with smaller support has zero amplitude. The ring and lattice distributive laws give the semiring laws component by component. Thus q is a unital semiring map. The notation $\widehat{0}$ always denotes e , not τ .

3.2 The full prime spectrum and its arithmetic retraction

Here Spec means prime ideals of the additive semiring: an ideal contains τ , is closed under addition, and absorbs multiplication by every element. It does not mean prime ideals of only the underlying multiplicative monoid. Put $S = S_{\mathbb{Z}}$, $X_L = \text{Spec } S$, and $Y = \text{Spec } \mathbb{Z}$.

Proposition 1 (All primes and all basic opens). *Write $\text{Spec}_{\text{lat}} L$ for the proper prime lattice ideals, where ideals are lower, contain 0_L , and are closed under finite joins. The points of X_L are exactly*

$$\begin{aligned} P_J &= \{z_\lambda : \lambda \in J\}, & J &\in \text{Spec}_{\text{lat}} L, \\ Q_{\mathfrak{p}} &= q^{-1}(\mathfrak{p}), & \mathfrak{p} &\in Y. \end{aligned} \quad (\text{ACF4})$$

Every P_J is strictly contained in $Q_{(0)}$, and $Q_{(0)} \subsetneq Q_{(p)}$ for every rational prime p . Within the support stratum inclusion of primes is exactly inclusion of J . With $A = \text{Spec}_{\text{lat}} L$, the basic opens are

$$\begin{aligned} D_S(z_\lambda) &= \{P_J : \lambda \notin J\}, \\ D_S(\widehat{n}) &= A \cup i(D_{\mathbb{Z}}(n)) \quad (n \in \mathbb{Z}), \\ D_S(e) &= A, & V_S(e) &= i(Y), & i(\mathfrak{p}) &= Q_{\mathfrak{p}}. \end{aligned} \quad (\text{ACF5})$$

In the second line $n = 0$ is allowed, and then the right side is A .

Proof. If an ideal I contains $\widehat{n}$ with $n \neq 0$, then multiplication by $\widehat{-1}$ and addition give $e \in I$. If it contains e initially the same conclusion is already available. Then $ez_\lambda = z_\lambda$ puts the entire

zero fibre in I . The amplitudes whose supported representatives belong to I form a ring ideal, so $I = q^{-1}(\mathfrak{a})$ for a unique ring ideal $\mathfrak{a}$. Otherwise I consists solely of z_λ for a proper lattice ideal J : addition gives join closure and $z_\mu z_\lambda = z_\mu$ when $\mu \leq \lambda$ gives downward closure. Conversely these constructions are ideals. The amplitude ideals are prime exactly when $\mathfrak{a}$ is prime, by surjectivity of q . For a support ideal, a product of two top-supported elements is still top-supported, even if its amplitude is zero. A product of a top-supported element with z_λ has support λ . For two zero elements primality is precisely $\lambda \wedge \mu \in J \Rightarrow \lambda \in J$ or $\mu \in J$. This proves exhaustion and primality. The strict cross-stratum inclusion follows because every $Q_{\mathfrak{p}}$ contains e and no P_J does. Membership in the displayed ideals proves every formula for the basic opens directly. $\square$

Proposition 2 (A retraction which retains the support stratum). *The map $i : Y \rightarrow X_L$ is a homeomorphism onto the closed subspace $V(e)$. There is a continuous retraction*

$$\begin{aligned} r : X_L &\longrightarrow Y, & r(P_J) &= \eta, & r(Q_{(0)}) &= \eta, & r(Q_{(p)}) &= (p), \\ & & ri &= \text{id}_Y. \end{aligned} \tag{ACF6}$$

Its fibres are $r^{-1}(\eta) = A \cup \{Q_{(0)}\}$ and $r^{-1}((p)) = \{Q_{(p)}\}$. For $n \neq 0$,

$$r^{-1}D_{\mathbb{Z}}(n) = D_S(\widehat{n}). \tag{ACF7}$$

For $n = 0$ the left side is empty, whereas $D_S(\widehat{0}) = A$.

Proof. The formulas in ACF5 give the homeomorphism onto $V(e)$ and show that ACF7 holds for every nonzero integer, since every nonempty basic open of Y contains η . The inverse image of the empty open is empty. These opens form a basis, proving continuity. All statements about fibres and the composite follow from the explicit assignments. In particular r is a quotient map: if $r^{-1}U$ is open, then $U = i^{-1}r^{-1}U$ is open. $\square$

For $0_L \neq 1_L$, this continuous retraction is not induced by a unital semiring homomorphism $\mathbb{Z} \rightarrow S$. Indeed any such homomorphism would send 1 to $\widehat{1}$. If u were the image of -1 , then $\widehat{1} + u = \tau$ would be required. Its left support is 1_L , which cannot equal 0_L . The continuous map and its sheaf pullback nevertheless exist exactly as proved.

3.3 The pullback sheaf and its exact comparison

Define a spherical-algebra sheaf on the actual full base by

$$\mathcal{F}_L = r^{-1}\mathcal{F} = (\Theta \circ r)^{-1}\mathcal{O}_{\mathbf{F}_1}. \tag{ACF8}$$

For a continuous map, the inverse image is the sheafification of the presheaf assigning to $W \subseteq X_L$ the colimit of $\mathcal{F}(V)$ over opens $V \subseteq Y$ for which $W \subseteq r^{-1}V$. This specifies the sheaf, not only a list of stalks.

Proposition 3 (Exact restriction, stalks, and direct image). *There are canonical isomorphisms*

$$i^{-1}\mathcal{F}_L \simeq \mathcal{F}, \quad r_*\mathcal{F}_L \simeq \mathcal{F}, \quad (\mathcal{F}_L)_x \simeq \mathcal{F}_{r(x)}. \tag{ACF9}$$

Consequently

$$(\mathcal{F}_L)_{P_J} = \mathbf{F}_1, \quad (\mathcal{F}_L)_{Q_{(0)}} = \mathbf{F}_1, \quad (\mathcal{F}_L)_{Q_{(p)}} = \mathbf{F}_1[T^{H_p^+}]. \tag{ACF10}$$

All generic-direction stalk maps from the last object to either of the first two are

$$\epsilon_p(0) = 0, \quad \epsilon_p(T^h) = 1 \quad (h \in H_p^+). \tag{ACF11}$$

Maps between stalks $\mathbf{F}_1$ in ACF10 are identities.

Proof. The stalk of the inverse-image presheaf at x is the colimit of $\mathcal{F}(V)$ over neighbourhoods of $r(x)$: every such V yields the neighbourhood $r^{-1}V$ of x , and every representative of the presheaf stalk is already represented on some such V . The two identifications respect equality of germs. Sheafification preserves this stalk, proving the third isomorphism. The first follows either by the same construction and $ri = \text{id}$ or by the canonical composition isomorphism for inverse images.

The second isomorphism holds for any sheaf E on Y in place of $\mathcal{F}$. The adjunction unit sends $E(U)$ into $(r^{-1}E)(r^{-1}U)$. Restriction along i gives a left inverse using $i^{-1}r^{-1}E = E$. If two sections over $r^{-1}U$ have the same restriction, their germs agree at every point $Q_{\mathfrak{p}}$ with $\mathfrak{p} \in U$. When U is nonempty it contains η . Each support point is a generization of $Q_{(0)}$, so the germ at that support point is the image of the germ at $Q_{(0)}$. The two sections therefore agree at all stalks and are equal. The case $U = \emptyset$ is automatic. This proves bijectivity, respecting all sheaf operations and restrictions.

The source has all endomorphisms $\text{Fr}_m(T^h) = T^{mh}$, including $m = 0$, with the absorbing zero fixed. Every sheaf generization map commutes with them. On $\mathbf{F}_1$, Fr_0 is the identity. Thus a generization g to $\mathbf{F}_1$ satisfies $g(T^h) = \text{Fr}_0 g(T^h) = g(T^0) = 1$. It sends the absorbing zero to zero. This proves ACF11 without selecting an unsupported alternative augmentation. All other displayed maps are unital maps $\mathbf{F}_1 \rightarrow \mathbf{F}_1$, hence identities. $\square$

Proposition 4 (The actual semiring structure sheaf). *Let $\mathcal{O}_X$ be the sheaf of locally represented fractions of S on X_L . For a lattice prime J define*

$$\begin{aligned} L_J &= L[(L \setminus J)^{-1}], \\ [\lambda] &= [\mu] \iff \text{there is } \nu \notin J \text{ with } \lambda \wedge \nu = \mu \wedge \nu. \end{aligned} \tag{ACF12}$$

Its operations are induced join and meet. The stalks are

$$(\mathcal{O}_X)_{P_J} = L_J, \quad (\mathcal{O}_X)_{Q_{(0)}} = G_L(\mathbb{Q}), \quad (\mathcal{O}_X)_{Q_{(p)}} = G_L(\mathbb{Z}_{(p)}). \tag{ACF13}$$

The generic-direction map from an arithmetic stalk to L_J is $(a, \lambda) \mapsto [\lambda]$. The maps between arithmetic stalks are the usual localization of amplitudes with support unchanged. If $J \subseteq K$, the map $L_K \rightarrow L_J$ is the canonical localization.

Proof. The locally represented fraction definition gives the stalk S_P at a prime P : a fraction near P has denominator outside P , and equality of fraction germs is the localization equality witnessed by another such denominator. At $Q_{\mathfrak{p}}$, every allowed denominator is $\hat{n}$ with $n \notin \mathfrak{p}$ and $n \neq 0$. Thus

$$(a, \lambda) / \hat{n} \longmapsto (a/n, \lambda)$$

is onto $G_L(\mathbb{Z}_{\mathfrak{p}})$. The usual localization cross-multiplication criterion proves injectivity: an additional denominator is top-supported, so equality preserves the supports exactly, and its amplitude is a nonzero integer. Operations match component by component.

At P_J , e is an allowed denominator. The relation $e^2 = e$ makes $e = 1$ after localization. Hence every (a, λ) equals its product with e , namely z_{λ} . This first localization is exactly L : $z_{\lambda} \mapsto \lambda$ is the isomorphism. The remaining allowed zero denominators are the z_{ν} with $\nu \notin J$, so the second localization is L_J . An invertible idempotent is 1; thus every fraction is represented by its numerator, and the usual fraction equality becomes precisely ACF12. This also proves the formula for all maps, because they are induced by the original localization maps. The top and bottom of L_J remain different: equality would require $\nu \notin J$ with $\nu = 0_L$, impossible for a lattice ideal. $\square$

Proposition 5 (Coefficient comparison over the whole base). *On Y define the sheaf B_L by $B_L(U) = G_L(\mathcal{O}_Y(U))$ for nonempty opens U , and the terminal semiring for the empty open. There is a canonical sheaf morphism*

$$\Phi : r^{-1}B_L \longrightarrow \mathcal{O}_X \tag{ACF14}$$

which is an isomorphism at every arithmetic stalk and a surjection

$$G_L(\mathbb{Q}) \longrightarrow L_J, \quad (a, \lambda) \longmapsto [\lambda] \quad (\text{ACF15})$$

at P_J . Its exact equality relation at that stalk is equality of the classes $[\lambda]$ and $[\mu]$ in ACF12, regardless of amplitudes.

Proof. Every two nonempty opens of Y meet, and their rings of sections embed in $\mathbb{Q}$. Compatible local sections therefore have one support label on all members of a nonempty cover and glue their amplitudes by the ring sheaf axiom. A nonzero glued amplitude forces top support. This proves the claimed sheaf description. Its stalks are $G_L(\mathbb{Z}_{(p)})$ and $G_L(\mathbb{Q})$: passage to the stalk keeps every support label and takes the corresponding filtered localization of the amplitude.

For $D_{\mathbb{Z}}(n)$ with $n \neq 0$, a section of B_L is represented by $(a/n^k, \lambda)$. Send it to the local fraction $(a, \lambda)/\widehat{n^k}$ over $D_S(\widehat{n})$, which is its full inverse image by ACF7. Equality and restriction are preserved by the fraction rules. These maps on a basis give $B_L \rightarrow r_*\mathcal{O}_X$ and hence ACF14. At arithmetic stalks the map is the isomorphism just proved. At a support stalk all nonzero integers have been inverted, and the subsequent inversion of e and z_ν gives ACF15. Every $[\lambda]$ is hit by z_λ . Formula ACF12 gives precisely its fibres, proving all assertions. $\square$

The functor used here is Connes–Consani’s H from *Absolute algebra and Segal’s Γ -rings: au dessous de $\text{Spec } \mathbb{Z}$* , original `F1_corr.tex`, formula `functsum` (lines 433–442) and Lemma `ssalg` (lines 519–533). That source was read completely, lines 1–1336, for the preceding coefficient derivation. Applying the semiring functor H degree by degree gives the corresponding spherical-algebra map $r^{-1}HB_L \rightarrow H\mathcal{O}_X$. Here $(HA)(k_+) = A^k$, a pointed map sums coordinates over its fibres, and multiplication sends $(a_i), (b_j)$ to $(a_i b_j)_{i,j}$. These formulas show directly that the comparison commutes with all pointed maps and multiplications. Stalks and inverse images commute with this finite coordinate construction. Thus ACF14 is also the exact connection with the split coefficient sheaf, rather than an identification with $\mathcal{F}_L$.

Proposition 6 (The spherical comparison is forced by the generic stalk). *There is exactly one unital spherical-algebra sheaf map*

$$\mathcal{F}_L \longrightarrow H\mathcal{O}_X. \quad (\text{ACF16})$$

At $Q_{(p)}$ it sends every T^h to 1 and the source absorbing zero to the target additive zero. At all other stalks it is the unit map from $\mathbf{F}_1$.

Proof. The original endomorphism Fr_0 of $\mathcal{O}_{\mathbf{F}_1} = \mathbf{F}_1[T]$ factors through the constant spherical algebra $\mathbf{F}_1$: every monomial goes to 1 and the absorbing zero to 0. This factorization is equivariant for every Fr_m , so it pulls back to a sheaf augmentation $\mathcal{F}_L \rightarrow \mathbf{F}_1$. Compose with the unit map to $H\mathcal{O}_X$ to obtain ACF16. For uniqueness, any such morphism has the unique unit map at $Q_{(0)}$. The generization $G_L(\mathbb{Z}_{(p)}) \rightarrow G_L(\mathbb{Q})$ is injective. Compatibility with ACF11 forces the image of every T^h at $Q_{(p)}$ to be the unique element whose image in $G_L(\mathbb{Q})$ is 1, namely 1. At support points the source is $\mathbf{F}_1$, whose unital map is unique. A spherical monoid-algebra map is determined by its values on monomials, as is also proved explicitly below. Hence the maps agree on every stalk and are equal as sheaf maps. $\square$

3.4 Every point of the finite-prime stalk, with every zero label

Put $S_{\mathbb{C}} = G_L(\mathbb{C})$. A spherical-algebra point of $\mathbf{F}_1[T^{H_p^+}]$ with values in $HS_{\mathbb{C}}$ is exactly a pointed unital multiplicative monoid map to $S_{\mathbb{C}}$. Explicitly, for a pointed monoid M , the source in degree k_+ is $M \wedge k_+$. A monoid map $f : M \rightarrow S_{\mathbb{C}}$ induces the map which sends (m, j) to the tuple with $f(m)$ in coordinate j and τ elsewhere. The wedge basepoint goes to the all- τ tuple. A pointed map between finite pointed sets moves this one occupied coordinate or sends it to the basepoint; the coordinate-sum formula of H does exactly the same. Products agree because f is

multiplicative. Conversely the maps $1_+ \rightarrow k_+$ selecting one coordinate force this formula for every natural map. Its degree-one component must preserve multiplication, the unit and the absorbing zero. This proves the source adjunction in the present case, including its zero convention.

Consequently a point is a character

$$\chi : H_p^+ \longrightarrow S_{\mathbb{C}}, \quad \chi(0) = 1, \quad \chi(h + h') = \chi(h)\chi(h'), \quad (\text{ACF17})$$

with the additional source absorbing zero mapped to τ . The exponent 0 and the absorbing zero are distinct source elements.

Proposition 7 (Complete root-sequence classification). *Define*

$$\mathcal{U}_p = \{(u_n)_{n \geq 0} \in (\mathbb{C}^\times)^\mathbb{N} : u_{n+1}^p = u_n\}. \quad (\text{ACF18})$$

Then the full monoid of points is the disjoint union

$$\begin{aligned} \mathcal{P}_{p,L} &= \mathcal{U}_p \sqcup \{Z_\lambda : \lambda \in L\}, \\ Z_\lambda(0) &= 1, \quad Z_\lambda(h) = z_\lambda \quad (h > 0). \end{aligned} \quad (\text{ACF19})$$

A sequence $u \in \mathcal{U}_p$ gives the character

$$\chi_u(a/p^n) = \widehat{u_n^a} \quad (a \in \mathbb{N}), \quad \chi_u(0) = 1. \quad (\text{ACF20})$$

Products and the amplitude map are exactly

$$\begin{aligned} \chi_u \chi_v &= \chi_{uv}, \quad \chi_u Z_\lambda = Z_\lambda, \quad Z_\lambda Z_\mu = Z_{\lambda \wedge \mu}, \\ q_* : \mathcal{P}_{p,L} &\longrightarrow \varprojlim_{x \mapsto x^p} \mathbb{C} = \mathcal{U}_p \sqcup \{\mathbf{0}\}, \\ q_*^{-1}(\mathbf{0}) &= \{Z_\lambda : \lambda \in L\}, \quad q_*^{-1}(u) = \{\chi_u\} \quad (u \in \mathcal{U}_p). \end{aligned} \quad (\text{ACF21})$$

This inverse limit is asserted as a multiplicative monoid, not as a perfectoid field or as the ordinary field $\mathbb{C}$.

Proof. Given a character put $s_n = \chi(1/p^n)$. Then $s_{n+1}^p = s_n$. Conversely, from any sequence with this property define $\chi(a/p^n) = s_n^a$, with $s_n^0 = 1$. To check independence of the representation, pass to a common denominator p^N : $s_n^a = s_N^{ap^{N-n}}$; equal rational exponents give equal integer powers. The same common-denominator calculation proves multiplicativity.

Write $s_n = (u_n, \lambda_n)$. Idempotence of lattice meet gives $\lambda_{n+1} = \lambda_n$ for every n . If one u_n is zero, the relations and the absence of nonzero nilpotents in $\mathbb{C}$ force all $u_n = 0$. Thus all $s_n = z_\lambda$ for a single arbitrary label λ , producing Z_λ . Otherwise all u_n are nonzero and every support is 1_L , producing exactly ACF18 and ACF20. All products follow by evaluating at the generators $1/p^n$. The unit is the constant character 1 in $\mathcal{U}_p$, and the absorbing point is Z_{0_L} . The amplitude formula follows coordinate by coordinate; a complex coherent sequence is either everywhere zero or everywhere nonzero. This proves exhaustion, injectivity, and every fibre statement. $\square$

The labelled zeros form a multiplicatively absorbing ideal of the point monoid, with multiplication given by meet. They remain distinct before the explicitly stated amplitude quotient. Each support prime P_J has a unique algebraic $HS_{\mathbb{C}}$ -valued stalk point, because its stalk is $\mathbf{F}_1$; these points are still indexed by their different base points P_J . The map on point sets induced by the stalk augmentation ACF11 sends this unique generic point to the constant unit character in ACF19. It does not select a zero branch.

Proposition 8 (Every Frobenius, including exponent zero). *For every integer $m \geq 1$, source Frobenius induces*

$$\text{Fr}_m^*(\chi_u) = \chi_{(u_n^m)_n}, \quad \text{Fr}_m^*(Z_\lambda) = Z_\lambda. \quad (\text{ACF22})$$

For $m = 0$, every point is sent to the constant unit character. The p -th Frobenius is a bijection, with inverse on the group part $(u_n) \mapsto (u_{n+1})$, and with every Z_λ fixed. More generally, write $m = p^a d$ with $(d, p) = 1$. On $\mathcal{U}_p$, Fr_m^* is surjective with kernel canonically isomorphic to $\mu_d(\mathbb{C})$ by the zeroth coordinate after removal of the p -power automorphism. For $d > 1$ it is not injective. All these maps commute with the sheaf generizations ACF11.

Proof. Precomposition sends $\chi(h)$ to $\chi(mh) = \chi(h)^m$, giving ACF22. For $m = 0$ it gives $\chi(0) = 1$ at every exponent, while the extra source absorbing zero is still mapped to τ . The inverse for p follows from $u_{n+1}^p = u_n$ and fixes the constant support sequences. For d prime to p , to lift a sequence u choose $v_0^d = u_0$. Having chosen v_n , choose a p -th root w of v_n . Then w^d/u_{n+1} is a p -th root of unity. Since raising to the d -th power permutes μ_p , multiply w by the unique suitable p -th root of unity to obtain v_{n+1} with both $v_{n+1}^p = v_n$ and $v_{n+1}^d = u_{n+1}$. This constructs a preimage. In the kernel every coordinate belongs to μ_d , and raising to p is an automorphism of that finite group, so v_0 determines the whole sequence uniquely. Combining with the p -power bijection proves the assertion for m . Finally $\epsilon_p(T^{mh}) = 1$ for every h, m , which proves the compatibility even at $m = 0$. $\square$

3.5 The source's actual locality condition

In `defnloc`, Connes–Consani define a local character using a homomorphism from the ordered group H into a complete Hausdorff topological group. It must be continuous for the topology induced by its place and then extends uniquely to the corresponding local field. For H_p the places are p and ∞ ; for the trivial group there are no such places (source Proposition `classpla`). We apply this definition exactly to the unit group of the present target.

Proposition 9 (The exact boundary excluded by group extension). *The units of $S_{\mathbb{C}}$ are exactly $\{\widehat{a} : a \in \mathbb{C}^\times\}$, identified with the usual topological group $\mathbb{C}^\times$. A character ACF17 extends to a group homomorphism $H_p \rightarrow S_{\mathbb{C}}^\times$ if and only if it belongs to $\mathcal{U}_p$. Every Z_λ fails this extension, including Z_{1_L} and Z_{0_L} .*

Proof. If $(a, \lambda)(b, \mu) = 1$, then $ab = 1$, so a, b are nonzero and both supports are 1_L . Conversely $\widehat{a}^{-1} = \widehat{a^{-1}}$. A character extending to a group must take every $h > 0$ to a unit, since its value at $-h$ is an inverse. This excludes every zero branch. A character from $\mathcal{U}_p$ extends by $\rho(h-k) = \chi(h)\chi(k)^{-1}$ for $h, k \in H_p^+$; if $h-k = h'-k'$, then $h+k' = h'+k$, proving independence of the representation. The group law and uniqueness follow from the same calculation. Thus the obstruction object is exactly the retained lattice ideal $\{Z_\lambda\}$ of ACF21. $\square$

This argument does not discard the algebraic boundary: it identifies its exact failure to satisfy the source's group-valued definition. For an explicit topological semiring realization give L the discrete topology and $S_{\mathbb{C}} \subseteq \mathbb{C} \times L$ the subspace topology. Its operations are continuous, since the ring operations are continuous and maps between finite products of discrete sets are continuous. Its unit group has exactly the stated topology. Even without requiring group extension, a zero-branch character is not continuous at exponent 0: at the archimedean place $1/p^n \rightarrow 0$, and at the p -adic place $p^n \rightarrow 0$, whereas its amplitudes are constantly 0 along these sequences and its amplitude at 0 is 1.

Proposition 10 (All archimedean local characters). *The archimedean local points form the family*

$$\chi_z(h) = \widehat{\exp(zh)}, \quad z \in \mathbb{C}, \quad h \in H_p^+. \quad (\text{ACF23})$$

Different z give different characters. The trivial point is $z = 0$. Source Frobenius Fr_m^ acts by $z \mapsto mz$.*

Proof. The group H_p is dense in $\mathbb{R}$: rounding $p^n t$ to the nearest integer approximates any real t with error at most $1/(2p^n)$. A continuous homomorphism $H_p \rightarrow \mathbb{C}^\times$ is uniformly continuous for the group uniformities. Completeness of $\mathbb{C}^\times$ for its group uniformity follows from $\mathbb{C}^\times \simeq \mathbb{R} \times S^1$, by logarithm of the modulus and phase. Therefore the homomorphism extends uniquely to $\mathbb{R}$: apply it to approximating Cauchy sequences, and use uniform continuity to prove independence and continuity of the limit. For a continuous homomorphism $f : \mathbb{R} \rightarrow \mathbb{C}^\times$, choose a small interval around zero on which f takes values in a neighbourhood of 1 with a continuous logarithm g and $g(0) = 0$. For sufficiently small $s, t, s+t$, the difference $g(s+t) - g(s) - g(t)$ belongs to $2\pi i\mathbb{Z}$ and is continuous; on the connected neighbourhood of $(0, 0)$ it is therefore zero. Continuous local additivity gives $g(t) = zt$: rational subdivisions first give this equality on rational multiples of a fixed sufficiently small nonzero real argument, and continuity gives it on the entire small interval. Subdividing any real t into sufficiently small equal pieces gives $f(t) = \exp(zt)$ everywhere. Conversely each such function is continuous and multiplicative. If two parameters give the same character on H_p , density makes them agree on $\mathbb{R}$; their local logarithms have equal slopes. Frobenius follows by substituting mh . $\square$

For the p -adic family fix its parameter convention explicitly. For $x \in \mathbb{Q}_p$, let $\{x\}_p \in \mathbb{Z}[1/p] \cap [0, 1)$ be its finite sum of negative p -adic digits. Its class modulo $\mathbb{Z}$ depends only on $x \bmod \mathbb{Z}_p$. Set

$$\psi_p(x) = \exp(2\pi i \{x\}_p). \quad (\text{ACF24})$$

The map $\mathbb{Z}[1/p]/\mathbb{Z} \rightarrow \mathbb{Q}_p/\mathbb{Z}_p$ is a bijection: every p -adic class has the finite negative-digit representative and its kernel is $\mathbb{Z}[1/p] \cap \mathbb{Z}_p = \mathbb{Z}$. It respects addition, so ACF24 is a continuous additive character with kernel $\mathbb{Z}_p$.

Proposition 11 (All p -adic local characters). *The p -adic local points form the family*

$$\chi_a(h) = \widehat{\psi_p(ah)}, \quad a \in \mathbb{Q}_p, \quad h \in H_p^+. \quad (\text{ACF25})$$

The parameter is unique, the trivial point is $a = 0$, and source Frobenius acts by $a \mapsto ma$.

Proof. The group H_p is dense in $\mathbb{Q}_p$: truncate the p -adic expansion of any number at successively larger nonnegative exponents. The completeness argument just used extends each continuous character uniquely to $\mathbb{Q}_p$. A continuous homomorphism $f : \mathbb{Q}_p \rightarrow \mathbb{C}^\times$ maps a sufficiently small open subgroup $p^N \mathbb{Z}_p$ into a neighbourhood of 1 containing no nontrivial subgroup. Such a neighbourhood exists: use a bounded open interval for $\log |z|$ and an open phase arc of length less than one half-turn; powers of any nonidentity element leave it. Continuity therefore gives $f(p^N \mathbb{Z}_p) = 1$. For each $x \in \mathbb{Q}_p$, some p -power multiple belongs to $p^N \mathbb{Z}_p$, so $f(x)$ is a root of unity of p -power order.

Write $x_n = f(1/p^n)$ and use the fixed isomorphism $\mathbb{Q}_p/\mathbb{Z}_p \rightarrow \mu_{p^\infty}(\mathbb{C})$ induced by ψ_p to write $x_n = \psi_p(b_n)$, where b_n is specified modulo $\mathbb{Z}_p$. The relation $x_{n+1}^p = x_n$ gives $pb_{n+1} \equiv b_n \pmod{\mathbb{Z}_p}$. Thus $a_n = p^n b_n$ is specified modulo $p^n \mathbb{Z}_p$ and satisfies $a_{n+1} \equiv a_n \pmod{p^n \mathbb{Z}_p}$. Choose representatives recursively; their differences have valuations at least n , so they form a Cauchy sequence in $\mathbb{Q}_p$. Its limit a satisfies $a \equiv a_n \pmod{p^n \mathbb{Z}_p}$ for every n . It follows that $x_n = \psi_p(a/p^n)$. Agreement on these generators gives agreement on H_p , and continuity gives agreement on $\mathbb{Q}_p$. If two parameters work, their difference belongs to $\bigcap_{n \geq 0} p^n \mathbb{Z}_p = \{0\}$, proving uniqueness. Conversely ACF24 shows that every a gives a continuous character. Replacing h by mh proves the Frobenius action. $\square$

Proposition 12 (The exact intersection of the two local families). *As subsets of the algebraic point set ACF19, their intersection is*

$$\{\chi_{2\pi i b} : b \in \mathbb{Z}[1/p]\} = \{\chi_b \text{ of ACF25} : b \in \mathbb{Z}[1/p]\}. \quad (\text{ACF26})$$

The place-labelled local space therefore has two copies of these characters when the labels p and ∞ are retained. Forgetting the place label identifies exactly these copies. The zero-labelled ideal is disjoint from both families.

Proof. If an archimedean character $\exp(zh)$ is also p -adic local, then its value at $h = 1$ is a root of unity of p -power order by the preceding proof. Therefore $\exp z = \exp(2\pi ic/p^n)$ for integers c, n , so $z = 2\pi ib$ with $b \in \mathbb{Z}[1/p]$. For such b and $h \in H_p$, the rational number bh belongs to $\mathbb{Z}[1/p]$ and differs from $\{bh\}_p$ by an integer. Hence $\exp(2\pi ibh) = \psi_p(bh)$. This proves both sufficiency and the exact match of parameters; uniqueness follows from the preceding propositions. Exclusion of the zero-labelled ideal was proved by the unit criterion. $\square$

3.6 Comparison with the foundation's resolved point diagram

The foundation's Definition `def:split-absolute-point-map` assigns, in the two-element support case, the ordered groups $\{0\}, \mathbb{Z}, H_p$ to $P_\tau, Q_{(0)}, Q_{(p)}$, respectively. Its Corollary `cor:resolved-stalk-chain` gives the forward maps $\{0\} \hookrightarrow \mathbb{Z} \hookrightarrow H_p$, and hence $\mathbf{F}_1 \rightarrow \mathbf{F}_1[T] \rightarrow \mathbf{F}_1[T^{H_p^+}]$. This is a point-and-stalk diagram explicitly described as such in the source. The sheaf pulled back along r instead has the stalks and the reverse, generic-direction maps in ACF10–ACF11. In particular it restricts exactly to CC's original sheaf on $V(e)$; its value at $Q_{(0)}$ is not $\mathbf{F}_1[T]$.

There are exact objectwise maps relating the two intermediate objects:

$$\mathbf{F}_1 \xrightarrow{\text{unit}} \mathbf{F}_1[T] \xrightarrow{T \mapsto 1} \mathbf{F}_1, \quad \text{whose composite is the identity.} \quad (\text{ACF27})$$

The following calculation records the precise defect when this augmentation is compared with the forward inclusion into the finite stalk. It does not assert a sheaf identification.

Proposition 13 (The coequalizer of the intermediate comparison). *Let $j, c : \mathbf{F}_1[T] \rightrightarrows \mathbf{F}_1[T^{H_p^+}]$ be the maps $j(T) = T^1$ and $c(T) = 1$. At the level of pointed multiplicative monoids, their coequalizer is*

$$\mathbf{F}_1[T^{H_p^+}] \longrightarrow \mathbf{F}_1[H_p/\mathbb{Z}], \quad T^h \longmapsto [h], \quad 0 \longmapsto 0. \quad (\text{ACF28})$$

The target is the group $H_p/\mathbb{Z}$ with an extra absorbing zero; its identity is $[0]$, distinct from that zero. Applying the spherical monoid-algebra functor gives the spherical-algebra coequalizer. Its $HS_{\mathbb{C}}$ -valued points are exactly the characters ACF25 with $a \in \mathbb{Z}_p$. They have no extra zero-support branches.

Proof. The imposed relation is $T^1 = 1$. Every $h = a/p^n \in H_p^+$ then satisfies $(T^h)^{p^n} = T^a = 1$, so all monomials become invertible. The resulting group is the quotient of the group completion H_p by $\mathbb{Z}$. More explicitly, if $h - k \in \mathbb{Z}$, multiply the relation $T^1 = 1$ by the appropriate nonnegative monomial to identify T^h and T^k . Conversely their classes in $H_p/\mathbb{Z}$ can be equal only under that condition. The extra absorbing zero remains separate, witnessed by the displayed map to the pointed group. Any monoid map equalizing j, c sends T^1 to 1, makes every image monomial a unit by the displayed power identity, and therefore factors uniquely through this quotient. This proves the universal property. The monoid-algebra adjunction transports that same universal property to spherical algebras: the coordinate-inclusion argument proving it above applies to every spherical target using its degree-one multiplication and its structure maps.

A point of this quotient is a coherent sequence of roots of unity $x_{n+1}^p = x_n$ with $x_0 = 1$. Since $x_n^{p^n} = 1$, the parameter construction in the proof of ACF25 yields a unique $a \in \mathbb{Z}_p$; conversely such a satisfies $\psi_p(a) = 1$ and gives a point. All monomials in a group must map to units, excluding every Z_λ . Thus the entire point set and the exclusion are exact. $\square$

This coequalizer is an additional explicitly stated quotient, not a replacement of ACF19. It records the relation required to make the two forward maps agree. Its p -Frobenius on points is $a \mapsto pa$ on $\mathbb{Z}_p$, which is injective and not surjective; in contrast ACF22 is bijective on the

unquotiented full point set. The loss of the inverse Frobenius is exact: $T^1 = 1$ does not imply $T^{1/p} = 1$.

The support-local semiring need not be Boolean. For the three-element chain $L = \{0 < a < 1\}$ and the prime ideal $J = \{0, a\}$, the only allowed lattice denominator is 1, so ACF12 gives $L_J = L$ with all three elements distinct. For $J = \{0\}$, inverting a identifies a with 1 and leaves two elements. Thus the actual support stalks retain the different local lattice geometries even though every spherical stalk in ACF10 over them is $\mathbf{F}_1$.

The comparison quotient also gives an exact Frobenius calculation. Under the character parameter its point multiplication is addition in $\mathbb{Z}_p$, because $\psi_p((a+b)h) = \psi_p(ah)\psi_p(bh)$. Writing $m = p^a d > 0$ with $(p, d) = 1$, its Frobenius therefore has

$$\begin{aligned} \ker(m : \mathbb{Z}_p \rightarrow \mathbb{Z}_p) &= 0, & \text{coker}(m : \mathbb{Z}_p \rightarrow \mathbb{Z}_p) &= \mathbb{Z}_p/p^a \mathbb{Z}_p \simeq \mathbb{Z}/p^a \mathbb{Z}, \\ \{\text{quotient characters}\} \cap \{\text{archimedean local characters}\} &= \{\chi_{2\pi i k} : k \in \mathbb{Z}\}. \end{aligned} \quad (\text{ACF32})$$

The first line follows because $\mathbb{Z}_p$ is an integral domain and d is a unit, with the last quotient identified by its first a p -adic digits. The second follows from ACF26 and $\mathbb{Z}_p \cap \mathbb{Z}[1/p] = \mathbb{Z}$. Thus the full algebraic point group has the prime-to- p kernel μ_d in ACF22, whereas this specific comparison quotient has the p -primary cokernel in ACF32. Both statements retain their respective original domains.

3.7 An exact diagram of the retained data

For $L = \{0, a, b, 1\}$ with $a \wedge b = 0$ and $a \vee b = 1$, there are two support primes $P_{\{0,a\}}$ and $P_{\{0,b\}}$, both below $Q_{(0)}$. The following diagram displays the maps and stalk values; every horizontal arrow points in the generization direction for stalk maps:

$$\begin{array}{ccccc} \mathbf{F}_1[T^{H_p^+}] & \xrightarrow{\epsilon_p} & \mathbf{F}_1 & \xrightarrow{\text{id}} & \mathbf{F}_1 \\ T^h \mapsto 1 \downarrow & & \downarrow \text{unit} & & \downarrow \text{unit} \\ HG_L(\mathbb{Z}_{(p)}) & \longrightarrow & HG_L(\mathbb{Q}) & \longrightarrow & HL_J. \end{array} \quad (\text{ACF29})$$

The three columns lie over $Q_{(p)}$, $Q_{(0)}$, P_J , and the last column is present separately for each of the two support primes. Its bottom horizontal map is support localization, not an amplitude map. For this Boolean lattice $L_J \cong \{0, 1\}$: localizing the unique nontrivial element outside J makes it 1, and its complementary element becomes 0. Before this localization the finite-prime point boundary consists of the four distinct characters Z_0, Z_a, Z_b, Z_1 . Amplitude collapses those four to one zero character; locality selects only the group part; neither operation silently changes the other. The source locators for the top row are FF Proposition [prop12](#) and its Frobenius structure; the bottom row is proved in ACF12–ACF16, and the four labelled characters in ACF19–ACF21.

3.8 The assembled point object and its exact local map

The base, all its spherical stalks, and all their algebraic characters can be assembled without any implicit identification. Define the set of base-labelled algebraic points, with its displayed projection to X_L , by

$$\begin{aligned} \mathfrak{P}_L &= \coprod_{J \in \text{Spec}_{\text{lat}} L} \{(P_J, *)\} \sqcup \{(Q_{(0)}, *)\} \\ &\sqcup \coprod_{p \text{ prime}} (\{Q_{(p)}\} \times (\mathcal{U}_p \sqcup L)). \end{aligned} \quad (\text{ACF30})$$

This is a set with explicitly specified fibre monoids; no additional topology on this set is asserted. Its structure is fixed by the sheaf $\mathcal{F}_L$, the maps ACF11, and the products ACF21. The point

map for a generization from $Q_{(p)}$ to $Q_{(0)}$ or P_J goes in the opposite direction and selects the constant unit character in the finite-prime fibre. The multiplicative point at each support prime retains its base index J even when the corresponding spherical stalk map is identical to that at another support prime.

Define the place-labelled local object, retaining the trivial character at every place, by

$$\begin{aligned} \mathfrak{P}_L^{\text{loc}} &= \coprod_{p \text{ prime}} ((\{Q_{(p)}, \infty\} \times \mathbb{C}) \sqcup (\{Q_{(p)}, p\} \times \mathbb{Q}_p)), \\ \ell : \mathfrak{P}_L^{\text{loc}} &\longrightarrow \mathfrak{P}_L, \quad (Q_{(p)}, \infty, z) \longmapsto (Q_{(p)}, \chi_z), \quad (Q_{(p)}, p, a) \longmapsto (Q_{(p)}, \chi_a). \end{aligned} \tag{ACF31}$$

All fibres of ℓ have size at most two. They have size two exactly at $\chi_{2\pi ib} = \chi_b$ with $b \in \mathbb{Z}[1/p]$, by ACF26; their two preimages are the displayed distinct place-labelled points. The map has no image in the zero-labelled boundary, and there are no place-labelled points above P_J or $Q_{(0)}$, because their group is $\{0\}$ and its place set is empty. These statements follow respectively from ACF19, the unit criterion, ACF23–ACF26, and the source place definition. Thus ACF30–ACF31 give the full object and its exact local subfamilies, rather than a collapse to its classical amplitude.

Every positive source Frobenius fixes the base, fixes every boundary label, and acts on the local parameters by $z \mapsto mz$ and $a \mapsto ma$. Exponent-zero Frobenius sends every finite-prime algebraic point to its constant unit character and sends both local parameters to zero while retaining their place labels. Consequently ℓ is equivariant for every $m \geq 0$, directly by ACF22–ACF25.

4 A common sheaf for spherical, tropical, and full support coefficients

4.1 Sources and the exact base

The human sources are Alain Connes and Caterina Consani, *Geometry of the Arithmetic Site*, original `arithmetic-site_Adv_final1.tex`, specifically Definition `site`, Theorem `structure2`, Remark `caseno`, Theorem `thmspz`, and Proposition `sheavesonspz0`; and their *On the Absolute Geometry of Spec Z and the Fargues–Fontaine curve*, original `FF.tex`, Proposition `prop12`, the monoid-algebra adjunction in Section 2.3, and the Frobenius formula at source lines 150–152. The arithmetic-site source was read here at lines 598–704 and 900–1017, without a claim to exhaustive reading; FF was previously read completely, lines 1–1764. The spherical functor H is the explicit coordinate functor of Connes–Consani’s *Absolute algebra and Segal’s Γ -rings*, original `F1_corr.tex`, formula `functsum` and Lemma `sssalg`. That source was previously read completely, lines 1–1336. The coefficient foundation is *The Clankers, Split Support Geometry*, Version 11, Definition `def:lattice-split` and its complete ideal and prime classification, in the complete pinned original source. The constructions below give the common diagram for the companion calculations ACF and ASL, with the required definitions and proofs included here.

Let L be a nontrivial bounded distributive lattice. It need not be finite or Boolean. For a nonzero commutative ring R retain

$$\begin{aligned} G_L(R) &= \{(0, \lambda) : \lambda \in L\} \cup \{(a, 1_L) : a \in R\}, \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu), \\ z_\lambda &= (0, \lambda), \quad \tau = z_{0_L}, \quad e = z_{1_L}, \quad 1 = (1_R, 1_L), \quad q(a, \lambda) = a, \quad \sigma(a, \lambda) = \lambda. \end{aligned} \tag{FSC1}$$

Put $S = G_L(\mathbb{Z})$, $X = \text{Spec } S$, $Y = \text{Spec } \mathbb{Z}$, and $A = \text{Spec}_{\text{lat}} L$. The additive semiring prime spectrum has points

$$P_J = \{z_\lambda : \lambda \in J\} \quad (J \in A), \quad Q_{\mathfrak{p}} = q^{-1}(\mathfrak{p}) \quad (\mathfrak{p} \in Y). \tag{FSC2}$$

For completeness, an ideal containing a top-supported element contains e : add its product by $(-1, 1_L)$ to itself if its amplitude is nonzero. Then $ez_\lambda = z_\lambda$ puts the whole zero fibre in it, and

its amplitudes form a ring ideal. All other ideals are exactly the proper lattice ideals in the zero fibre. For an amplitude ideal primality is ring primality; for a support ideal it is meet primality. This proves FSC2 directly. Their basic opens are

$$D_S(z_\lambda) = D_L(\lambda), \quad D_S((n, 1_L)) = A \cup i(D_{\mathbb{Z}}(n)), \quad A = D_S(e), \quad i(Y) = V_S(e). \quad (\text{FSC3})$$

Here $i(\mathfrak{p}) = Q_{\mathfrak{p}}$ is the closed inclusion and $j : A \hookrightarrow X$ the open inclusion. Every open meeting $i(Y)$ is uniquely $W_U = A \cup i(U)$ for a nonempty open $U \subseteq Y$; the other opens are exactly the opens $V \subseteq A$. This follows from FSC3 because every nonempty open of Y contains its generic point and all support points are generizations of every arithmetic point. There is a continuous retraction

$$r(P_J) = r(Q_{(0)}) = \eta, \quad r(Q_{(p)}) = (p), \quad ri = \text{id}. \quad (\text{FSC4})$$

Indeed $r^{-1}D_{\mathbb{Z}}(n) = D_S((n, 1_L))$ for $n \neq 0$; $r^{-1}\emptyset = \emptyset$. These constitute the basis test. In particular $r^{-1}U = W_U$ for every nonempty open U .

4.2 The support sheaf, its direct image, and all their sections

For a lattice prime J define

$$L_J = L[(L \setminus J)^{-1}], \quad [\lambda]_J = [\mu]_J \iff \exists \nu \notin J : \lambda \wedge \nu = \mu \wedge \nu. \quad (\text{FSC5})$$

Every invertible idempotent is the unit; hence this formula follows from the localization equality, and every fraction is represented by its numerator. It makes explicit the possible loss of a label in an individual support stalk.

Lemma 14 (Sections of the support sheaf). *The restriction $\mathcal{O}_A = j^{-1}\mathcal{O}_X$ of the actual localization sheaf of S satisfies*

$$\mathcal{O}_A(D_L(\lambda)) = \downarrow \lambda, \quad a|_{D_L(\mu)} = a \wedge \mu \quad (\mu \leq \lambda), \quad (\mathcal{O}_A)_J = L_J. \quad (\text{FSC6})$$

In particular $\Gamma(A, \mathcal{O}_A) = L$. For every open $V \subseteq A$ its sections are exactly

$$\{(a_\lambda)_{D(\lambda) \subseteq V} : a_\lambda \leq \lambda, \quad a_\lambda \wedge \mu = a_\mu \text{ whenever } \mu \leq \lambda \text{ and } D(\lambda) \subseteq V\}. \quad (\text{FSC7})$$

The two operations are coordinatewise join and meet.

Proof. First $S[e^{-1}] = L$ because $e^2 = e$ makes $e = 1$ and $e(a, \lambda) = z_\lambda$. Then $L[\lambda^{-1}] = \downarrow \lambda$ by $a \mapsto a \wedge \lambda$, with unit λ . Indeed the equality of localized elements is exactly equality after meeting with λ . To verify the sheaf assertion, the prime-ideal separation theorem for bounded distributive lattices gives an injective map $\lambda \mapsto D_L(\lambda)$ preserving finite joins and meets, and these basic opens are compact. Here is the needed argument. If an ideal and filter are disjoint, an ideal maximal subject to avoiding the filter exists by Zorn's lemma and is prime: if it contains $a \wedge b$ but contains neither a nor b , adjoining either meets the filter, and distributivity makes the meet of those two filter elements belong to the original ideal, a contradiction. Apply this with the ideal $\downarrow \mu$ and filter $\uparrow \lambda$ when $\lambda \not\leq \mu$ to obtain separation. For compactness, a cover $D(\lambda) \subseteq \bigcup_i D(\mu_i)$ without a finite subcover would leave the ideal generated by all μ_i disjoint from $\uparrow \lambda$. The same prime would contradict the cover. For a finite cover $\lambda = \bigvee_i \lambda_i$, sections $a_i \leq \lambda_i$ agree on overlaps precisely when $a_i \wedge \lambda_j = a_j \wedge \lambda_i$. Their unique gluing is $\bigvee_i a_i$, by distributivity. Compactness reduces any cover of a basic open to this case. Thus FSC6 defines the localization sheaf on a basis and has the stalk FSC5. Restricting a section to every basis open gives FSC7; conversely those compatible values glue and are unique. This proves the formula for all opens, including $V = A$. $\square$

Set $\mathcal{D} = j_*\mathcal{O}_A$. There is an actual unital restriction map $d : \mathcal{O}_X \rightarrow \mathcal{D}$. Its sections and stalks are

$$\begin{array}{c|cc} & W_U = A \cup i(U) & V \subseteq A \\ \hline \mathcal{D} & L & \mathcal{O}_A(V) \\ \mathcal{O}_X & G_L(\mathcal{O}_Y(U)) & \mathcal{O}_A(V) \end{array} \quad \mathcal{D}_{Q_p} = L, \quad \mathcal{D}_{P_J} = L_J. \quad (\text{FSC8})$$

On W_U , d is σ ; on V , it is the identity. To justify the remaining entry for $\mathcal{O}_X$, localizing at an arithmetic prime gives $G_L(\mathbb{Z}_p)$, including $G_L(\mathbb{Q})$ at $Q_{(0)}$: all denominators there are nonzero integer amplitudes with top support, so labels never change. A section over W_U has one rational amplitude and one label at $Q_{(0)}$. Generization from each $Q_{(p)}$ injects $G_L(\mathbb{Z}_{(p)}) \hookrightarrow G_L(\mathbb{Q})$, forcing its amplitude to lie in $\mathcal{O}_Y(U)$ and its label to be the same. At all support points the section is determined by the germ at $Q_{(0)}$, and equals the localization of that label. Conversely the corresponding element of $G_L(\mathcal{O}_Y(U))$ gives these local fractions. This proves FSC8. Every neighbourhood of an arithmetic point contains all of A , so the direct-image stalk there is L ; at a support point it is the ordinary support stalk. Restriction from W_U to V sends (a, λ) to the section represented by λ , whose component on $D(\mu) \subseteq V$ is $\lambda \wedge \mu$.

Write $\underline{L}_X$ for the constant sheaf: its sections are the locally constant maps into the discrete set L , with pointwise lattice operations. There is a canonical unital sheaf map

$$\ell : \underline{L}_X \longrightarrow \mathcal{D}, \quad \ell_{Q_p} = \text{id}_L, \quad \ell_{P_J}(\lambda) = [\lambda]_J. \quad (\text{FSC9})$$

On W_U a locally constant map is constant: every point of A is a generization of $Q_{(0)}$ and every other arithmetic point specializes from $Q_{(0)}$, whereas a locally constant map into a discrete space is constant along either relation. Thus its sections there are L . On $V \subseteq A$ send a locally constant label function to the section whose germ at J is the localized label there; it is locally represented by a single label and therefore is a section of $\mathcal{O}_A$. This proves FSC9 on sections and under restrictions.

4.3 Tropical coefficients, labels, and Frobenius zero

For each rational prime p , put $H_p = \mathbb{Z}[1/p]$ and $K_p = H_{p, \max} = H_p \cup \{-\infty\}$ with operations maximum and ordinary addition. Its semiring zero is $\mathbf{0} = -\infty$, its semiring unit is $\mathbf{1} = 0 \in H_p$. Define

$$K_{p,L} = K_p \otimes_{\mathbb{B}} L, \quad \kappa(\lambda) = \mathbf{1} \otimes \lambda, \quad \rho \left(\bigoplus_i h_i \otimes \lambda_i \right) = \bigvee_{h_i \neq -\infty} \lambda_i. \quad (\text{FSC10})$$

The map $K_p \rightarrow \mathbb{B}$ sending every finite exponent to 1 and $-\infty$ to 0 is a semiring homomorphism. Tensoring it proves that ρ is well defined and a homomorphism; the tensor relations prove the same for κ , and $\rho\kappa = \text{id}$. Consequently κ is injective without collapsing any label. Positive Frobenius multiplies every finite h_i by its integer index n , while fixing $-\infty$. Exponent-zero Frobenius sends every finite exponent to 0 and fixes $-\infty$, so

$$\text{Fr}_0 = \kappa\rho, \quad \rho \text{Fr}_n = \rho \quad (n \geq 0), \quad \text{Fr}_n \kappa = \kappa. \quad (\text{FSC11})$$

These are semiring maps: scaling preserves maximum and exponent addition, and the zero action preserves which coefficients are absorbing zero. The last assertions also follow directly from the tensor formulas.

The source geometric morphism $\Theta : Y \rightarrow \widehat{\mathbb{N}}_0^\times$ has $\Theta(\eta) = \{0\}$ and $\Theta(p) = H_p$. Let $\mathcal{T}_L = \Theta^*(\mathcal{O}^0 \otimes_{\mathbb{B}} \underline{L})$. The source Proposition `sheavesonspz0`, including the zero Frobenius convention in Remark `caseno`, gives

$$(\mathcal{T}_L)_\eta = L, \quad (\mathcal{T}_L)_p = K_{p,L}. \quad (\text{FSC12})$$

The tensor commutes with inverse images: its finite products, coproducts, and coequalizers imposing the semiring relations are preserved by those functors. More explicitly, for every nonempty open $U \subseteq Y$ the complete section description is

$$\begin{aligned} \mathcal{T}_L(U) &= \{(\lambda, (w_p)_{p \in U}) : \lambda \in L, w_p \in K_{p,L}, \rho(w_p) = \lambda, \\ &w_p = \kappa(\lambda) \text{ for all but finitely many primes } p \in U\}. \end{aligned} \quad (\text{FSC13})$$

Operations are componentwise and restrictions discard removed prime coordinates. To prove this rather than infer a sheaf from stalks, use the source description: an original coefficient section is either absorbing zero everywhere, or finite exponents with value 0 outside a finite set of primes. Each local tensor expression therefore has a single generic label λ and is $\kappa(\lambda)$ outside a finite set. Every open of Y is quasi-compact, since a single nonempty member of a cover misses only finitely many points. A finite cover of local expressions gives precisely FSC13. Conversely, for each prime in its finite exceptional set, express w_p as a finite tensor sum; extend each finite coefficient to the source section with that value at p and exponent 0 elsewhere. On the open excluding the other exceptional primes this realizes w_p and is $\kappa(\lambda)$ elsewhere. These sections agree on overlaps and glue uniquely. Zero coefficients can be omitted, including the all-zero case. This proves both directions and every stated restriction map.

Set $\mathcal{E} = r^{-1}\mathcal{T}_L$. It has maps $\rho : \mathcal{E} \rightarrow \underline{L}_X$ and $\kappa : \underline{L}_X \rightarrow \mathcal{E}$ with $\rho\kappa = \text{id}$. Its sections on the two types of open, and its stalks, are

$$\begin{aligned} \mathcal{E}(W_U) &= \mathcal{T}_L(U), & \mathcal{E}(V) &= \text{LC}(V, L) \quad (V \subseteq A), \\ \mathcal{E}_{Q(p)} &= K_{p,L}, & \mathcal{E}_{Q(0)} &= L, \quad \mathcal{E}_{P_J} = L. \end{aligned} \quad (\text{FSC14})$$

Here LC denotes locally constant maps. For the first identity the inverse-image adjunction gives a map in one direction and restriction along i a left inverse. Sections with equal arithmetic restriction have equal germs at $Q_{(0)}$, hence at every support generalization, so the left inverse is also injective. For the second, $r|_A$ factors through the point η ; inverse image to that point is the stalk L , and pulling it back to A is the constant sheaf. The stalk formula follows either way. Restriction from W_U to V remembers the constant label λ in FSC13. Finally put

$$t = \ell\rho : \mathcal{E} \longrightarrow \mathcal{D}. \quad (\text{FSC15})$$

This is the promised unital tropical-to-support map: at arithmetic points it is ρ , at P_J it is $L \rightarrow L_J$, and on all sections it is fixed by every Frobenius, including zero.

4.4 The actual fibre product and its constrained carrier

Form the following fibre product of sheaves of unital semirings:

$$\mathcal{C} = \mathcal{O}_X \times_{\mathcal{D}} \mathcal{E}, \quad \mathcal{C}(W) = \{(s, w) : d(s) = t(w) \text{ in } \mathcal{D}(W)\}. \quad (\text{FSC16})$$

All operations are componentwise. Compatible pairs glue uniquely in the two factor sheaves, and the equality in $\mathcal{D}$ can be checked locally, so FSC16 is already a sheaf. The projections are unital homomorphisms. For any other sheaf with maps to both factors whose composites to $\mathcal{D}$ agree, their pair is the unique map to FSC16. This proves the fibre-product universal property.

Its sections on every open and its stalks are as follows:

$$\begin{aligned} \mathcal{C}(W_U) &= \{(a, \lambda, (w_p)) : a \in \mathcal{O}_Y(U), (\lambda, (w_p)) \in \mathcal{T}_L(U), a \neq 0 \Rightarrow \lambda = 1_L\}, \\ \mathcal{C}(V) &= \text{LC}(V, L) \quad (V \subseteq A). \end{aligned} \quad (\text{FSC17})$$

In the second line a label function f corresponds to the pair $(\ell f, f)$, not only to its localized value ℓf . Restrictions among the W_U localize the amplitude and remove prime coordinates. Restriction

to $V \subseteq A$ forgets the amplitude and prime coordinates and keeps the constant function λ . On the arithmetic finite stalk, abbreviate $R_p = \mathbb{Z}_{(p)}$. Then

$$\begin{aligned} \mathcal{C}_{Q_{(p)}} &\simeq \{(a, w) \in R_p \times K_{p,L} : a \neq 0 \Rightarrow \rho(w) = 1_L\}, \\ \mathcal{C}_{Q_{(0)}} &\simeq G_L(\mathbb{Q}), \quad \mathcal{C}_{P_J} \simeq L. \end{aligned} \tag{FSC18}$$

The finite-stalk pair (a, w) corresponds to $((a, \rho(w)), w)$ in the original fibre product. Its sum and product are $(a + b, w \oplus v)$ and (ab, wv) , with their exact original amplitudes and tropical coefficients. The support stalk λ corresponds to $([\lambda]_J, \lambda)$. To prove these assertions, substitute FSC8 and FSC14 into FSC16. For stalks, every finite set of section representatives can be put on one common neighbourhood, and an equality of their germs holds on a smaller common neighbourhood. Thus a stalk commutes with this finite fibre product. At a support prime the two maps to L_J are the identity on L_J and the localization $L \rightarrow L_J$, so their fibre product is L , by precisely the pair displayed above. At an arithmetic point the two maps are σ and ρ , which give the first two lines. This proves all sections and stalks rather than identifying the three input sheaves.

Write $\mathcal{R} = i_*\mathcal{O}_Y$. On W_U it is $\mathcal{O}_Y(U)$ and on every $V \subseteq A$ it is the zero ring (the singleton ring with $0 = 1$). The amplitude of an actual local fraction gives a unital map $q : \mathcal{O}_X \rightarrow \mathcal{R}$. There is an injective unital comparison

$$\mathcal{C} \hookrightarrow \mathcal{R} \times \mathcal{E}, \quad (s, w) \longmapsto (q(s), w). \tag{FSC19}$$

At finite arithmetic stalks its image is exactly the constrained subset in FSC18; at the generic arithmetic stalk it is $G_L(\mathbb{Q}) \subset \mathbb{Q} \times L$; at support stalks it is all of $\{0\} \times L$. Injectivity follows directly from these stalk descriptions: on arithmetic stalks the amplitude and $\rho(w)$ recover s , while on support stalks s is the localization of w . A sheaf map injective on stalks is injective on sections, since equality of germs gives local equality and then equality. The additional elements in the unrestricted finite-stalk ambient product are exactly

$$\{(a, w) : a \neq 0, \rho(w) \neq 1_L\}. \tag{FSC20}$$

They are not present in $\mathcal{C}$. At the generic stalk the corresponding extra pairs are $a \neq 0$ with support below 1_L . Thus the correspondence never silently enlarges the original nonzero-amplitude support constraint.

4.5 Original coefficients, the tropical ideal, and their maps

Let B_L be the coefficient sheaf on Y with $B_L(U) = G_L(\mathcal{O}_Y(U))$ for nonempty U . A compatible cover has one support label because all nonempty intersections contain η ; the ring amplitudes glue and retain top support when nonzero. This proves the sheaf assertion. Preserve the actual inverse image under the separate name

$$\mathcal{P} = r^{-1}B_L.$$

Its sections on W_U are $G_L(\mathcal{O}_Y(U))$, while its sections on $V \subseteq A$ are $\text{LC}(V, G_L(\mathbb{Q}))$. Its arithmetic stalks are $G_L(R_p)$ and $G_L(\mathbb{Q})$, and its stalk at *every* support point is also $G_L(\mathbb{Q})$. These follow from the inverse-image stalk formula and the factorization of $r|_A$ through η , whose B_L -stalk is $G_L(\mathbb{Q})$. Thus this original pullback still retains rational amplitudes on the support open. It is not the fixed sheaf constructed next.

Define the separate sheaf

$$\mathcal{B} = \mathcal{O}_X \times_{\mathcal{D}} \underline{L}_X.$$

Its sections on W_U are $G_L(\mathcal{O}_Y(U))$, represented by $((a, \lambda), \lambda)$; its sections on $V \subseteq A$ are $\text{LC}(V, L)$, represented by $(\ell f, f)$. Its stalks are $G_L(R_p)$, $G_L(\mathbb{Q})$, and L at support points. These formulas

are direct substitutions from FSC8–FSC9, with the same sheaf and finite-stalk argument as for FSC16–FSC18. There is an exact sequence of comparison maps (no additive exactness terminology is imposed)

$$\mathcal{P} \xrightarrow{\alpha} \mathcal{B} \xrightarrow{\Phi} \mathcal{O}_X. \quad (\text{FSC33})$$

Here Φ is the first projection. On arithmetic stalks both maps are the identity under the displayed identifications; at P_J they are exactly

$$G_L(\mathbb{Q}) \xrightarrow{\sigma} L \longrightarrow L_J, \quad (a, \lambda) \longmapsto \lambda \longmapsto [\lambda]_J. \quad (\text{FSC34})$$

To construct them on sections, let α be the identity on W_U and the coordinatewise support map on locally constant sections $V \rightarrow G_L(\mathbb{Q})$. Restriction of an arithmetic-open section to V is its constant rational-amplitude germ in $\mathcal{P}$, and its constant support label in $\mathcal{B}$; hence these formulas commute with that restriction as well as restrictions within each kind of open. Operations are preserved by σ . This proves α as an actual unital sheaf map and proves FSC34. Both maps are epimorphisms because all their stalk maps are surjective. The equality relation of α at support points is equality of support labels, with amplitudes arbitrary within the original constrained carrier; at arithmetic points it is equality of the whole element. The equality relation of Φ at support points is FSC5, and at arithmetic points again equality. Their composite is the previous full coefficient-to-localization map, with the support-only loss of amplitude occurring at the first arrow and lattice localization at the second. All three sheaves remain explicit.

There is an injective unital map and a unital retraction

$$\begin{aligned} \nu : \mathcal{B} &\hookrightarrow \mathcal{C}, & (a, \lambda) &\longmapsto (a, \kappa(\lambda)), \\ \beta : \mathcal{C} &\longrightarrow \mathcal{B}, & (a, w) &\longmapsto (a, \rho(w)), & \beta\nu &= \text{id}. \end{aligned} \quad (\text{FSC21})$$

The displayed formulas apply at arithmetic stalks and to the corresponding prime coordinates of sections; at support points both maps are the identity of L . They respect every restriction formula already proved. Their operations follow because κ and ρ are homomorphisms and $\rho\kappa = \text{id}$. The composite $\mathcal{B} \rightarrow \mathcal{C} \rightarrow \mathcal{O}_X$ is the isomorphism at arithmetic stalks and the localization $L \rightarrow L_J$ at support stalks. These are the exact places where the fixed coefficient comparison and the actual localization sheaf differ.

Frobenius on the common object is

$$F_n(s, w) = (s, \text{Fr}_n w) \quad (n \geq 0), \quad F_0 = \nu\beta, \quad \text{Fix}(F_0) = \nu(\mathcal{B}). \quad (\text{FSC22})$$

It is a sheaf endomorphism because $\rho \text{Fr}_n = \rho$. Every amplitude is retained unchanged; no claim is made that ordinary ring powering is an additive endomorphism. The last two assertions follow from FSC11 and FSC21. They also recover the original constrained $G_L(R_p)$ exactly at every arithmetic stalk. As sheaves of semirings, F_0 is an actual idempotent with the indicated image, not an assumed identification of the spherical, tropical and arithmetic structures.

There is a second exact map

$$\delta : \mathcal{E} \longrightarrow \mathcal{C}, \quad \delta(w) = (z_{\rho(w)}, w) \text{ ext at arithmetic stalks}, \quad \delta(\lambda) = \lambda \text{ at support stalks}. \quad (\text{FSC23})$$

On W_U it sets the amplitude equal to zero and retains all tropical coordinates. On V it is the identity on locally constant labels. These descriptions commute with restrictions and give an injective map preserving zero, addition, and multiplication. It is nonunital: its unit image has arithmetic amplitude zero. More precisely let $b = \delta(1_{\mathcal{E}})$, the global section whose pair in FSC16 is $(e, 1_{\mathcal{E}})$. Then

$$b^2 = b, \quad b\mathcal{C} = \delta(\mathcal{E}) = \ker(\mathcal{C} \longrightarrow \mathcal{R}), \quad \delta : \mathcal{E} \xrightarrow{\sim} b\mathcal{C}. \quad (\text{FSC24})$$

The last map is unital when the principal ideal is given its own identity b . Indeed multiplication by b sends (a, w) to $(0, w)$, which proves both equalities and the asserted identity on the ideal; at

support points $b = 1$ and the same formula holds in the zero-amplitude ambient description. All inverse maps are given by the tropical projection. There cannot be a unital sheaf map $\mathcal{E} \rightarrow \mathcal{C}$: addition is idempotent in $\mathcal{E}$, whereas $1_{\mathcal{C}} + 1_{\mathcal{C}} \neq 1_{\mathcal{C}}$ at the arithmetic generic stalk, since its amplitudes are 2 and 1 in $\mathbb{Q}$. Thus FSC23–FSC24 are the exact replacement for the invalid unital inclusion, while retaining the entire ideal.

The amplitude map is the ring reflection of this common sheaf:

$$\mathcal{C}/\sim_{\delta\mathcal{E}} \simeq \mathcal{R}, \quad x \sim_{\delta\mathcal{E}} y \iff \text{locally } x + u = y + v \text{ for } u, v \in \delta\mathcal{E}. \quad (\text{FSC25})$$

It is surjective on sections of W_U , since a has a lift $(a, \kappa(1_L))$, and on sections of support opens, whose target is one element. If two arithmetic-open sections (a, w) and (a, w') have the same amplitude, then

$$(a, w) + \delta(w') = (a, w') + \delta(w)$$

in the full pair presentation: supports join on both sides and the tropical sums agree. At support opens the whole semiring is the ideal. Conversely adding ideal sections does not change amplitude. This proves the exact congruence on all opens, hence its sheaf version. Any unital map to a sheaf of rings annihilates this ideal because its elements are additively idempotent; it therefore factors uniquely through FSC25. This proves the universal property, without any unsupported stable-spectrum consequence.

4.6 The spherical map into the entire common object

Let $\mathcal{F}_L = r^{-1}\Theta^{-1}\mathbf{F}_1[T]$, with $\mathcal{F}_\eta = \mathbf{F}_1$ and $\mathcal{F}_p = \mathbf{F}_1[T^{H_p^+}]$ as in FF Proposition prop12. On the source topos the pointed multiplicative map

$$\mathbf{F}_1[T] \longrightarrow H(\mathbb{Z}_{\max} \otimes_{\mathbb{B}} L), \quad T^n \longmapsto n \otimes 1_L, \quad 0 \longmapsto \mathbf{0} \quad (\text{FSC26})$$

preserves the unit: T^0 goes to exponent 0, the tropical unit, not to the absorbing $-\infty$. It preserves multiplication because $n + m$ is the tropical product, and it commutes with every source Frobenius including zero. To see its extension to spherical algebras explicitly, a monoid-algebra element in degree k_+ has a single monomial and one nonbasepoint coordinate. Put its image in that coordinate and the additive zero elsewhere. Pointed maps move or discard that coordinate, agreeing with the fibre-sum formula of H ; products agree by the proved multiplicative formula. Conversely coordinate inclusions force this construction. Thus FSC26 is an actual spherical-algebra morphism. Inverse image gives

$$u : \mathcal{F}_L \longrightarrow H\mathcal{E}, \quad u_{Q_{(p)}}(T^h) = h \otimes 1_L, \quad u_{P_J} : \mathbf{F}_1 \rightarrow HL \text{ the unit map.} \quad (\text{FSC27})$$

The same unit formula holds at $Q_{(0)}$.

There is also the augmentation $v : \mathcal{F}_L \rightarrow H\mathcal{O}_X$, with $T^h \mapsto 1$ and the absorbing zero mapped to the target additive zero. It exists by pulling back the equivariant source augmentation $T^n \mapsto 1$. It is unique: any such map is the unit map at $Q_{(0)}$, whose source is $\mathbf{F}_1$; the injection $G_L(R_p) \rightarrow G_L(\mathbb{Q})$ then forces every monomial's image at $Q_{(p)}$ to be 1. The source generalization has $T^h \mapsto 1$, because it commutes with Fr_0 , which is the identity on $\mathbf{F}_1$ and sends T^h to T^0 . At support primes the source is also $\mathbf{F}_1$, so the unit map is unique there. Equality on stalks proves equality of sheaf maps.

The composites $Hd \circ v$ and $Ht \circ u$ are equal. On every finite stalk both send each monomial to 1_L and the absorbing zero to 0_L ; at support points both give the unit map to HL_J . The monoid-algebra coordinate argument makes this equality hold in every finite pointed degree. Since H takes the semiring fibre product to the spherical fibre product degree by degree, there is a unique map

$$\begin{aligned} \gamma : \mathcal{F}_L &\longrightarrow HC, \\ \gamma_{Q_{(p)}}(T^h) &= (1, h \otimes 1_L), \quad \gamma_{Q_{(p)}}(0) = (0, \mathbf{0}), \\ \gamma_{Q_{(0)}}(1) &= (1, 1_L), \quad \gamma_{Q_{(0)}}(0) = (0, 0_L), \quad \gamma_{P_J} : \mathbf{F}_1 \rightarrow HL \text{ the unit map.} \end{aligned} \quad (\text{FSC28})$$

Here $(1, h \otimes 1_L)$ denotes exactly the constrained pair of FSC18. This map is injective. At a finite stalk, distinct h stay distinct: a lattice prime exists by the separation proof above, and its Boolean character gives an evaluation $K_{p,L} \rightarrow K_p$ taking $h \otimes 1_L$ to h . The absorbing zero has a different arithmetic amplitude. At the generic and support stalks the two values are also distinct, since L is nontrivial. In each finite pointed degree the coordinate description has at most one nonzero entry, preserving its coordinate and value. Thus it is injective in every degree and every stalk, hence as a sheaf map. This proves a faithful spherical comparison retaining positive exponents, even though its projection to the ordinary split structure sheaf is the forced augmentation. Finally $F_n \gamma = \gamma \text{Fr}_n$ for all $n \geq 0$ by FSC22 and the displayed monomial formulas; no ordinary amplitude Frobenius has been introduced.

The complete diagram is

$$\begin{array}{ccccc}
\mathcal{F}_L & \xrightarrow{\gamma} & HC & \longrightarrow & H\mathcal{E} \\
v \downarrow & & \downarrow & & \downarrow Ht \\
HO_X & \xlongequal{\quad} & HO_X & \xrightarrow{Hd} & H\mathcal{D}.
\end{array} \tag{FSC29}$$

The right square is a fibre product, all arrows are actual unital spherical sheaf maps, and the upper-left map is injective. The ideal map δ of FSC23 is deliberately outside this diagram of unital maps; it becomes a unital isomorphism only onto the ideal with identity b in FSC24. Exact proof locators for the topological base and bottom row are FSC2–FSC9; for the right-hand maps FSC10–FSC16; and for the spherical monomial maps FSC26–FSC28.

4.7 The precise obstruction to a global section and its replacement

The projection $\pi : \mathcal{C} \rightarrow \mathcal{O}_X$ is an epimorphism of sheaves, but for the full lattice its arithmetic section need not extend over the support open. Its exact stalk fibres are

$$\begin{aligned}
\pi_{Q_{(p)}}^{-1}(a, \lambda) &= \{w \in K_{p,L} : \rho(w) = \lambda\}, & \pi_{Q_{(0)}} &= \text{id}_{G_L(\mathbb{Q})}, \\
\pi_{P_J}^{-1}([\mu]_J) &= \{\lambda \in L : \exists \nu \notin J, \lambda \wedge \nu = \mu \wedge \nu\}.
\end{aligned} \tag{FSC30}$$

The finite fibres are nonempty, containing $\kappa(\lambda)$; the support fibres are nonempty because localization is represented by a numerator. Thus it is surjective on stalks. To see directly that this implies the sheaf epimorphism assertion, any section of the target has a preimage germ at each point; choose local representative sections of the domain and shrink so their images equal the given target section. These local lifts prove local surjectivity, the epimorphism condition for sheaves of sets with algebraic operations.

A unital sheaf section of π exists if and only if L is the two-element lattice. First suppose one exists. Compose it with β to obtain a section of $\Phi : \mathcal{B} \rightarrow \mathcal{O}_X$ from FSC21. At $Q_{(0)}$, Φ is the identity, so this section is forced to be the identity there. Compatibility with generization to P_J forces its support-stalk component to satisfy

$$s_J([\lambda]_J) = \lambda \quad \text{for every } \lambda \in L. \tag{FSC31}$$

Indeed apply the compatibility square to z_λ at $Q_{(0)}$: in $\mathcal{O}_X$ it generizes to $[\lambda]_J$, whereas in $\mathcal{B}$ it generizes to the unlocalized label λ . If $0_L < \lambda < 1_L$, prime-ideal separation supplies a prime J avoiding λ . In L_J , $[\lambda]_J = [1_L]_J$, which contradicts FSC31. Thus no intermediate element exists and $L = \{0_L, 1_L\}$. Conversely for this two-element lattice its sole prime is $\{0_L\}$, and localization there is the identity. Therefore Φ is an isomorphism at every stalk, hence a sheaf isomorphism: local inverses exist and glue uniquely by injectivity. The map $\nu\Phi^{-1}$ then is a unital section of π . This proves both directions.

The obstruction therefore defines the exact replacement rather than removing the support geometry: $\mathcal{B}$ is the coefficient sheaf which retains the unlocalized support labels, embeds by ν into

$\mathcal{C}$, and maps onto the actual $\mathcal{O}_X$ by precisely FSC31's localization quotient. On the arithmetic closed stratum,

$$i^{-1}\mathcal{C} = B_L \times_{\underline{L}_Y} \mathcal{T}_L, \quad B_L \hookrightarrow i^{-1}\mathcal{C}, \quad (a, \lambda) \longmapsto (a, \kappa\lambda). \quad (\text{FSC32})$$

To verify this, inverse image commutes with finite products and equalizers, or one may substitute the already proved sections and stalks and use the same finite-representative germ argument as in FSC18. Here $i^{-1}\mathcal{O}_X = B_L$, $i^{-1}\mathcal{D} = \underline{L}_Y$, and $i^{-1}\mathcal{E} = \mathcal{T}_L$. Thus the original constrained coefficient sheaf survives exactly, while its globally extendable replacement and the actual localization map remain visible.

For the three-element chain $0 < a < 1$, the support primes are $J_0 = \{0\}$ and $J_a = \{0, a\}$. At J_0 , localization sends a to 1 and gives $L_{J_0} = \{0, 1\}$; at J_a , only 1 is inverted and $L_{J_a} = L$. Accordingly $\mathcal{C}$ has the three-element stalk L at both support points, whereas $\mathcal{O}_X$ has respectively two and three elements. The fibre over $1 \in L_{J_0}$ in FSC30 is exactly $\{a, 1\}$. This proves the obstruction concretely without replacing the chain lattice by independent Boolean coordinates. At a finite arithmetic stalk, $w = \mathbf{1} \otimes a$ has $\rho(w) = a$. The pair $(1, w)$ therefore exists in the unrestricted ambient product but is excluded by FSC20, whereas $(0, w)$ exists in $\mathcal{C}$. The original three zero labels appear exactly as $(0, \kappa 0)$, $(0, \kappa a)$, $(0, \kappa 1)$ in its fixed coefficient subobject. These examples describe the map, its fibre, and the excluded amplitudes using the same original lattice throughout.

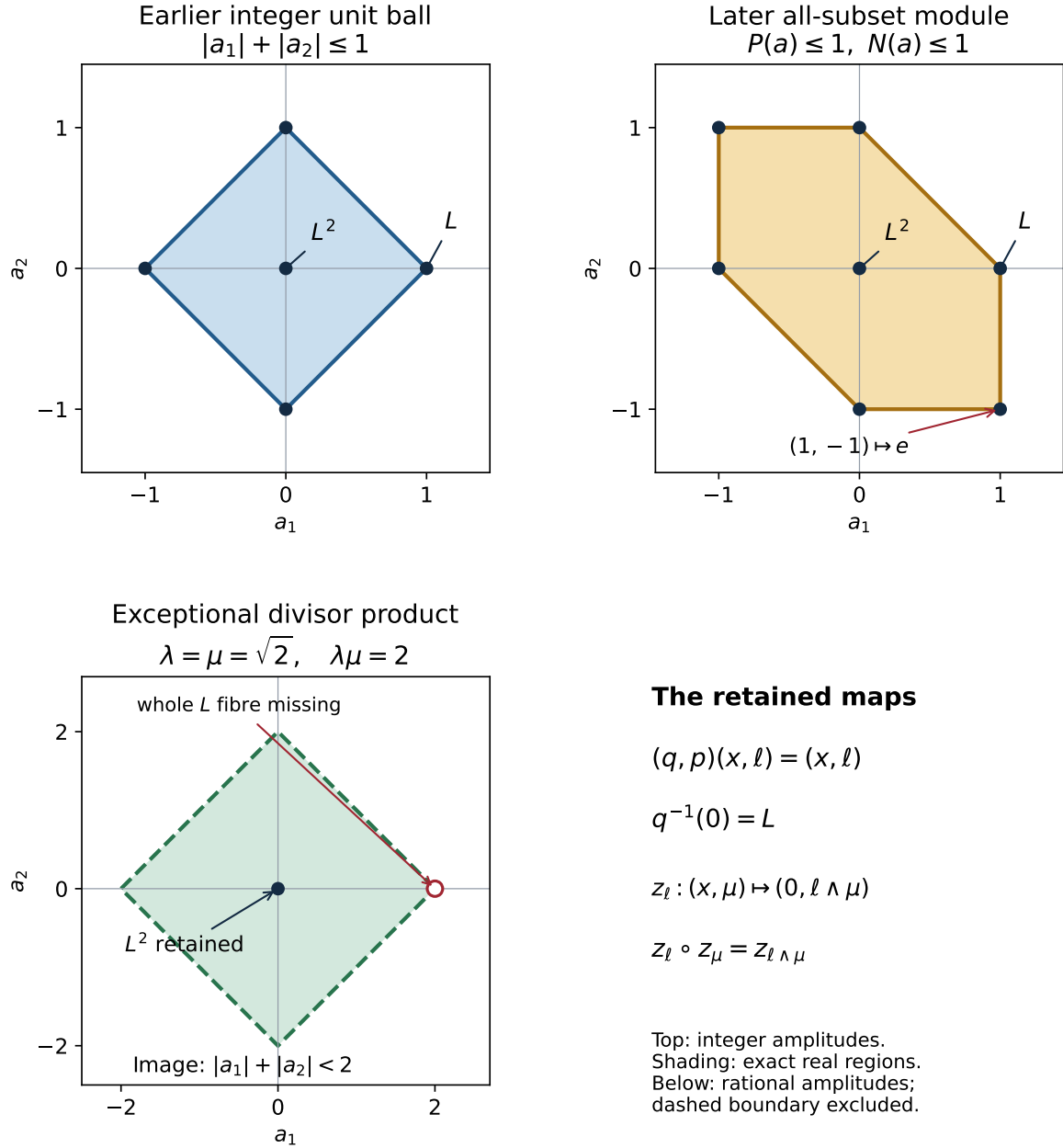


Figure 2: The exact older and later section bounds and the exceptional divisor product. Every zero-amplitude coordinate retains its full L fibre. Complete proofs: HG39–41 and SDM6–8.

5 The full Split-Zero coefficient object as a Connes–Consani Γ -algebra

The construction used here is Alain Connes and Caterina Consani, *Absolute algebra and Segal’s Γ -rings: au dessous de $\overline{\text{Spec } \mathbb{Z}}$* .

The original author source is `F1_corr.tex`, associated with arXiv:1502.05585v2. The whole source, including its bibliography and final empty lines, was read: lines 1–1336. The read file has SHA-256

3b1319ec1df4ed6e2412ace960c2813b683a
36c88509234c9abd0a39f98e8913.

The local source has duplicate labels `sssalg1`; below the line numbers distinguish the full-faithfulness proposition (line 536) from the Boolean-semiring lemma (line 575). The formula `functsum`, lines 433–442, and Lemma `sssalg`, lines 519–533, specify the actual functor and multiplication. Proposition `sssalg2`, lines 905–932, specifies the norm construction. The conclusions below are direct calculations with those formulas, not an identification of every spherical algebra with a semiring.

Let R be a nonzero commutative ring with identity and let L be a bounded distributive lattice. Write its bottom and top as $0_L, 1_L$. Keep the entire coefficient object

$$\begin{aligned} G &= G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\}, \\ (r, \lambda) + (s, \mu) &= (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu), \\ \tau &= (0, 0_L), \quad e = (0, 1_L), \quad \hat{r} = (r, 1_L), \quad z_\lambda = (0, \lambda). \end{aligned} \tag{HG1}$$

Thus $\hat{0} = e$. No occurrence of e in this fragment is silently replaced by τ . The semiring identity is $\hat{1}$. The formulas close on the stated carrier, and associativity and distributivity follow coordinate by coordinate from the ring and lattice laws. Regard L itself as the semiring with addition $\vee$ and multiplication $\wedge$.

5.1 Every finite pointed set, every fold, and multiplication

For a finite pointed set X let $X^\circ = X \setminus \{*\}$. The base point of $HG(X)$ is the tuple constantly equal to τ . For $X = 0_+$ it is the unique empty tuple. The exact formula is

$$HG(X) = \{((r_x, \lambda_x))_{x \in X^\circ} : r_x \neq 0 \implies \lambda_x = 1_L\}. \tag{HG2}$$

For a pointed map $f : X \rightarrow Y$ set

$$(HG(f)(r, \lambda))_y = \left(\sum_{x \in X^\circ : f(x)=y} r_x, \quad \bigvee_{x \in X^\circ : f(x)=y} \lambda_x \right), \quad y \in Y^\circ. \tag{HG3}$$

An empty sum is 0_R and an empty join is 0_L . Coordinates sent to the base point of Y are discarded, rather than added to a retained coordinate.

Theorem 15. *Equations (HG2)–(HG3), together with*

$$\mu_{X,Y}((r, \lambda), (s, \mu))_{(x,y)} = (r_x s_y, \lambda_x \wedge \mu_y), \quad (x, y) \in X^\circ \times Y^\circ, \tag{HG4}$$

define the Connes–Consani commutative $\mathbb{S}$ -algebra HG . Its unit $\mathbb{S}(X) = X \rightarrow HG(X)$ sends $x \neq *$ to the tuple with $\hat{1}$ at x and τ elsewhere, and sends $*$ to the all- τ tuple.

Proof. If a sum in (HG3) is nonzero, at least one summand is nonzero, so its support is 1_L and the output join is 1_L . Thus the output satisfies (HG2), even when other terms cancel. Identity maps act identically. For $g : Y \rightarrow Z$ and $z \neq *$, the fibres $f^{-1}(y)$ for $g(y) = z$ partition $(gf)^{-1}(z)$; no point discarded by f can reappear, since $g(*) = *$. Associativity of finite sums and joins now gives $HG(g)HG(f) = HG(gf)$.

Formula (HG4) lands in (HG2) because a nonzero product requires both amplitudes to be nonzero. An all- τ input produces an all- τ output, so this map factors through the smash product of pointed sets. Naturality in $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ is the equality, at (x', y') , between the sum over $f^{-1}(x') \times g^{-1}(y')$ of the products and the product of the two fibre sums. The amplitude equality is ring distributivity; the support equality is

$$\bigvee_{x,y} (\lambda_x \wedge \mu_y) = \left(\bigvee_x \lambda_x \right) \wedge \left(\bigvee_y \mu_y \right).$$

This natural bifunctor map is precisely the multiplication map for the Day smash product, by the source's definition preceding `colim`, lines 445–464. The two triple products agree coordinatewise, as do the switched products. The unit rule is natural because each unit tuple has at most one non- τ coordinate; the product with that tuple has its original value on its single selected coordinate and τ elsewhere. This proves all claimed algebra axioms directly. $\square$

5.2 Amplitude and lattice support form one exact object

Define $q : G \rightarrow R$ and $p : G \rightarrow L$ by

$$q(r, \lambda) = r, \quad p(r, \lambda) = \lambda. \quad (\text{HG5})$$

Both maps preserve zero, identity, addition and multiplication. Consequently $Hq : HG \rightarrow HR$ and $Hp : HG \rightarrow HL$ are morphisms of $\mathbb{S}$ -algebras, not merely maps on level 1.

Theorem 16. *The joint map*

$$(Hq, Hp) : HG \hookrightarrow HR \times HL \quad (\text{HG6})$$

is injective at every finite pointed set. Its image is exactly

$$\{(r, \lambda) \in R^{X^\circ} \times L^{X^\circ} : r_x \neq 0 \implies \lambda_x = 1_L\}. \quad (\text{HG7})$$

For fixed amplitude r and fixed support λ , respectively, the fibres are

$$\begin{aligned} (Hq)_X^{-1}(r) &\cong \prod_{x:r_x=0} L, \\ (Hp)_X^{-1}(\lambda) &\cong \prod_{x:\lambda_x=1_L} R. \end{aligned} \quad (\text{HG8})$$

All omitted support coordinates in the first line are 1_L ; all omitted amplitude coordinates in the second line are 0_R . These descriptions also cover $0_L = 1_L$, when $G_L(R) = R$.

Proof. A coefficient is exactly its amplitude and support pair, proving injectivity and the image claim. The naturality of the two projections follows from (HG3): the first takes fibre sums and the second takes fibre joins, which are the defining actions of HR and HL . Equation (HG4) proves multiplication compatibility and the unit tuples give unit compatibility. At each zero amplitude coordinate every support is allowed; at each nonzero amplitude coordinate only top is allowed. Conversely, at a support different from top the amplitude must be zero, and at top every amplitude is allowed. Independence of the coordinates proves (HG8). $\square$

The pointwise inverse image of zero under Hq is the embedded full lattice object

$$\iota : HL \longrightarrow HG, \quad (\lambda_x)_x \longmapsto (z_{\lambda_x})_x, \quad Hp \circ \iota = \text{id}_{HL}. \quad (\text{HG9})$$

This is a natural injective map of Γ -sets and preserves products, but when $R \neq 0$ it is not a unital map of $\mathbb{S}$ -algebras: its unit is sent to e rather than $\widehat{1}$. Its image is an ideal for (HG4), with the explicit action

$$(r_x, \lambda_x) z_{\mu_y} = z_{\lambda_x \wedge \mu_y}. \quad (\text{HG10})$$

The map Hp has only the base point in its zero fibre when $0_L \neq 1_L$, yet it is not injective: e and $\widehat{1}$ have the same support. No argument appropriate only to abelian-group exact sequences can erase these fibres.

Proposition 17. *Fix $r \in HR(X)$ and $f : X \rightarrow Y$. Write $r' = HR(f)r$ and $I_y = \{x \in X^\circ : f(x) = y\}$. The image of the entire fibre $(Hq)_X^{-1}(r)$ in $(Hq)_Y^{-1}(r')$ consists exactly of the following support choices, independently at each y :*

$$\lambda'_y \in \begin{cases} \{0_L\}, & I_y = \emptyset, \\ \{1_L\}, & \text{some } x \in I_y \text{ has } r_x \neq 0, \\ L, & I_y \neq \emptyset \text{ and all } r_x = 0 \text{ on } I_y. \end{cases} \quad (\text{HG11})$$

In particular nontrivial cancellation in a fold produces e , not τ , while a fold of labelled zeros produces their exact lattice join.

Proof. The first case is the empty-join convention. In the second case at least one input support is top, so the output support is top regardless of whether r'_y vanishes. In the third case all input supports are unrestricted. Every $\lambda \in L$ occurs as the join by assigning it to any one input coordinate and assigning bottom to the others. Distinct I_y are disjoint, so all these choices can be made independently. The last assertions are exactly (HG3). $\square$

5.3 The failed section and its complete correction object

There is always the pointed set section

$$b : R \rightarrow G, \quad b(0) = \tau, \quad b(r) = \widehat{r} \ (r \neq 0), \quad q \circ b = \text{id}_R. \quad (\text{HG12})$$

Let $b_X : R^{X^\circ} \rightarrow G^{X^\circ}$ apply this formula coordinatewise. This is a family of pointed maps, not yet a natural transformation.

Theorem 18. *Assume $0_L \neq 1_L$. Every natural transformation of Γ -sets $HR \rightarrow HG$ is the constant base-point map. In particular (HG12) does not lift to a Γ -map, and Hq has no Γ -section. The pointed section b is multiplicative precisely when R has no nonzero zero divisors.*

Proof. Let $\eta : HR \rightarrow HG$ be natural and write $u = \eta_{1_+}$. For each $x \in X^\circ$, the pointed projection $p_x : X \rightarrow 1_+$ takes x to 1 and every other point to 0. Naturality forces $(\eta_X(r))_x = u(r_x)$. Apply this first to $X = 2_+$ and then to its fold $1, 2 \mapsto 1$. One obtains $u(a + b) = u(a) + u(b)$, with $u(0) = \tau$. This is the same recovery of addition used by Connes–Consani in (addition) and the proof at lines 536–558, here proved for a Γ -map without assuming it preserves multiplication.

If $v + w = \tau$ in G , their supports satisfy $p(v) \vee p(w) = 0_L$, so both supports are bottom. Since $0_L \neq 1_L$, the carrier (HG1) then forces $v = w = \tau$. Thus τ is the only additively invertible element of G . For every $r \in R$ the equality $u(r) + u(-r) = u(0) = \tau$ forces $u(r) = \tau$. The coordinate projections now force every η_X to be constant.

For multiplicativity of b , if either input vanishes the identity follows from absorption by τ . If both are nonzero and their product is nonzero, both sides equal the top-supported product. If $r, s \neq 0$ but $rs = 0$, the left side $b(r)b(s)$ is e and the right side $b(rs)$ is τ . This proves exactly the asserted criterion. $\square$

For the obstruction formulas (HG13)–(HG16), take $0_L \neq 1_L$. When L has one element, $G_L(R) = R$, b is the identity, and each correction is the unique zero element of HL . To calculate the obstruction at every arity, use the two-element subset $\{0_L, 1_L\}$ inside L . For $a \in R^{X^\circ}$ put $\nu(a)_x = 1_L$ when $a_x \neq 0$ and 0_L otherwise. For a pointed map $f : X \rightarrow Y$ define

$$C_f(a)_y = 1_L \iff \sum_{x:f(x)=y} a_x = 0 \text{ and } \text{some } x \in X^\circ \text{ with } f(x) = y \text{ has } a_x \neq 0. \quad (\text{HG13})$$

otherwise put $C_f(a)_y = 0_L$. This defines an actual element of $HL(Y)$, including empty fibres and discarded coordinates. The exact failure-of-naturality equation is

$$HG(f)b_X(a) = b_Y(HR(f)a) + \iota_Y(C_f(a)). \quad (\text{HG14})$$

Indeed both amplitudes are the fibre sum. If that sum is nonzero, both supports are top. If it is zero, the left support is top exactly when at least one input amplitude is nonzero, which is precisely (HG13). In particular, for $a \neq 0$ the binary fold of $(b(a), b(-a))$ is e and its correction is z_{1_L} .

$$\begin{aligned} (a, -a) &\xrightarrow{b_{2+}} (\widehat{a}, \widehat{-a}) \xrightarrow{HG(\text{fold})} e = (0, 1_L), \\ (a, -a) &\xrightarrow{HR(\text{fold})} 0 \xrightarrow{b_{1+}} \tau = (0, 0_L). \end{aligned}$$

Figure 3: The two exact paths compared by naturality, for any $a \neq 0$ and $0_L \neq 1_L$. Their endpoints are different. Equation (HG14) supplies their full correction inside the embedded HL , and (HG15) computes its behavior under further folds. The fold operation is Connes–Consani’s **functsum**, source lines 433–442. This is a diagram of the calculated noncommutation, not a commuting square.

The correction itself has an exact composition rule. Put $Z(a)_x = 1_L$ when $a_x = 0$, and bottom otherwise. For $g : Y \rightarrow Z$ one has

$$C_{gf}(a) = C_g(HR(f)a) \vee (Z(HR(gf)a) \wedge HL(g)(C_f(a))), \quad (\text{HG15})$$

where meet and join are coordinatewise. To prove it, a final coordinate with nonzero amplitude has zero on both sides. At a final coordinate with zero amplitude, an original nonzero summand is either in an intermediate fibre whose amplitude is nonzero, or in an intermediate fibre that cancelled to zero. The first event is exactly the first term on the right, and the second is exactly its second term. This includes their possible simultaneous occurrence and proves (HG15).

There is also a complete multiplication correction for general rings with zero divisors:

$$\begin{aligned} D(a, b)_{x,y} = 1_L &\iff a_x \neq 0, b_y \neq 0, a_x b_y = 0, \\ \mu(b_X(a), b_Y(b)) &= b_{X \wedge Y}((a_x b_y)_{x,y}) + \iota_{X \wedge Y}(D(a, b)), \end{aligned} \quad (\text{HG16})$$

with $D = 0_L$ at all other coordinates. At a nonzero product both sides have top support; at a zero product only the case where both factors were nonzero contributes e . This proves the formula without a domain assumption.

5.4 The exact spherical replacement and universal ring image

Let $M(R)$ be the multiplicative monoid of R , pointed by its ring zero. The source’s spherical monoid algebra is $\mathbb{S}M(R)(X) = M(R) \wedge X$; see **mono2sss** and **monomono**, lines 489–511. The pointed family

$$\widetilde{b}_X : M(R) \wedge X \rightarrow HG(X), \quad (r, x) \mapsto [b(r) \text{ at } x, \tau \text{ elsewhere}] \quad (\text{HG17})$$

is a Γ -map for every R : an ordinary pointed map either retains the single selected coordinate, carrying its value with it, or discards it entirely. Its compatibility with the multiplication of spherical algebras is equivalent to $b(rs) = b(r)b(s)$ at level 1, hence holds exactly under the criterion in Theorem 18. It is a unital $\mathbb{S}$ -algebra map for a domain. When $0_L \neq 1_L$ it is not such a map for a ring with nonzero zero divisors; when L has one element it is the source's usual map $\mathbb{S}M(R) \rightarrow HR$ for every ring.

For every ring, including rings with zero divisors, the required multiplicative replacement retains the two zero states. Let

$$M^+(R) = R \sqcup \{\perp\}, \quad \perp m = m \perp = \perp, \quad (\text{HG18})$$

with the original ring product on R , identity 1_R , and distinguished pointed zero $\perp$. The original element $0_R \in R$ is not $\perp$. Define

$$c : M^+(R) \rightarrow G, \quad c(\perp) = \tau, \quad c(r) = \widehat{r}, \quad d : M^+(R) \rightarrow M(R), \quad d(\perp) = 0, \quad d(r) = r. \quad (\text{HG19})$$

Both are pointed unital multiplicative maps, including products of two nonzero zero divisors, because such a product lands at the original 0_R and hence at e under c . The source adjunction gives the actual commuting diagram

$$\begin{array}{ccc} \mathbb{S}M^+(R) & \xrightarrow{\tilde{c}} & HG \\ \text{\textsf{Sd}} \downarrow & & \downarrow Hq \\ \mathbb{S}M(R) & \xrightarrow{\rho_R} & HR. \end{array} \quad (\text{HG20})$$

Here ρ_R inserts one ring coefficient into its selected coordinate, and $\tilde{c}$ does the same with c . Their naturality follows from the single-coordinate argument just given; their multiplicativity and units follow from (HG19), so (HG20) is proved coordinatewise. This is a precise multiplicative replacement, with its correction state visible.

Amplitude also has a universal characterization. For any ring A , every unital semiring homomorphism $u : G \rightarrow A$ factors uniquely through $q : G \rightarrow R$. Indeed every z_λ is additively idempotent, so $u(z_\lambda) = 0$ in the additive group of A . Therefore $v(r) = u(\widehat{r})$ satisfies $v(0) = 0$, $v(1) = 1$, and preserves sums and products by (HG1), giving a ring map $v : R \rightarrow A$ with $u = vq$. More generally this same proof, omitting multiplication, shows that q is the universal map from the additive monoid of G into an abelian group. Thus

$$G_{\text{additive}}^{\text{gp}} \cong (R, +), \quad \text{Hom}_{\text{semiring}}(G, A) \cong \text{Hom}_{\text{ring}}(R, A). \quad (\text{HG21})$$

This explains exactly which coefficient information a ring-valued realization forgets: the full ideal (HG9). It does not provide a Γ -section in the reverse direction.

Proposition 19. *Degreewise additive Grothendieck group completion gives the exact natural isomorphism*

$$(HG(X))^{\text{gp}} \xrightarrow{\cong} HR(X), \quad [(r, \lambda)] \mapsto r. \quad (\text{HG21a})$$

It preserves all pointed-set maps and the bilinear multiplication after group completion. The additive congruence that effects this quotient is exactly

$$(r, \lambda) \sim (s, \mu) \iff r = s \iff (r, \lambda) + (e)_{x \in X^\circ} = (s, \mu) + (e)_{x \in X^\circ}. \quad (\text{HG21b})$$

Thus group completion identifies all support choices over each fixed amplitude tuple and makes no further identifications.

Proof. For a single coordinate, additive idempotence gives $[z_\lambda] = [z_\lambda] + [z_\lambda]$, hence $[z_\lambda] = 0$ in a group. The identity $\widehat{r} + \widehat{-r} = e$ gives $[\widehat{-r}] = -[\widehat{r}]$. The map $r \mapsto [\widehat{r}]$ is additive, because the sum of any two top-supported coefficients is the top-supported coefficient of their ring sum, including cancellation. Its value at zero is $[e] = 0$. It is inverse to the map induced by q . For finitely many coordinates, every tuple is the sum of its one-coordinate insertions; the same argument therefore gives the inverse $r \mapsto \sum_x [\widehat{r}_x \text{ at } x, \tau \text{ elsewhere}]$ and proves (HG21a) with its universal property. More explicitly, any additive map from G^{X° to an abelian group kills every tuple of labelled zeros and is determined uniquely by these coordinate classes, so factors uniquely through q^{X° .

Adding e at a coordinate replaces its support by 1_L without changing its amplitude. This proves (HG21b). Conversely the congruence must identify all labelled zeros because their classes vanish; its remaining quotient is already the group R^{X° by the displayed inverse, so no larger congruence is present.

Applying a pointed-set map to a tuple class replaces the amplitude at y by $\sum_{f(x)=y} r_x$ by (HG3), which is exactly $HR(f)$. The isomorphism is therefore natural on the entire category of finite pointed sets, not only on projections. The product on the additive completions is determined by its values on coefficient classes, where (HG4) becomes $r_x s_y$; its bilinear extension is precisely the usual product of HR . This proves the asserted compatibility. $\square$

The conclusion is the following pair of proved maps, with the operation on the right explicitly stated:

$$HL \xrightarrow{\iota} HG_L(R) \xrightarrow{Hq} HR \cong (HG_L(R))_{\text{degree-wise}}^{\text{gp}}. \quad (\text{HG21c})$$

It is not a claim that an arbitrary stable realization of $HG_L(R)$ is equivalent to the Eilenberg–Mac Lane spectrum of R . It is also not an unmentioned group completion of the bounded objects introduced next: those bounds need not be closed under coordinatewise addition.

5.5 The original seminorm constraint retains the full support ideal

Give R a submultiplicative seminorm with the exact axioms in Connes–Consani (`subverti1`), source lines 895–903: $|0| = 0$, $|1| = 1$, $|r + s| \leq |r| + |s|$, and $|rs| \leq |r||s|$. Define

$$|(r, \lambda)|_G = |r|. \quad (\text{HG22})$$

This is a submultiplicative seminorm on G , as each required inequality reduces to its amplitude counterpart. For $c > 0$ define the closed and strict functors

$$\begin{aligned} B_c^L(X) &= \{(r, \lambda) \in HG(X) : \sum_{x \in X^\circ} |r_x| \leq c\}, \\ B_{< c}^L(X) &= \{(r, \lambda) \in HG(X) : \sum_{x \in X^\circ} |r_x| < c\}. \end{aligned} \quad (\text{HG23})$$

Theorem 20. B_1^L is a unital $\mathbb{S}$ -subalgebra of HG . Both functors in (HG23) are modules over B_1^L and the multiplication maps restrict to

$$B_c^L \wedge B_d^L \rightarrow B_{cd}^L, \quad B_{< c}^L \wedge B_d^L \rightarrow B_{< cd}^L \quad (c, d > 0). \quad (\text{HG24})$$

Under amplitude the fibres are still exactly (HG8), for every amplitude tuple obeying its bound. Under support the fibre at λ is exactly

$$\{(r_x)_{x: \lambda_x = 1_L} : \sum_{x: \lambda_x = 1_L} |r_x| \leq c\}, \quad (\text{HG25})$$

with the corresponding strict inequality for $B_{< c}^L$. The entire HL embeds into every one of these functors, and is precisely the zero-amplitude fibre. In particular every support label has zero seminorm cost.

Proof. The fibre-sum inequality in the source becomes, with all supports retained,

$$\sum_{y \in Y^\circ} \left| \sum_{x: f(x)=y} r_x \right| \leq \sum_{x: f(x) \neq *} |r_x| \leq \sum_{x \in X^\circ} |r_x|. \quad (\text{HG26})$$

Thus (HG3) preserves both bounds. The exact support joins remain those of (HG3), since no inequality has changed any coefficient. Likewise

$$\sum_{x,y} |r_x s_y| \leq \left(\sum_x |r_x| \right) \left(\sum_y |s_y| \right) \quad (\text{HG27})$$

proves (HG24), including the strict case since $d > 0$. Unit tuples have seminorm sum 1, so are in B_1^L . The module identities are the restrictions of the already proved algebra identities. Since (HG23) depends only on amplitude, it removes none of the support choices over any allowed tuple. Substituting the support fibre from (HG8) gives (HG25). Zero tuples have total seminorm zero, so all of HL remains in every positive closed or strict bound. $\square$

Let $N = \{r \in R : |r| = 0\}$. The triangle and product inequalities make N an ideal: closure under additive inverses follows from $|-r| \leq |-1||r| = 0$. At the closed zero bound the exact object is

$$B_0^L(X) = \{(r, \lambda) \in HG(X) : r_x \in N \text{ for all } x\}. \quad (\text{HG28})$$

It equals HL precisely when $N = \{0\}$. This distinction prevents a degenerate seminorm from silently erasing nonzero amplitudes. The strict unit ball is not unital, because the unit tuple has seminorm sum exactly 1; it still retains every element of HL .

Theorem 21. *Assume $0_L \neq 1_L$ and let $B_c(R)$ denote the source's classical amplitude functor with the same bound $c > 0$. A Γ -section of $q : B_c^L \rightarrow B_c(R)$ exists exactly when*

$$|r| + |-r| > c \quad \text{for every nonzero } r \in R. \quad (\text{HG34})$$

When it exists it is unique and is the coordinatewise map b_X . For strict balls $B_{<c}$ the exact criterion is $|r| + |-r| \geq c$ for all nonzero r . At $c = 1$, an existing Γ -section is a unital $\mathbb{S}$ -algebra section exactly when, in addition,

$$r, s \neq 0, \quad |r| \leq 1, \quad |s| \leq 1 \implies rs \neq 0. \quad (\text{HG35})$$

Proof. Projection to any coordinate is a map of the bounded functors, because discarding terms cannot increase the sum of seminorms. Thus any section is determined coordinatewise by its value at 1_+ . A nonzero amplitude has only one preimage in G , and pointedness forces the zero amplitude to go to τ . Therefore b_X is the only possible section.

If $r \neq 0$ and $|r| + |-r| \leq c$, the tuple $(r, -r)$ is admissible; its two paths in the figure following (HG14) have endpoints e and τ . Naturality fails. Conversely, suppose (HG34). A nonzero correction $C_f(a)_y$ for an admissible tuple would give a zero-sum fibre with some nonzero summand a_i . In the ring, $-a_i = \sum_{x \in I_y \setminus \{i\}} a_x$. The triangle inequality gives

$$|a_i| + |-a_i| \leq \sum_{x \in I_y} |a_x| \leq \sum_{x \in X^\circ} |a_x| \leq c,$$

contrary to (HG34). Hence every correction vanishes and (HG14) proves naturality. The strict case has the same proof with the final strict inequality; its failed pair condition is correspondingly $|r| + |-r| < c$.

Unit preservation holds for the coordinatewise b , and (HG16) shows that multiplication is preserved if (HG35) holds, because all individual amplitudes in a unit-ball tuple have seminorm at most 1. If (HG35) fails, place the two nonzero zero divisors at level 1. Each is allowed by the unit bound and their product produces e instead of τ , proving failure of multiplication. This proves both directions of the criterion. $\square$

5.6 The sheaf calculation on the source's compactified base

The indexing space here is the source's $X = \overline{\text{Spec } \mathbb{Z}}$. Its topology is defined in `sectstructsh`, lines 938–946. This section constructs a coefficient sheaf on that specified space; no identification with a prime spectrum of $G_L(\mathbb{Z})$ is used. For a nonempty open U put

$$R_U = \{r \in \mathbb{Q} : v_p(r) \geq 0 \text{ for every finite prime } p \in U\}. \quad (\text{HG29})$$

Define a sheaf candidate, with empty-open value the one-point algebra, by

$$\mathcal{A}_L(U)(K) = \begin{cases} HG_L(R_U)(K), & \infty \notin U, \\ \{(r, \lambda) \in HG_L(R_U)(K) : \sum_{k \in K^\circ} |r_k| \leq 1\}, & \infty \in U. \end{cases} \quad (\text{HG30})$$

The absolute value is the usual one on $\mathbb{Q}$. The restriction map for nonempty $V \subset U$ includes R_U into R_V and retains each support coordinate unchanged.

Theorem 22. *Equation (HG30) is a sheaf of unital $\mathbb{S}$ -algebras, with amplitude morphism to the exact Connes–Consani sheaf $\mathcal{O}_X$ of `(subsalg)` and Proposition `propspecz`, lines 944–977. Its amplitude-zero sheaf is the constant HL on nonempty opens, including opens containing ∞ . Its stalks are*

$$\begin{aligned} (\mathcal{A}_L)_\eta &= HG_L(\mathbb{Q}), \\ (\mathcal{A}_L)_p &= HG_L(\mathbb{Z}_{(p)}) \quad (p \text{ finite}), \\ (\mathcal{A}_L)_\infty &= B_1^L(\mathbb{Q}). \end{aligned} \quad (\text{HG31})$$

All these identities retain the full lattice L .

Proof. Every value in (HG30) is a unital algebra by Theorem 15 or Theorem 20; the restrictions are the stated algebra inclusions. Every pair of nonempty opens intersects in the generic point. Thus compatible local sections over a cover have literally the same rational amplitudes and the same support coordinates in the ambient $HG_L(\mathbb{Q})(K)$. An amplitude integral at all primes in each covering open is integral at every prime of their union. If the union contains ∞ , one covering open does too, and imposes the norm bound on that common tuple. The common tuple therefore belongs to (HG30) on the union, and is its unique gluing. This proves the sheaf condition at every K , with multiplication and units already compatible. Amplitude removes precisely the support coordinates and gives the source sheaf formulas. Its zero fibre is HL on every nonempty open, with identity restriction maps.

At the generic point one can take opens excluding ∞ and any finite set of denominator primes, so every rational tuple occurs. At a finite prime p the same procedure allows all denominator primes other than p and excludes ∞ , giving exactly $\mathbb{Z}_{(p)}$. At ∞ every open retains the norm bound, but any finite collection of rational denominators is allowed after removing its finite prime divisors. Hence the union of these bounded sets is exactly $B_1^L(\mathbb{Q})(K)$ at each K . The restrictions never change supports, so the stalk descriptions include all of L . $\square$

The global sections at a finite pointed set k_+ are especially explicit. Since $R_X = \mathbb{Z}$, the inequality $\sum_{i=1}^k |r_i| \leq 1$ allows either the zero amplitude tuple or a single coordinate with value $+1$ or -1 . Consequently

$$\mathcal{A}_L(X)(k_+) = L^k \sqcup \coprod_{i=1}^k \coprod_{\varepsilon \in \{+1, -1\}} \{r_i = \varepsilon, \lambda_i = 1_L, r_j = 0, \lambda_j \in L \ (j \neq i)\}. \quad (\text{HG32})$$

These are disjoint subsets; the first is the entire zero-amplitude fibre, whose own pointed zero is the all-bottom tuple. If L is finite of cardinality N , then for $k \geq 1$

$$|\mathcal{A}_L(X)(k_+)| = N^k + 2kN^{k-1}. \quad (\text{HG33})$$

For $k = 0$ there is one section. At level 1, the global sections are all z_λ together with $\widehat{1}$ and $\widehat{-1}$, so their number is $N + 2$ when L is finite. This is a direct application of the original unit bound, with no added positivity assumption.

Proposition 23. *For $0_L \neq 1_L$, the global-sections amplitude map*

$$\mathcal{A}_L(X) \longrightarrow \mathcal{O}_X(X) \quad (\text{HG36})$$

has a unique unital $\mathbb{S}$ -algebra section, given by b in every coordinate. Nevertheless the sheaf amplitude map $\mathcal{A}_L \rightarrow \mathcal{O}_X$ has no section even as a sheaf of Γ -sets. Already on the single open $U = X \setminus \{2\}$ there is no Γ -section. There is likewise no Γ -section at the infinity stalk.

Proof. For a nonzero integer r , $|r| + |-r| = 2|r| \geq 2 > 1$, and $\mathbb{Z}$ has no nonzero zero divisors. Theorems 20 and 21 therefore prove the unique algebra section on global sections. Its image has all unused coordinates equal to τ , leaving the other labelled-zero points outside this chosen section.

On $U = X \setminus \{2\}$ the coefficient ring is $\mathbb{Z}[1/2]$ and the archimedean bound is still exactly 1. Thus $(1/2, -1/2)$ is an allowed tuple and

$$HG(\text{fold})(\widehat{1/2}, \widehat{-1/2}) = e, \quad b(HR(\text{fold})(1/2, -1/2)) = \tau. \quad (\text{HG37})$$

Pointedness and the coordinate projections force any putative section to use those two lifts, so there is no alternative choice that could repair this failure. A section of sheaves would evaluate to a section on U , which is impossible. The same allowed tuple in $B_1(\mathbb{Q})$, the source infinity stalk from (HG31), proves its asserted failure directly. $\square$

For an arbitrary bound $c > 0$, the same proof replaces (HG32) by all integer vectors satisfying $\sum_i |r_i| \leq c$, each with fibre $L^{\{i:r_i=0\}}$. For $0 < c < 1$ the entire global bounded object is exactly HL ; at the strict bound $c = 1$ it is again exactly HL . At a rational or complex archimedean coefficient ring, a pair $r, -r$ with $r \neq 0$ and $2|r| \leq 1$ is allowed by the unit ball. Folding its two top-supported coordinates gives e , whereas folding the zero tuple gives τ . Thus cancellation-labelled zero survives the actual archimedean restriction at precisely the original norm bound.

The complete retained object is therefore (HG7) with (HG3), (HG4), and (HG23), and its nontrivial zero-amplitude ideal is (HG9)–(HG10). The original seminorm does not force that ideal to vanish. Ring-valued realization removes it by (HG21), while the Γ -algebra and every positive archimedean bound retain it explicitly.

5.7 Exact comparison with the later all-subset Riemann–Roch module

The following second source uses a different, explicitly stated module: Alain Connes and Caterina Consani, *Riemann–Roch for the ring $\mathbb{Z}$* , arXiv:2306.00456, original `RRZ.tex`, equation `normdefn`, Lemma `xlem`, and equation `xdefn`, source lines 185–208. That source records the older sum-of-absolute-values bound and then defines the new module by all subset sums. These definitions give an exact inclusion without identifying the two modules or changing either original bound.

For an integer $m \geq 0$ set

$$E_{L,m}(k_+) = \{((a_i, \lambda_i)) \in HG_L(\mathbb{Z})(k_+) : -m \leq \sum_{i \in I} a_i \leq m \quad \text{for every } I \subseteq \{1, \dots, k\}\}. \quad (\text{HG38})$$

This is the same complete support lift as (SRR2) in the accompanying programme calculation, here defined in full so that the comparison needs none of its dimension claims. An output subset under a pointed map has as preimage a subset of input coordinates, so (HG3) preserves (HG38). Thus it is a Γ -subfunctor. Its amplitude version is denoted E_m .

Proposition 24. For $P(a) = \sum_{a_i > 0} a_i$ and $N(a) = \sum_{a_i < 0} (-a_i)$, the two exact bounds are

$$\begin{aligned} (a, \lambda) \in B_m^L(\mathbb{Z})(k_+) &\iff P(a) + N(a) \leq m, \\ (a, \lambda) \in E_{L,m}(k_+) &\iff P(a) \leq m \text{ and } N(a) \leq m. \end{aligned} \tag{HG39}$$

There are natural inclusions preserving every amplitude and support coordinate:

$$B_m^L(\mathbb{Z}) \hookrightarrow E_{L,m} \hookrightarrow B_{2m}^L(\mathbb{Z}). \tag{HG40}$$

For $m \geq 1$ both inclusions are strict; for $m = 0$ all three objects are exactly HL . The amplitude diagrams for these inclusions commute, and their zero-amplitude fibres are identically HL , with no quotient of the lattice labels.

Proof. The first equivalence is the identity $\sum_i |a_i| = P(a) + N(a)$. Every subset sum is between $-N(a)$ and $P(a)$, and these endpoints are attained by taking all negative or all positive coordinates respectively. This proves the second equivalence, including empty subsets, and is the calculation in the source's subsequent `xdefnr`. The numerical implications $P + N \leq m \Rightarrow (P \leq m, N \leq m) \Rightarrow P + N \leq 2m$ prove (HG40). All three use exactly the same support constraint (HG2) and the same pushforward (HG3), proving naturality and compatibility with amplitude.

The tuple $(\widehat{m}, \widehat{-m})$ belongs to $E_{L,m}$ but has absolute-value sum $2m > m$, so is outside the first ball. The one-coordinate tuple $\widehat{2m}$ belongs to the second ball but its singleton subset sum exceeds m , so is outside $E_{L,m}$. At $m = 0$, each individual coordinate is a subset sum, hence every amplitude is zero; all support tuples remain, giving HL . The same observation proves the assertion about zero-amplitude fibres for every m . $\square$

Proposition 25. For $0_L \neq 1_L$ and every $m \geq 1$, the map $E_{L,m} \rightarrow E_m$ has no Γ -section. In particular, the unique global Γ -algebra section on the older $B_1(\mathbb{Z})$ from (HG36) does not extend to the later E_1 module, even though their level-1 point sets agree.

Proof. Pointed coordinate projections are available in the all-subset module. Therefore any section must send zero to τ and every nonzero coordinate a to its unique lift $\widehat{a}$, exactly as in the proof of Theorem 21. For every $m \geq 1$, the tuple $(1, -1)$ belongs to $E_m(2_+)$: its subset sums are $0, 1, -1, 0$, all in $[-m, m]$. The required naturality equation would be

$$e = HG(\text{fold})(\widehat{1}, \widehat{-1}) = b(E_m(\text{fold})(1, -1)) = b(0) = \tau, \tag{HG41}$$

which is false for $0_L \neq 1_L$. At $m = 1$ the tuple $(1, -1)$ is exactly an additional tuple supplied by the first strict inclusion in (HG40); it was disallowed by the older absolute-value sum bound. At level 1 both bounds require only $|a| \leq 1$, so their point sets do agree there. The failure therefore occurs in their actual higher-level fold structures, as calculated, and does not change any coordinate or bound. $\square$

6 The full support contribution to the bounded section module

Alain Connes and Caterina Consani, *Riemann–Roch for the ring $\mathbb{Z}$* , arXiv:2306.00456, define $(HA)_X$ by requiring every partial sum to belong to X (Lemma `xlem`, equation `xdefn`). Their Definition `generator` defines dimension over the spherical base $\mathbb{S}$ by the smallest number of distinct generators, with coefficients in $\{0, 1\}$ and with the stated tolerance relation. We apply those exact definitions to the additive monoid of the full Split-Zero semiring. The following calculation concerns that explicitly constructed module. It does not presume a new definition of H^1 or a Riemann–Roch identity for an unspecified cohomology theory.

6.1 All levels, all fold maps, and the amplitude fibre

Let L be a bounded distributive lattice with $0_L \neq 1_L$. Its full carrier and operations are those of *Split Support Geometry*, version 11, by The Clankers, Definition `def:lattice-split`. Keep

$$M = G_L(\mathbb{Z}) = \{z_\lambda = (0, \lambda) : \lambda \in L\} \cup \{\hat{a} = (a, 1_L) : a \in \mathbb{Z}\}, \quad \tau = z_{0_L}, \quad e = z_{1_L} = \hat{0}. \quad (\text{SRR1})$$

The operations remain $(a, \lambda) + (b, \mu) = (a + b, \lambda \vee \mu)$ and $(a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu)$. The additive identity is τ , and the amplitude map is $q(a, \lambda) = a$. Fix an integer $m \geq 0$ and put $X_m = q^{-1}([-m, m] \cap \mathbb{Z})$.

For the finite pointed set $r_+ = \{*, 1, \dots, r\}$ define

$$\mathcal{E}_{L,m}(r_+) = \left\{ (x_1, \dots, x_r) \in M^r : \sum_{j \in Z} x_j \in X_m \text{ for every } Z \subseteq \{1, \dots, r\} \right\}. \quad (\text{SRR2})$$

An empty sum is τ . A pointed map $f : r_+ \rightarrow s_+$ acts by

$$(x_j = (a_j, \lambda_j))_{j=1}^r \mapsto \left(\sum_{f(j)=i} a_j, \bigvee_{f(j)=i} \lambda_j \right)_{i=1}^s. \quad (\text{SRR3})$$

An empty join is 0_L . The pointed value in every level is the tuple whose entries are all τ . For a subset W of the target indices, the sum of the resulting entries is the sum over the disjoint union $\bigcup_{i \in W} f^{-1}(i)$. Thus (SRR3) preserves (SRR2). Associativity of finite sums and joins proves composition, and the identity map acts identically. These are all the structure maps of the resulting Γ -set, or $\mathbb{S}$ -module.

Write $\mathcal{E}_m = (H\mathbb{Z})_{[-m, m]}$, with the same all-subset definition used by Connes–Consani. Its full relation to (SRR2) is

$$\begin{aligned} \mathcal{E}_{L,m}(r_+) &= \{((a_j), (\lambda_j)) \in \mathcal{E}_m(r_+) \times L^r : a_j \neq 0 \implies \lambda_j = 1_L\}, \\ q_r : \mathcal{E}_{L,m}(r_+) &\longrightarrow \mathcal{E}_m(r_+), \quad ((a_j), (\lambda_j)) \longmapsto (a_j). \end{aligned} \quad (\text{SRR4})$$

Indeed the amplitude of a subset sum is its integer sum, independently of its retained join support. This proves the equality. The maps q_r are surjective: assign 1_L to every nonzero coordinate and 0_L to every zero coordinate. Formula (SRR3) proves naturality. This last choice is a set-theoretic section in each level. For $m \geq 1$ no pointed natural section exists: the only possible lifts of 1 and -1 are $\hat{1}$ and $\widehat{-1}$. Naturality for the two coordinate projections forces their two-coordinate lift to be that pair. It is admissible, but its fold is e , while pointedness forces the lift of the integer zero to be τ . Since $e \neq \tau$, fold naturality fails. This proves the exact obstruction to a section of (SRR4).

For a fixed integer tuple $a \in \mathcal{E}_m(r_+)$, its fibre is exactly $L^{\{j:a_j=0\}}$: all the other supports are forced to be 1_L . In particular

$$q_r^{-1}(0) = \{(z_{\lambda_1}, \dots, z_{\lambda_r}) : (\lambda_j) \in L^r\}. \quad (\text{SRR5})$$

The maps induced on this fibre are finite joins along f -fibres. It is therefore exactly the Γ -set $H(L, \vee, 0_L)$, with its explicit inclusion into $\mathcal{E}_{L,m}$. No support label is identified at this stage.

6.2 The original dimension definition on this module

Give $\mathcal{E}_{L,m}(1_+) = X_m$ the equality tolerance relation. For a finite set $F \subset X_m$ and a subset $Z \subseteq F$, the Connes–Consani sum $\sum_Z x$ exists precisely when every partial sum of Z has amplitude in $[-m, m]$. This follows because the projection maps $\delta(j, r)$ in their equation `defnsum` force the tuple in (SRR2) to be exactly the listed summands, while the fold map is (SRR3). With equality

tolerance, F generates if and only if every $x \in X_m$ is one of these admissible sums. The empty set generates only τ .

Let $J(L)$ denote the set of nonzero join-irreducible elements: $j \neq 0_L$ is in $J(L)$ if $j = u \vee v$ implies $j = u$ or $j = v$. Put $J_<(L) = J(L) \setminus \{1_L\}$. Every element of a finite lattice is a join of join-irreducible elements below it. To prove this, induct on the number of elements below x . The bottom is the empty join. If x is join-irreducible there is nothing further to prove; otherwise $x = u \vee v$ with both $u, v < x$, and the induction hypothesis on those two smaller lower sets gives the required finite join. If a finite join equals a join-irreducible j , at least one of its terms equals j , by repeated application of its defining binary property.

Theorem 26. *For finite L , the dimension defined above is exactly*

$$\dim_{\mathbb{S}} \mathcal{E}_{L,m} = \begin{cases} |J(L)|, & m = 0, \\ n + 1 + |J_<(L)|, & 2^{n-1} \leq m < 2^n, \quad m \geq 1. \end{cases} \quad (\text{SRR6})$$

In the second line $n \geq 1$ is the unique integer with the displayed inequalities. In particular the number $n + 1$ is the unchanged Connes–Consani value for $\mathcal{E}_m$; the additional term is the exact contribution of the proper nonzero join-irreducible supports.

Proof. When $m = 0$, every point is some z_λ and every tuple is admissible. Sums are exactly joins. The preceding finite-lattice argument proves that the points z_j for $j \in J(L)$ generate, and that every generating set must contain each one. This proves the first line.

Now let $m \geq 1$. Partition any generating set into its nonzero amplitude elements and its zero amplitude elements. A sum containing a nonzero amplitude summand has support 1_L , even when its amplitudes cancel. Thus any representation of z_j , for $j \in J_<(L)$, uses only zero amplitude summands. Their labels have join j , so some summand must be z_j itself. Every such generator is therefore compulsory.

Project the nonzero amplitude generators to a subset $A \subset [-m, m] \setminus \{0\}$. This is a bijection on those generators, because nonzero amplitudes have unique lifts. For each integer $a \in [-m, m]$, project an admissible representation of $\hat{a}$ and delete its zero amplitude terms. Every partial sum of the remaining terms is still in $[-m, m]$, so A generates $\mathcal{E}_m$. The empty subset represents integer zero. Connes–Consani, Lemma [sequencebleft\(ii\)](#), proves that the least number of such integer generators is $n + 1$. Their counting lower bound can also be seen directly here: $2^{|A|}$ subsets must represent $2m + 1 > 2^n$ different integers, hence $|A| \geq n + 1$. Together with the compulsory distinct zero generators this proves the lower bound in (SRR6).

For the upper bound take an optimal integer generating set A_m from that same lemma. It can be chosen to admit a nonempty admissible zero sum. We verify this extra property, which is needed to generate the supported zero e rather than just integer zero. For $n > 4$ the source construction is $A_m = \{-m\} \cup \{\alpha_1, \dots, \alpha_n\}$, where the distinct α_i are positive, sum to m , and have subset sums covering $[0, m]$. The full set has sum zero, and every partial sum lies in $[-m, m]$, since the positive total is m and its only negative summand is $-m$. This proves the required property in that range, using the exact source construction.

For the remaining values the following are the source’s sets, with their values of m stated explicitly:

m	A_m	m	A_m
1	$\{-1, 1\}$	2	$\{-2, 1, 2\}$
3	$\{-3, 1, 2\}$	4	$\{-3, -1, 1, 3\}$
5	$\{-4, -1, 2, 3\}$	6	$\{-6, 1, 2, 3\}$
7	$\{-7, 1, 2, 4\}$	8	$\{-7, -1, 1, 2, 5\}$
9	$\{-8, -1, 1, 3, 5\}$	10	$\{-10, 1, 2, 3, 4\}$
11	$\{-11, 1, 2, 3, 5\}$	12	$\{-12, 1, 2, 3, 6\}$
13	$\{-13, 1, 2, 3, 7\}$	14	$\{-14, 1, 2, 4, 7\}$
15	$\{-15, 1, 2, 4, 8\}$		

(SRR7)

Except at $m = 2$, each whole set has sum zero, positive total m and negative total $-m$; hence its full zero sum is admissible. At $m = 2$, the subset $\{-2, 2\}$ is the required admissible zero sum. The source proves these sets generate their respective integer intervals with the stated admissibility condition.

Lift A_m to $\hat{A}_m \subset M$ and adjoin $\{z_j : j \in J_<(L)\}$. All nonzero amplitude targets are generated by the lifted admissible integer sums. Every proper support zero is generated by its join-irreducible decomposition, and τ is the empty sum. Finally the nonempty admissible zero sum just verified generates e , since all its summands have top support. Thus this set generates every point of X_m and has exactly $n + 1 + |J_<(L)|$ elements. This attains the lower bound. $\square$

For infinite L , the construction and exact maps (SRR1)–(SRR5) remain unchanged, and no finite set generates $\mathcal{E}_{L,m}$ for any $m \geq 0$. Indeed a generating set of size t has at most 2^t subset sums, whereas the target contains the distinct points z_λ for every $\lambda \in L$. If one extends the minimum-cardinality definition to arbitrary generating sets with finitely supported sums, its value is exactly $|L|$: an infinite set of size κ has κ finite subsets, forcing $\kappa \geq |L|$, and the full point set X_m has cardinality $|L|$ and generates by its singletons. This states the cardinal convention explicitly; the failure of finite generation needs no such extension of the convention.

6.3 Exact consequences for the stated archimedean bound

Retain a real parameter $\lambda > 0$ and the original bound $[-\lambda, \lambda] \cap \mathbb{Z}$. Put $m = \lfloor \lambda \rfloor$; this equality identifies precisely the same subset of integers and does not rescale its metric or change its endpoints. Define $\mathcal{E}_{L,\lambda}$ by (SRR2) using that interval. Then

$$\dim_{\mathbb{S}} \mathcal{E}_{L,\lambda} = \begin{cases} |J(L)|, & 0 < \lambda < 1, \\ n + 1 + |J_<(L)|, & 2^{n-1} \leq \lambda < 2^n, \quad \lambda \geq 1. \end{cases} \quad (\text{SRR8})$$

For the second line, since each 2^{n-1} and 2^n is an integer, the displayed inequalities on λ are equivalent to the same inequalities on m . Formula (SRR6) proves (SRR8). For the particular divisor $D = \delta\{\infty\}$ used in the source’s calculation, keep $\lambda = e^\delta$ exactly. Thus the full-support module over its integer sections has dimension $|J(L)|$ for $\delta < 0$, and $n + 1 + |J_<(L)|$ for $(n - 1) \log 2 \leq \delta < n \log 2$. This is a calculation of the specified module and its surjection (SRR4), with the entire support fibre (SRR5) retained.

For the two-element lattice, $J(L) = \{1_L\}$: the dimension is one when $0 < \lambda < 1$ and equals the original value $n + 1$ for $\lambda \geq 1$. For a chain with four elements $0_L < u < v < 1_L$, $J(L) = \{u, v, 1_L\}$: the two respective values are 3 and $n + 3$. For the Boolean lattice of subsets of $\{u, v\}$, $J(L) = \{\{u\}, \{v\}\}$ and its top is not join-irreducible: the two values are 2 and $n + 3$. These follow from the definition of join-irreducibility; they show why retaining only the two-element support lattice would miss the exact dimension contribution.

7 Divisor multiplication and degree with the full lattice fibre

The source construction is Alain Connes and Caterina Consani, *Absolute algebra and Segal’s Γ -sets*, original `F1_corr.tex`, §5.2: Proposition `propspecz`, equations `propsmash1`, `propsmash2`, and Proposition `proppic`, source lines 938–1053. Their divisor modules and exceptional multiplication rule are the source results. Here we calculate the rule for the full lattice carrier and replace the global-section test by an intrinsic test that retains every labelled zero. The split carrier is from The Clankers, *Split Support Geometry*, v11, Definition `def:lattice-split`.

7.1 The original divisor and its entire support lift

Fix a nontrivial bounded distributive lattice L and write $z_\ell = (0, \ell)$, $\tau = z_{0_L}$, $e = z_{1_L}$, and $\hat{x} = (x, 1_L)$ for $x \neq 0$. Retain the original divisor

$$D = \sum_p n_p[p] + a[\infty], \quad n_p \in \mathbb{Z}, \quad n_p = 0 \text{ for all but finitely many } p, \quad a \in \mathbb{R}. \quad (\text{SDM1})$$

The underlying space is the source's $X = \overline{\text{Spec } \mathbb{Z}}$, whose nonempty opens are complements of finite sets of closed primes, possibly including ∞ . This section constructs sheaves on that specified space; it does not identify it with $\text{Spec } G_L(\mathbb{Z})$. For nonempty U define

$$\begin{aligned} R_U &= \{x \in \mathbb{Q} : v_p(x) \geq 0 \text{ for finite } p \in U\}, \\ I_D(U) &= \{x \in \mathbb{Q} : v_p(x) + n_p \geq 0 \text{ for finite } p \in U\}, \\ G_L(I_D(U)) &= \{(0, \ell) : \ell \in L\} \cup \{(x, 1_L) : x \in I_D(U)\}. \end{aligned} \quad (\text{SDM2})$$

Here $v_p(0) = +\infty$. Addition joins the supports, and the scalar action of $G_L(R_U)$ multiplies amplitudes and meets supports. For a finite pointed set K , with $K^\circ = K \setminus \{*\}$, put

$$\mathcal{M}_L(D)(U)(K) = \left\{ (x_j, \ell_j)_{j \in K^\circ} \in G_L(I_D(U))^{K^\circ} : \infty \in U \implies \sum_j |x_j| \leq e^a \right\}. \quad (\text{SDM3})$$

The empty-open value is the singleton. A pointed map sums the amplitudes and joins the supports over each nonbasepoint fibre; an empty fibre has value τ . Restriction includes the rational amplitudes and leaves every support unchanged. Write $\mathcal{A}_L = \mathcal{M}_L(0)$. Also define $\mathcal{M}_L^<(D)$ by replacing $\leq e^a$ by $< e^a$ on opens containing ∞ , with no change elsewhere.

Theorem 27. *These are sheaves of $\mathcal{A}_L$ -modules. The amplitude-zero subsheaf is HL on every nonempty open, for both inequalities. Their stalks are*

$$\begin{aligned} \mathcal{M}_L(D)_\eta &= HG_L(\mathbb{Q}), \\ \mathcal{M}_L(D)_p &= HG_L(p^{-n_p} \mathbb{Z}_{(p)}), \\ \mathcal{M}_L(D)_\infty &= B_{e^a}^L(\mathbb{Q}). \end{aligned} \quad (\text{SDM4})$$

For the strict sheaf the last stalk is $B_{<e^a}^L(\mathbb{Q})$. The notation H on an additive module denotes its finite-tuple Γ -set; it does not assert that $I_D(U)$ is a ring.

Proof. The valuation inequalities make $I_D(U)$ an R_U -module. Addition and scalar action on the displayed carrier satisfy their axioms by distributivity in L and in $\mathbb{Q}$. A nonzero product has both input supports equal to 1_L , as required. The triangle inequality proves closure under pointed maps, including maps that discard input coordinates. For the external module action the coordinates are $(r_i x_j, \lambda_i \wedge \ell_j)$; its bound is $\sum_{i,j} |r_i x_j| = (\sum_i |r_i|)(\sum_j |x_j|) \leq e^a$, and is strict when the second bound is strict. The valuation inequalities add. This proves the module identities coordinatewise.

Every two nonempty opens intersect. Compatible local sections therefore have exactly the same rational amplitudes and lattice labels. All valuation conditions required on the union hold on a covering open; if the union contains infinity, one such open supplies the required bound. This proves existence and uniqueness of gluing. Every zero-amplitude tuple satisfies either positive bound, proving the subsheaf assertion. At a finite prime p , all other primes q where one of the finitely many inequalities $v_q(x_j) + n_q \geq 0$ fails can be excluded, together with infinity. This is a finite set: it includes denominator primes and may also include primes with negative divisor coefficient. At infinity every such finite prime can be excluded, while the original bound remains. At the generic point every such finite prime and infinity can be excluded. Taking these unions gives (SDM4) without changing any label. $\square$

7.2 The exact tensor law, including its boundary

Let $D' = \sum_p n'_p[p] + a'[\infty]$, and set $\lambda = e^a$, $\mu = e^{a'}$. Multiplication defines

$$m_{D,D'} : \mathcal{M}_L(D) \wedge_{\mathcal{A}_L} \mathcal{M}_L(D') \longrightarrow \mathcal{M}_L(D + D'), \quad (x_i, \ell_i), (y_j, t_j) \longmapsto (x_i y_j, \ell_i \wedge t_j). \quad (\text{SDM5})$$

The relative smash product is the sheafification of the module coequalizer for the Day smash product. The displayed product is balanced and respects all pointed maps by the two distributive laws.

Theorem 28. *The map (SDM5) is injective as a map of sheaves and identifies its source with*

$$\begin{cases} \mathcal{M}_L^<(D + D'), & \lambda, \mu \notin \mathbb{Q} \text{ and } \lambda\mu \in \mathbb{Q}, \\ \mathcal{M}_L(D + D'), & \text{otherwise.} \end{cases} \quad (\text{SDM6})$$

In particular its image contains the whole HL in every case. The exceptional missing stalk tuples are exactly those at infinity whose rational amplitudes satisfy $\sum_j |x_j| = \lambda\mu$; every permissible choice of support on their zero coordinates is missing with them. The finite stalks have no such defect.

Proof. We give the module calculation, rather than infer it from level one. At a finite prime, multiplication by the rational p^{-n_p} gives an isomorphism from $HG_L(\mathbb{Z}_{(p)})$ to the second object in (SDM4). The map multiplies every amplitude and fixes every label; its inverse multiplies by p^{n_p} . The same holds for D' . The unit identity $A \wedge_A A \simeq A$ now proves the claimed isomorphism on that stalk. At the generic point both modules are the coefficient module itself.

At infinity put $A = B_1^L(\mathbb{Q})$. For each rational $r > 0$,

$$s_r : A \xrightarrow{\sim} B_r^L(\mathbb{Q}), \quad (x_j, \ell_j)_j \longmapsto (rx_j, \ell_j)_j \quad (\text{SDM7})$$

is an A -module isomorphism, with inverse $s_{1/r}$ on the indicated codomain. For $0 < r \leq t$ the inclusion $B_r^L \subset B_t^L$ corresponds under these isomorphisms to multiplication by $\widehat{r/t}$ in $A(1_+)$. Thus the multiplication map induces $B_r^L \wedge_A B_s^L \simeq B_{rs}^L$ and the transition maps are exactly the coordinatewise inclusions, including all zero labels. For completeness, the inverse of the balanced-unit multiplication is $A \simeq A \wedge \mathbb{S} \rightarrow A \wedge A \rightarrow A \wedge_A A$, using the unit $\mathbb{S} \rightarrow A$ in the middle map. One composite is the identity by the unit law; the other is the identity because the balanced relation transfers the first A -factor to the second. A tuple $(z_{\ell_j})_j$ is represented by that tuple paired with $\widehat{1}$ and is recovered with supports $\ell_j \wedge 1_L = \ell_j$.

For any real $c > 0$ the full bounded module is the filtered union

$$B_c^L(\mathbb{Q}) = \operatorname{colim}_{r \in \mathbb{Q}, 0 < r \leq c} B_r^L(\mathbb{Q}). \quad (\text{SDM8})$$

Indeed each tuple has rational total absolute amplitude. If that total is positive it may be used as r ; if it is zero any rational $0 < r \leq c$ works, retaining its arbitrary lattice labels. The relative smash product preserves colimits in each variable: the Day smash is a left adjoint in each variable, and the defining balanced coequalizer commutes with colimits. Hence the product at infinity is the union of B_{rs}^L over positive rational $r \leq \lambda$, $s \leq \mu$. A tuple of rational total amplitude $t < \lambda\mu$ belongs to this union by choosing r, s sufficiently close to their upper endpoints. The same is true for $t = 0$. The endpoint $t = \lambda\mu$ is attained precisely when both endpoints λ, μ are rational: since $r/\lambda, s/\mu \leq 1$, their product is one only when both equal one. If exactly one endpoint is irrational the product is irrational, so no rational tuple total equals it. If both are irrational and their product is irrational, the same observation applies. This proves (SDM6) at infinity. All these are injections into the same tuple module.

Stalks commute with the tensor construction used here. To verify this directly, the stalk is a filtered colimit over neighborhoods; each Day representative uses finite pointed sets and finitely

many sections, so all its entries occur on a common neighborhood. Every defining relation between representatives also occurs on a common neighborhood. The same finite-representative argument applies to the balanced coequalizer. Sheafification leaves stalks unchanged. Consequently the stalk computations prove the sheaf statement. $\square$

7.3 Recovering degree without deleting labelled zeros

For a pointed Γ -set M , call $x \in M(1_+)$ additively idempotent when there exists $w \in M(2_+)$ with

$$M(\pi_1)w = x = M(\pi_2)w, \quad M(\nabla)w = x, \quad (\text{SDM9})$$

where π_1, π_2 are the two pointed coordinate projections and ∇ is the fold. This property uses only the given Γ -structure and is preserved by isomorphisms.

Lemma 29. *For every nonempty open in (SDM3), its additively idempotent level-one elements are exactly $\{z_\ell : \ell \in L\}$, for either inequality. The same holds for the image sheaf in (SDM6).*

Proof. The two projections force $w = (x, x)$. If $x = (u, \ell)$, the fold equation says $2u = u$ and $\ell \vee \ell = \ell$. In $\mathbb{Q}$ the first equality is equivalent to $u = 0$. Conversely (z_ℓ, z_ℓ) always satisfies the valuation conditions and either positive bound, and its fold is z_ℓ . The image description (SDM6) supplies the last assertion. $\square$

Keep all the coefficients of (SDM1), and define

$$c_D = \prod_p p^{-n_p} > 0, \quad \deg D = a + \sum_p n_p \log p = a - \log c_D. \quad (\text{SDM10})$$

These formulas do not replace D by another divisor. The actual global amplitude module $I_D(X)$ is $c_D \mathbb{Z}$: for a rational x , all $v_p(x/c_D) \geq 0$ hold exactly when $x/c_D \in \mathbb{Z}$. For $D_b = b[\infty]$, set $\mathcal{T}_b = \mathcal{M}_L(D) \wedge_{\mathcal{A}_L} \mathcal{M}_L(D_b)$. Define the following invariant of an abstract $\mathcal{A}_L$ -module sheaf of the displayed form:

$$\alpha_L(\mathcal{M}_L(D)) = \inf \{b \in \mathbb{R} : \mathcal{T}_b(X)(1_+) \text{ contains a non-idempotent element}\}. \quad (\text{SDM11})$$

Theorem 30. *One has the exact formula and classification*

$$\alpha_L(\mathcal{M}_L(D)) = \log c_D - a = -\deg D, \quad \mathcal{M}_L(D) \simeq \mathcal{M}_L(D') \iff D' - D \text{ is principal.} \quad (\text{SDM12})$$

The invariant in (SDM11) uses the complete support sheaf and no quotient of its zero fibre.

Proof. By (SDM6), a non-idempotent global section is a nonzero element of $c_D \mathbb{Z}$ with $|x| \leq e^{a+b}$, with strict inequality in the exceptional case. Its least possible absolute value is exactly c_D . Thus it does not exist for $b < \log c_D - a$ and does exist for every $b > \log c_D - a$. At equality it exists precisely when the tensor law is not exceptional, a distinction that leaves the infimum unchanged. More precisely, the set in (SDM11) is $[-\deg D, \infty)$ when $e^a \in \mathbb{Q}$ and $(-\deg D, \infty)$ otherwise: at equality the product of the two radius bounds is $c_D \in \mathbb{Q}$, and $e^b = c_D/e^a$ is rational exactly when e^a is rational. This proves the first formula with the endpoint fully accounted for. An isomorphism of module sheaves induces an isomorphism after tensoring with $\mathcal{M}_L(D_b)$; it preserves (SDM9), and therefore preserves this infimum. It follows that isomorphic sheaves have $\deg D = \deg D'$.

Put $q = \prod_p p^{n'_p - n_p} \in \mathbb{Q}_{>0}$. Equality of degrees gives $a' - a = -\log q$, while $v_p(q) = n'_p - n_p$ for every p . Thus $D' = D + (q)$ with the original principal-divisor convention $(q) = \sum_p v_p(q)[p] - \log |q|[\infty]$. Conversely, for any $q \in \mathbb{Q}^\times$ and $D' = D + (q)$, the explicit map

$$\mathcal{M}_L(D)(U)(K) \longrightarrow \mathcal{M}_L(D')(U)(K), \quad (x_j, \ell_j)_j \longmapsto (q^{-1}x_j, \ell_j)_j \quad (\text{SDM13})$$

is an isomorphism. Indeed $v_p(q^{-1}x_j) + n_p + v_p(q) = v_p(x_j) + n_p$, and the new archimedean bound is $e^a/|q|$. Its inverse multiplies by q . It commutes with all restrictions, pointed maps and module actions, and fixes every z_ℓ . This proves both implications. $\square$

For nontrivial L , replacing “non-idempotent” in (SDM11) by “different from the pointed zero” would give $-\infty$, since $e \neq \tau$ is always present. This is the precise defect of transporting the source’s nonzero-section test without its lattice fibre. The intrinsic test proves (SDM12) while preserving that fibre. The tensor calculation concerns the original sum-of-absolute-values divisor sheaves. It does not identify them with the later all-subset modules of *Riemann–Roch for the ring $\mathbb{Z}$* ; their exact comparison is (HG38)–(HG41). No weight or cohomological purity statement is inferred from these module calculations.

7.4 All abstract divisor-module morphisms and their full zero fibre

Retain the original divisors, lattice, space, coefficient sheaf and modules of SDM1–4. In particular, no coefficient of either divisor is changed. The divisors are

$$D = \sum_p n_p[p] + a[\infty], \quad D' = \sum_p n'_p[p] + a'[\infty].$$

The construction of these modules is that of Alain Connes and Caterina Consani, *Absolute algebra and Segal’s Γ -sets*, arXiv:1502.05585v2, Propositions **propspecz** and **proppic**, with the full lattice lift calculated in SDM1–13. The following proof determines every morphism in this lifted construction, including the maps with zero amplitude.

Define the subset of the original generic scalar semiring $G_L(\mathbb{Q})$ by

$$C_L(D, D') = \{z_\ell : \ell \in L\} \cup \left\{ \widehat{r} : r \in \mathbb{Q}^\times, \quad v_p(r) + n'_p - n_p \geq 0 \text{ for every finite } p, \quad |r| \leq e^{a' - a} \right\}. \quad (\text{SDM14})$$

It is a set of permitted scalars; no closure under addition at the archimedean bound is asserted. For every such scalar, define the coordinatewise map f_u on every nonempty open and every finite pointed set by multiplication by u . Its exact coordinate formulas are

$$\begin{aligned} f_{\widehat{r}}(x, \mu) &= (rx, \mu), \\ f_{z_\ell}(x, \mu) &= (0, \ell \wedge \mu). \end{aligned} \quad (\text{SDM15})$$

In particular the supported-zero scalar e gives $(x, \mu) \mapsto (0, \mu)$, whereas τ gives the constant map to the actual pointed zero. Both occur and are distinct.

Theorem 31. *Multiplication gives the canonical bijection of pointed sets*

$$\begin{aligned} C_L(D, D') &\xrightarrow{\sim} \text{Hom}_{\mathcal{A}_L}(\mathcal{M}_L(D), \mathcal{M}_L(D')), \\ u &\longmapsto f_u. \end{aligned} \quad (\text{SDM16})$$

$$C_L(D, D') = \mathcal{M}_L(D' - D)(X)(1_+).$$

The equality on the right uses the full labelled-zero fibre. For a third original divisor D'' and $v \in C_L(D', D'')$, the exact composition law is

$$f_v \circ f_u = f_{vu}. \quad (\text{SDM17})$$

Thus multiplication of zero scalars meets their lattice labels; it does not collapse them to the actual zero.

Proof. First prove that every permitted scalar defines the displayed module morphism. For $u = \widehat{r}$ and a local amplitude x satisfying the source valuation inequalities, one has, for every finite p in the open,

$$v_p(rx) + n'_p = (v_p(x) + n_p) + (v_p(r) + n'_p - n_p) \geq 0.$$

On an open containing infinity, its finite tuple satisfies

$$\sum_j |rx_j| = |r| \sum_j |x_j| \leq |r|e^a \leq e^{a'}.$$

These are the original inequalities, with their constants retained. For $u = z_\ell$, every output amplitude is zero, so every valuation and positive archimedean bound is satisfied. Formula SDM15 gives its support exactly.

Multiplication by a fixed scalar commutes with all pointed pushforwards. For the zero scalar this uses the explicit identity

$$\ell \wedge \bigvee_{j \in F} \mu_j = \bigvee_{j \in F} (\ell \wedge \mu_j)$$

for every finite fibre F , including the empty fibre; its empty join is 0_L . For the nonzero scalar it uses distributivity of rational multiplication over each amplitude sum, with unchanged support joins. The maps commute with restrictions because those are inclusions of the same amplitudes and labels. They commute with the external $\mathcal{A}_L$ -module action by associativity and commutativity of scalar multiplication and lattice meet. They therefore define morphisms of the required sheaves of modules.

We now prove that these are every abstract morphism; no preservation of an amplitude projection or of an individual zero label is assumed. Let

$$f : \mathcal{M}_L(D) \longrightarrow \mathcal{M}_L(D')$$

be any morphism of $\mathcal{A}_L$ -module sheaves. The canonical generic-stalk identifications in SDM4 identify its source, target and coefficient algebra with $A_\eta = HG_L(\mathbb{Q})$. Write $u = f_\eta(1_+)(\widehat{1})$, where the input is the actual multiplicative unit $\widehat{1} = (1, 1_L)$ of that generic module. For every finite pointed set K and every $x \in A_\eta(K)$, the external action of x on the module unit at 1_+ returns the same tuple x , using $K \wedge 1_+ \simeq K$. Module compatibility therefore gives

$$f_\eta(K)(x) = x u.$$

This proves the scalar formula at every finite arity, not only at level one.

For every nonempty open U , the map from $\mathcal{M}_L(D')(U)(K)$ to its generic stalk is injective: all restriction maps are coordinatewise inclusions in the common tuple carrier $G_L(\mathbb{Q})^{K^\circ}$, and the stalk is their filtered union. Naturality with respect to these maps now forces $f_U(K)$ to be the restriction of multiplication by that same generic scalar u . This proves uniqueness once its permitted values have been determined.

Every element of $G_L(\mathbb{Q})$ is either z_ℓ for a unique $\ell \in L$, or $\widehat{r}$ for a unique nonzero rational r . The zero-amplitude elements have already been shown to work without any further inequality. Suppose $u = \widehat{r}$. At the finite stalk p , apply the map to the literal input $\widehat{p^{-n_p}}$. It belongs to the source module $HG_L(p^{-n_p}\mathbb{Z}_{(p)})$. Its output must belong to $HG_L(p^{-n'_p}\mathbb{Z}_{(p)})$, so

$$v_p(r) - n_p \geq -n'_p.$$

This is exactly the finite inequality in SDM14.

At infinity, every positive rational $t \leq e^a$ is a permitted level-one amplitude. If $|r|e^a > e^{a'}$, density of the rationals provides

$$\frac{e^{a'}}{|r|} < t < e^a.$$

Its image has absolute amplitude greater than $e^{a'}$, a contradiction. Hence $|r| \leq e^{a'-a}$ is necessary, including the original endpoint. The sufficiency of precisely these inequalities was proved at the start. This proves the bijection. Its basepoint is preserved because τ acts as the actual constant-zero morphism.

The equality with global sections in SDM16 is now obtained by substituting the coefficients $n'_p - n_p$ and $a' - a$ of the actual difference divisor into SDM2–3. The same full zero lattice appears on both sides. Finally, composition of the constructed maps is scalar multiplication by vu . If both amplitudes are nonzero, their finite inequalities add and their archimedean bounds multiply. If either amplitude is zero, the product is its exact lattice-zero scalar. In either case $vu \in C_L(D, D'')$, proving SDM17. $\square$

Corollary 32. *Every abstract isomorphism of the displayed $\mathcal{A}_L$ -module sheaves is the scaling in SDM13 and fixes each labelled zero. More precisely the isomorphisms are exactly*

$$\begin{aligned} f_{\widehat{r}} \quad (r \in \mathbb{Q}^\times), \quad n'_p - n_p = -v_p(r) \quad \text{for every } p, \quad a' - a = \log |r|, \\ D' = D + (r^{-1}), \quad f_{\widehat{r}}(z_\ell) = z_\ell \quad (\ell \in L). \end{aligned} \quad (\text{SDM18})$$

For every original D , its complete endomorphism monoid and automorphism group are

$$\begin{aligned} \text{End}_{\mathcal{A}_L}(\mathcal{M}_L(D)) &\simeq \{z_\ell : \ell \in L\} \cup \{\widehat{1}, \widehat{-1}\}, \\ \text{Aut}_{\mathcal{A}_L}(\mathcal{M}_L(D)) &= \{f_{\widehat{1}}, f_{\widehat{-1}}\}. \end{aligned} \quad (\text{SDM19})$$

The first isomorphism retains multiplication of the original split scalars; it does not assert an additive ring structure on this bounded set.

Proof. An inverse pair induces inverse generic scalar multiplications, by SDM16–17. A zero-amplitude scalar cannot be invertible, since its product with any scalar has zero amplitude. A nonzero generic scalar $\widehat{r}$ has the unique inverse $\widehat{r^{-1}}$ in $G_L(\mathbb{Q})$. Applying the finite inequalities of SDM14 to the map and its inverse yields

$$v_p(r) + n'_p - n_p \geq 0, \quad -v_p(r) + n_p - n'_p \geq 0.$$

Their left sides sum to zero, so both equal zero. Applying the two archimedean inequalities gives

$$|r| \leq e^{a'-a}, \quad |r|^{-1} \leq e^{a-a'}.$$

The second is $|r| \geq e^{a'-a}$; hence equality holds. These are exactly the conditions in SDM18. Conversely those equalities put both scalars in the required sets SDM14, so SDM17 proves that they are inverse sheaf maps. The principal-divisor convention in SDM12 gives

$$(r^{-1}) = \sum_p (-v_p(r)) [p] + \log |r| [\infty],$$

which proves the claimed identity of divisors with the signs unchanged. Formula SDM15 proves fixation of every z_ℓ .

For $D' = D$, the nonzero scalars in SDM14 are rational numbers with every finite valuation nonnegative and with $|r| \leq 1$. Such a rational number is an integer: a reduced denominator greater than one would have a prime divisor giving a negative valuation. The only nonzero integers of absolute value at most one are 1 and -1 . All zero scalars remain allowed. This proves the endomorphism assertion and, by the preceding inverse calculation, its exact automorphism subgroup. $\square$

7.5 The complete category of the original divisors

Define a category Div_L retaining each original divisor D of SDM1 as an object. Its morphisms, composition and identities are the specified data

$$\begin{aligned} \text{Hom}_{\text{Div}_L}(D, D') &= \mathcal{M}_L(D' - D)(X)(1_+) = C_L(D, D'), \\ (v, u) &\longmapsto v \circ u := vu, \quad \text{id}_D = \widehat{1}. \end{aligned} \quad (\text{SDM20})$$

The product on the second line is the actual split multiplication: $\widehat{s}\widehat{r} = \widehat{s}r$, $\widehat{r}z_\ell = z_\ell\widehat{r} = z_\ell$ for $r \neq 0$, and $z_\ell z_\mu = z_{\ell \wedge \mu}$. Every z_ℓ is a morphism between every ordered pair of divisors; no such label has been suppressed.

These data satisfy the category axioms. Closure of composition is the scalar inequality calculation in the proof of SDM17. The identity scalar belongs to $C_L(D, D)$ by SDM14, since its finite valuations are zero and its absolute value is one. Associativity and the two identity laws follow directly from associativity and the unit of $G_L(\mathbb{Q})$. The distinguished zero morphism is τ ; its products on both sides are τ . Other z_ℓ remain separate morphisms and have the meet-composition law just displayed. No additive closure or ring enrichment is asserted: already in $\text{End}_{\text{Div}_L}(D)$, the amplitude sum $1 + 1 = 2$ fails the archimedean bound in SDM19.

Let ModDiv_L be the full subcategory of $\mathcal{A}_L$ -module sheaves with objects the displayed $\mathcal{M}_L(D)$ for original divisors D . Then

$$\begin{aligned} \mathcal{F} : \text{Div}_L &\longrightarrow \text{ModDiv}_L, \\ D &\longmapsto \mathcal{M}_L(D), \\ u &\longmapsto f_u \end{aligned} \tag{SDM21}$$

is fully faithful and essentially surjective. Indeed SDM17 proves preservation of composition and SDM15 proves preservation of identities. The bijection SDM16 is exactly the bijection on every Hom set required for full faithfulness. Every object of ModDiv_L is the image of its displayed original divisor, proving essential surjectivity. If the full subcategory is enlarged to contain all module sheaves isomorphic to those displayed objects, the same argument proves essential surjectivity onto that enlargement as well: its defining isomorphism is an isomorphism to an image of $\mathcal{F}$.

Consequently the original divisor category, with its complete lattice of zero-amplitude morphisms, is equivalent to the specified full divisor-module subcategory. The functor retains the original divisor coefficients and takes each z_ℓ to its exact action $(x, \mu) \mapsto (0, \ell \wedge \mu)$. Its isomorphisms and automorphisms are precisely SDM18–19; passing to this category does not identify the labelled zero morphisms or assume an additional purity property.

8 Full-lattice unit orbits and the Cartier valuation sheaf

The human source for the Cartier quotient and its arithmetic-site comparison is Alain Connes and Caterina Consani, *Geometry of the arithmetic site*, arXiv:1502.05580v1, original author source `arithmetic-site_Adv_final1.tex`, Lemma `hyper`, source lines 987–1001, and Proposition `sheavesonspz`, source lines 1003–1024, in the subsection identifying the structure sheaf with Cartier divisors on the cyclotomic spectrum. Their quotient $\mathcal{K}/\mathcal{O}^\times$, normalization $v(p) = p - 1$ and cyclotomic value group are retained. The local equal-value sum requires a residue-field correction, proved below. The full split carrier and its two kinds of prime stalks are those of The Clankers, *Split Support Geometry*, version 11, Definition `def:lattice-split` and its prime-localization theorems, calculated explicitly in FSC2–9 of this reader. No purity hypothesis is made in this construction.

8.1 The genuine unit quotient

Let K be a field and $v : K^\times \rightarrow \Gamma$ a surjective valuation to an ordered abelian group, written additively, with $v(0) = +\infty$. Write

$$\begin{aligned} \mathcal{O}_v &= \{x : v(x) \geq 0\}, & \mathfrak{m}_v &= \{x : v(x) > 0\}, \\ \kappa_v &= \mathcal{O}_v / \mathfrak{m}_v, & U_v &= \mathcal{O}_v^\times = \{u : v(u) = 0\}. \end{aligned} \tag{FCQ1}$$

The trivial valuation is permitted. Fix the original nontrivial bounded distributive lattice L and retain

$$\begin{aligned} S = G_L(K) &= \{z_\ell = (0, \ell) : \ell \in L\} \cup \{\widehat{x} = (x, 1_L) : x \in K^\times\}, \\ \tau &= z_{0_L}, \quad e = z_{1_L}, \\ (x, \ell) + (y, m) &= (x + y, \ell \vee m), \quad (x, \ell)(y, m) = (xy, \ell \wedge m). \end{aligned} \tag{FCQ2}$$

The notation for a zero-amplitude pair in the last line always means its displayed z -label. The actual multiplicative units of $G_L(\mathcal{O}_v)$ are precisely $\widehat{u}$ with $u \in U_v$: an invertible element must have nonzero amplitude, and its amplitude and inverse must both belong to $\mathcal{O}_v$. Conversely $\widehat{u}^{-1} = \widehat{u^{-1}}$ verifies the assertion.

These units act by multiplication on S . Their action is

$$\widehat{u}\widehat{x} = \widehat{ux}, \quad \widehat{u}z_\ell = z_\ell. \tag{FCQ3}$$

It fixes every labelled zero. Multiplication by a unit is an additive automorphism; it is not being called a unital semiring automorphism, since it need not fix the multiplicative unit. Let $\mathcal{Q}_L(K, v)$ be its orbit set. Nonzero amplitudes have the same orbit exactly when their valuations agree: $v(x) = v(y)$ is equivalent to $x/y \in U_v$. Thus the exact orbit set is

$$\mathcal{Q}_L(K, v) = \{z_\ell : \ell \in L\} \sqcup \{\mathbf{a} : a \in \Gamma\}, \quad \mathbf{a} = [\widehat{x}] \iff a = -v(x). \tag{FCQ4}$$

In particular the symbol $\mathbf{0}$ is the nonzero-amplitude unit orbit. It is neither τ nor e .

Define orbit multiplication and multivalued addition by

$$[x][y] = [xy], \quad [x] \boxplus [y] = \{\widehat{u}x + \widehat{w}y : u, w \in U_v\}. \tag{FCQ5}$$

Changing either representative changes neither set: absorb its unit into u or w . Multiplication is well defined because the product of two such unit factors is a unit. The operations on arbitrary subsets below mean union over their elements.

8.2 The complete addition table and the residue-field correction

The operations retaining every lattice element are

$$\begin{aligned} z_\ell z_m &= z_{\ell \wedge m}, & z_\ell \mathbf{a} &= z_\ell, & \mathbf{a}\mathbf{b} &= \mathbf{a} + \mathbf{b}, \\ z_\ell \boxplus z_m &= \{z_{\ell \vee m}\}, & z_\ell \boxplus \mathbf{a} &= \{\mathbf{a}\}, \\ \mathbf{a} \boxplus \mathbf{b} &= \{\max(\mathbf{a}, \mathbf{b})\} \quad (a \neq b). \end{aligned} \tag{FCQ6}$$

For the equal-value case, the exact formula is

$$\mathbf{a} \boxplus \mathbf{a} = \begin{cases} \{e\} \cup \{\mathbf{b} : b < a\}, & |\kappa_v| = 2, \\ \{e\} \cup \{\mathbf{b} : b \leq a\}, & |\kappa_v| > 2. \end{cases} \tag{FCQ7}$$

The set $b < a$ may be empty for the trivial valuation.

Proof. The first two lines of FCQ6 follow by the original split laws. For different valuations the nonarchimedean equality $v(x + y) = \min(v(x), v(y))$ gives its last line for every choice of unit representatives.

For equal values, divide both representatives by an element of valuation $-a$. The possible sums reduce to sums of two units. Their valuations are nonnegative, or $+\infty$. Every strictly positive value $\gamma \in \Gamma$ occurs: choose $t \in K^\times$ with $v(t) = \gamma$, and write $t = 1 + (t - 1)$. Both 1 and $t - 1$ are units. Exact cancellation is $1 + (-1) = 0$. In the split carrier, this cancellation has support $1_L \vee 1_L = 1_L$, and hence gives e , never τ .

A sum of two units has valuation zero exactly when the sum of their two nonzero residues is nonzero. Over $\mathbf{F}_2$ the only two nonzero residues are both 1, whose sum is zero, so valuation zero cannot occur. If $|\kappa_v| > 2$, choose a nonzero residue different from -1 and lift it to a unit u . Then $1 + u$ has nonzero residue and valuation zero. Multiplying all these constructions by the chosen element of valuation $-a$ proves both directions of FCQ7. $\square$

This corrects the generality of Lemma **hyper** in the source. For $K = \mathbb{Q}$ at the prime 2, with its required scale $v(2) = 2 - 1 = 1$, the units have odd numerator and odd denominator in reduced form. The sum of two units has even numerator and odd denominator and cannot be a unit. Consequently

$$\mathbf{0} \boxplus \mathbf{0} = \{e, -1, -2, \dots\}, \quad \mathbf{0} \notin \mathbf{0} \boxplus \mathbf{0}. \quad (\text{FCQ8})$$

Every displayed negative value occurs by $1 + (2^j - 1) = 2^j$, $j \geq 1$, and e occurs by cancellation. Replacing e by the ordinary zero gives the same counterexample to the source's claimed equal-value interval including its upper endpoint. The correction concerns that finite-stage lemma, not an assertion that the cyclotomic endpoint fails.

8.3 The exact multivalued algebra and its projections

The orbit object is a commutative associative multivalued additive monoid with identity τ . Its multiplication is a commutative monoid with identity $\mathbf{0}$ and absorber τ , and it distributes over the multivalued addition with equality of sets:

$$c(a \boxplus b) = ca \boxplus cb. \quad (\text{FCQ9})$$

Here these axioms specify the resulting multivalued algebra; no additive-inverse axiom is tacitly included.

Proof. Commutativity follows from FCQ5. An iterated three-term orbit sum on either parenthesization is exactly

$$\{[\hat{u}x + \hat{w}y + \hat{t}z] : u, w, t \in U_v\}.$$

To verify this assertion, a representative of an intermediate orbit differs from its defining two-term sum by a unit, which multiplies both corresponding coefficients. Conversely one may take that extra unit to be 1. This proves associativity including every possible zero-amplitude intermediate sum. The identity equation follows because a unit multiple of τ is τ and addition by τ is the original identity. The multiplicative monoid laws descend from FCQ2. For distributivity, fixing a representative c , both sides of FCQ9 are exactly

$$\{[c(\hat{u}x + \hat{w}y)] : u, w \in U_v\} = \{[\hat{u}cx + \hat{w}cy] : u, w \in U_v\}.$$

This uses original distributivity and proves equality, not merely one inclusion. $\square$

It is not an ordinary hyperring with zero τ . Indeed $\tau \in a \boxplus b$ forces both amplitudes to vanish and their support join to be 0_L , so $a = b = \tau$. Thus no other element has an additive inverse to τ . The zero-amplitude part $Z_L = \{z_\ell : \ell \in L\}$ is closed under addition and absorbs multiplication into itself. Its addition and multiplication are exactly join and meet.

There is nevertheless an exact ordinary hyperfield inside this object, with its different additive zero:

$$\mathcal{H}(K, v) = \{e\} \sqcup \{\mathbf{a} : a \in \Gamma\} \cong K/U_v. \quad (\text{FCQ10})$$

On this subset, e is the additive identity and the multiplicative absorber. Every $\mathbf{a}$ has multiplicative inverse $-\mathbf{a}$. Since $-1 \in U_v$, each is its own additive opposite, with e in its self-sum; FCQ6 shows that no other value can be its opposite. For completeness, reversibility of this hyperaddition follows directly from a field equation. The assertion $[c] \in [a] \boxplus [b]$ in K/U_v means $wc = ua + tb$ for some $u, t, w \in U_v$. Solving for a gives $a = u^{-1}wc - u^{-1}tb$, which is precisely $[a] \in [c] \boxplus [-b]$. Solving back gives the converse. Associativity and distributivity were already proved. This verifies all the hyperfield axioms of FCQ10 without assuming the incorrect residue-independent addition formula.

Write $0_{\mathcal{H}}$ for e when it is regarded as the zero of FCQ10. There are exact projections

$$\begin{aligned} \pi : \mathcal{Q}_L(K, v) &\longrightarrow \mathcal{H}(K, v), & \pi(z_\ell) &= 0_{\mathcal{H}}, & \pi(\mathbf{a}) &= \mathbf{a}, \\ \sigma : \mathcal{Q}_L(K, v) &\longrightarrow L, & \sigma(z_\ell) &= \ell, & \sigma(\mathbf{a}) &= 1_L. \end{aligned} \quad (\text{FCQ11})$$

Both preserve multiplication and their respective units and zeros. For addition they preserve the complete image sets: $\pi(a \boxplus b) = \pi(a) \boxplus \pi(b)$ and $\sigma(a \boxplus b) = \{\sigma(a) \vee \sigma(b)\}$. These statements follow in each case from FCQ6–7. The second formula retains the support of cancellation.

The two maps together are injective and have precisely the following image:

$$\mathcal{Q}_L(K, v) \xrightarrow{\sim} \{(h, \ell) \in \mathcal{H}(K, v) \times L : h \neq 0_{\mathcal{H}} \Rightarrow \ell = 1_L\}. \quad (\text{FCQ12})$$

The inverse sends $(0_{\mathcal{H}}, \ell)$ to z_ℓ and $(\mathbf{a}, 1_L)$ to $\mathbf{a}$. Under this bijection the sum is

$$(h, \ell) \boxplus (h', m) = \{(t, \ell \vee m) : t \in h \boxplus h'\},$$

and the product is $(hh', \ell \wedge m)$. These sets obey the displayed admissibility condition: if a sum has nonzero amplitude at least one input has nonzero amplitude and hence top support. A product of nonzero amplitudes has both supports top. Thus FCQ12 is an exact construction of the full object from its ordinary Cartier hyperfield and its complete support lattice, not an identification of their distinct zero elements.

The lattice inclusion $\ell \mapsto z_\ell$ preserves join, meet and the actual zero, but sends the lattice unit to e , not to $\mathbf{0}$. The inclusion of FCQ10 preserves its full operations and its multiplicative unit, but sends its additive zero to e , not to τ . Their composites with the corresponding projections are identities. The original orbit map $q : G_L(K) \rightarrow \mathcal{Q}_L(K, v)$ preserves products exactly and satisfies $q(x + y) \in q(x) \boxplus q(y)$; its unit-saturated image of sums is exactly FCQ5. The commutative square

$$\begin{array}{ccc} G_L(K) & \longrightarrow & \mathcal{Q}_L(K, v) \\ \downarrow \text{amplitude} & & \downarrow \pi \\ K & \longrightarrow & K/U_v \end{array} \quad (\text{FCQ13})$$

uses the ordinary field quotient on the bottom row.

Every bounded lattice homomorphism $\epsilon : L \rightarrow L'$ induces the exact operation-preserving map $z_\ell \mapsto z_{\epsilon(\ell)}$, $\mathbf{a} \mapsto \mathbf{a}$. It commutes with both projections in FCQ11 and with FCQ13. In particular this gives all Boolean-branch maps, without replacing the original L by a chosen branch.

8.4 The actual valuation scale and the divisor modules

At a prime $\wp$ of a number field K above the rational prime p , retain the source's scale $v_\wp(p) = p - 1$. If $e_\wp$ is its ramification index and $\text{ord}_\wp$ takes value one at a uniformizer, then

$$v_\wp = \alpha_\wp \text{ord}_\wp, \quad \alpha_\wp = \frac{p-1}{e_\wp}, \quad \Gamma_\wp = \alpha_\wp \mathbb{Z}. \quad (\text{FCQ14})$$

This follows because $\text{ord}_\wp(p) = e_\wp$ and the required scale fixes the multiplier. If the residue degree is $f_\wp$, the residue field has $p^{f_\wp}$ elements. The second line of FCQ7 applies except exactly when $p = 2$ and $f_\wp = 1$; in that case its first line applies. All intervals in the source's negative-valuation coordinate are therefore intervals in the actual group $\alpha_\wp \mathbb{Z}$, with the specified endpoint rule.

For a discrete valuation, let ϖ be a uniformizer and $\alpha = v(\varpi) > 0$. The original fractional ideal $\varpi^{-n} \mathcal{O}_v$ and its full split lift have the following exact orbit set under U_v :

$$G_L(\varpi^{-n} \mathcal{O}_v)/U_v = \{z_\ell : \ell \in L\} \sqcup \{\mathbf{a} : a \in \Gamma, a \leq n\alpha\}. \quad (\text{FCQ15})$$

Indeed membership of a nonzero amplitude is $v(x) \geq -n\alpha$, which is exactly the displayed inequality for $a = -v(x)$. Every such value occurs by the definition of the value group. The

whole zero lattice occurs and is fixed by the unit action. The addition table FCQ6–7 preserves this bound, and multiplication by the coefficient quotient, whose nonzero values satisfy $a \leq 0$, preserves it as well.

For the finite stalk at a rational prime in SDM4, $I_D = p^{-n_p} \mathbb{Z}_{(p)}$, the formula is

$$G_L(I_D)/\mathbb{Z}_{(p)}^\times = \{z_\ell : \ell \in L\} \sqcup \{\mathbf{a} : a \in (p-1)\mathbb{Z}, a \leq (p-1)n_p\}. \quad (\text{FCQ16})$$

The coefficient n_p has not been replaced: the multiplier $p-1$ records the source's valuation scale. For the principal change $D' = D + (q)$, the actual map $x \mapsto q^{-1}x$ in SDM13 sends

$$\mathbf{a} \mapsto \mathbf{a} + \mathbf{v}_p(\mathbf{q}), \quad z_\ell \mapsto z_\ell, \quad v_p(q) = (p-1) \text{ord}_p(q).$$

Its target bound is $(p-1)n'_p = (p-1)n_p + v_p(q)$, exactly the translated source bound. This proves the local Cartier receiver of the original divisor-module isomorphism, with all signs and all support labels retained.

8.5 Why the cyclotomic stalk has the full closed interval

Let $K^{\text{cyc}} = \mathbb{Q}^{\text{cyc}}$ be the union of the cyclotomic number fields, and let v extend the p -adic valuation with $v(p) = p-1$. Let $\mathcal{O}_v$ now be its valuation ring in this full field. Its residue field is

$$\kappa_v = \overline{\mathbb{F}}_p. \quad (\text{FCQ17})$$

Here the equality means an isomorphism inside a fixed algebraic residue-field closure.

Proof. Every residue of an algebraic element of K^{cyc} is algebraic over $\mathbb{F}_p$, so the residue field is an algebraic extension of $\mathbb{F}_p$. For every $f \geq 1$ the field contains all roots of unity of order $n = p^f - 1$. Their reductions are distinct. To prove this, if a nontrivial n -th root ζ reduced to 1, then the equation $1 + \zeta + \dots + \zeta^{n-1} = 0$ would reduce to $n = 0$ in characteristic p , contrary to $p \nmid n$. Applying this to the ratio of any two roots proves distinctness. Their n distinct nonzero reductions are all the roots of $X^n - 1$; together with zero they are the roots of $X^{p^f} - X$. In characteristic p this root set is closed under addition, subtraction, multiplication and inversion of nonzero elements, because the p^f -power map is a field homomorphism. It is therefore a copy of $\mathbb{F}_{p^f}$ inside the residue field. Taking every f proves FCQ17. $\square$

In particular this residue field always has more than two elements. At $p = 2$, the correction is already visible after adjoining a primitive cube root of unity: its residue lies in $\mathbb{F}_4 \setminus \mathbb{F}_2$, and its sum with 1 is nonzero. This supplies the equal-value endpoint missing over the finite field $\mathbb{F}_2$.

The source's cyclotomic value-group calculation, with its scale unchanged, is

$$v((K^{\text{cyc}})^\times) = H(p) = \mathbb{Z}[1/p]. \quad (\text{FCQ18})$$

Its local-field input, explicitly used in the proof of Proposition **sheavesonspz**, is that the local cyclotomic extension is obtained by adjoining the unramified roots of unity of order prime to p and the totally ramified p -power roots of unity. The source cites Jean-Pierre Serre, *Corps locaux*, second edition (1968), for this local-field statement. We retain that cited input rather than assert an unreferenced new ramification theorem. Its consequences for the value group can be checked with the exact constants as follows.

For a primitive p^m -th root ζ , $\Phi_{p^m}(1+T)$ is Eisenstein at p : its constant term is p , and its reduction modulo p is $T^{(p-1)p^{m-1}}$. The latter identity follows from $\Phi_{p^m}(X) = \sum_{j=0}^{p-1} X^{jp^{m-1}}$ by reducing in characteristic p and substituting $X = 1+T$. The valuation of $\pi = \zeta - 1$ is positive, since a p -power root of unity reduces to 1 in residue characteristic p . In its Eisenstein equation, all intermediate terms then have value strictly greater than $v(p)$ if the leading term has value $v(p)$; more directly, equality of the minimal values in that equation forces

$$((p-1)p^{m-1})v(\zeta - 1) = v(p) = p-1.$$

Indeed, if the leading value were smaller than $v(p)$, it would be the unique minimum, since an intermediate coefficient has value at least $v(p)$; if it were larger, the constant term would be the unique minimum. Thus $v(\zeta - 1) = p^{-(m-1)}$. The unramified extensions do not enlarge the value group; the finite totally ramified cyclotomic stage has index $(p-1)p^{m-1}$, so its group is exactly $p^{-(m-1)}\mathbb{Z}$. Taking their union proves FCQ18. This is also why replacing $v(p)$ by 1 without recording the multiplier would give the wrong displayed group.

Combining FCQ7, FCQ17 and FCQ18 proves the full cyclotomic stalk, rather than importing an assumed tropical addition:

$$\begin{aligned}\mathcal{Q}_L(K^{\text{cyc}}, v) &= \{z_\ell : \ell \in L\} \sqcup \{\mathbf{a} : a \in \mathbb{Z}[1/p]\}, \\ \mathbf{a} \boxplus \mathbf{a} &= \{e\} \cup \{\mathbf{b} : b \in \mathbb{Z}[1/p], b \leq a\}.\end{aligned}\tag{FCQ19}$$

Collapsing the full zero lattice by π gives precisely the source's $H(p)_{\text{hmax}}$, with its ordinary zero written $-\infty$. The same closed interval is not valid at every finite residue- $\mathbf{F}_2$ stage, as FCQ8 shows.

For an extension of valued fields with the same stated scale, the induced map of quotient orbit sets is injective: the nonzero orbits are determined by the unchanged valuation and the zero labels are fixed. Its additive image is contained in the target hyperaddition, but equality need not hold. New value-group elements or a larger residue field can add new target sums. At each cyclotomic stalk, every element, unit witness and finite sum relation occurs in a finite cyclotomic subfield, after adjoining the finitely many elements involved. Thus the filtered union of these exact finite-stage quotient relations is the relation FCQ19. This proves how the corrected finite-stage rule yields the source's final cyclotomic hyperfield.

8.6 The quotient sheaf over the original arithmetic scheme

Let A be an integral domain with fraction field K , and $X = \text{Spec } A$. On this irreducible space define the split meromorphic sheaf $G_L(\underline{K})$ by the constant full carrier on each nonempty open, with its coordinatewise operations. The ordinary unit sheaf $\mathcal{O}_X^\times$ acts by FCQ3. Let

$$\mathcal{C}_L(X) = (G_L(\underline{K})/\mathcal{O}_X^\times)^\#, \quad \mathcal{C}(X) = (\underline{K}/\mathcal{O}_X^\times)^\#.\tag{FCQ20}$$

The superscript means sheafification of the orbit presheaf. The second sheaf is the source's Cartier hyperring sheaf. In the first, the relation of multivalued addition is the sheafification of the orbit-sum relation FCQ5; equivalently, membership is locally witnessed by unit representatives. The multiplication and the support and amplitude maps are compatible with restriction, so they give sheaf maps.

At every point $x \in X$, the exact stalk is

$$\mathcal{C}_L(X)_x = G_L(K)/\mathcal{O}_{X,x}^\times.\tag{FCQ21}$$

To prove the orbit assertion, an equality of two stalk orbits is witnessed by one unit germ, and that unit and its inverse are represented on some common neighborhood. After shrinking, their product is the identity, so the germ is represented by an actual section of the unit sheaf there. The converse follows by taking germs. The same argument with the finitely many representatives and unit witnesses in FCQ5 proves the stalk formula for hyperaddition. Sheafification leaves these stalks unchanged. Thus no relation is introduced merely by asserting a stalk calculation.

When $\mathcal{O}_{X,x}$ is a valuation ring, FCQ21 has the full table FCQ6–7. At the generic point its unit group is $K^\times$, and the quotient has just one nonzero-amplitude orbit, together with the full L :

$$\mathcal{C}_L(X)_\eta = \{z_\ell : \ell \in L\} \sqcup \{\mathbf{0}\}, \quad \mathbf{0} \boxplus \mathbf{0} = \begin{cases} \{e\}, & K = \mathbf{F}_2, \\ \{e, \mathbf{0}\}, & |K| > 2. \end{cases}\tag{FCQ22}$$

This follows either from the same residue argument for the trivial valuation or directly by summing two nonzero field elements. For the number fields and cyclotomic field here, the second case applies and the collapsed generic stalk is the Krasner hyperfield. The map from a valuation stalk to the generic stalk sends every $\mathbf{a}$ to $\mathbf{0}$ and fixes every z_ℓ . If the valuation is nontrivial, its group has an element less than any given a , which supplies the nonzero generic self-sum even in the residue- $\mathbf{F}_2$ case. If it is trivial, FCQ22 already gives the exact rule. This arithmetic generalization preserves hyperaddition in the inclusion sense, and is not in general strong. For distinct $a, b \in \Gamma$, the image of $\mathbf{a} \boxplus \mathbf{b}$ is the singleton $\{\mathbf{0}\}$, whereas the generic sum $\mathbf{0} \boxplus \mathbf{0}$ is $\{e, \mathbf{0}\}$ for the fields in this application. The larger generic unit group has added a cancellation. Thus the sheaf relations and their restriction maps retain this distinction; no equality of these two sets is assumed.

There is an exact sheaf version of FCQ12:

$$\mathcal{C}_L(X) \cong \{(h, \ell) \in \mathcal{C}(X) \times \underline{L} : h \neq 0 \implies \ell = 1_L\}. \quad (\text{FCQ23})$$

The condition is understood locally, or equivalently on germs, and the operations are the componentwise formulas following FCQ12. The natural map sends an orbit to its ordinary Cartier orbit and its retained support. It is bijective on every stalk by FCQ21 and FCQ12, so it is an isomorphism of sheaves; the stalk calculation also proves exact preservation of their multivalued relations. This is the explicit sheaf map to the original Cartier construction, retaining all of its zero-fibre data.

For $A = \mathcal{O}_{\mathbb{Q}^{\text{cyc}}}$, a prime above p has local ring equal to the valuation ring used in FCQ19. Indeed both are the union of the compatible finite localizations at that prime: every element and its denominator occur in a finite cyclotomic subfield. The ordinary collapsed stalks in FCQ23 are consequently $H(p)_{\text{hmax}}$ at closed primes and the Krasner hyperfield at the generic point, exactly as in the source's Proposition **sheavesonspz**, now with the corrected finite-stage addition. Applying FCQ23 to that source comparison gives the full supported sheaf over its arithmetic-site pullback. The finite-support valuation sections and the normalization remain those constructed in that proposition; no numerical value has been assigned to a lattice label.

8.7 Extension to the actual spectrum of the split carrier

The arithmetic scheme X in FCQ20 is not the whole prime spectrum of $S_A = G_L(A)$. To retain that additional geometry, put $Y = \text{Spec } S_A$, with its local-fraction structure sheaf $\mathcal{O}_Y$. The original prime classification and localization give

$$\begin{aligned} Q_{\mathfrak{q}} &= \{z_\ell : \ell \in L\} \cup \{\widehat{a} : a \in \mathfrak{q}, a \neq 0\}, & P_{\mathfrak{p}} &= \{z_\ell : \ell \in \mathfrak{p}\}, \\ (\mathcal{O}_Y)_{Q_{\mathfrak{q}}} &= G_L(A_{\mathfrak{q}}), & (\mathcal{O}_Y)_{P_{\mathfrak{p}}} &= L[(L \setminus \mathfrak{p})^{-1}], \\ V_Y(e) &\cong X, & D_Y(e) &\cong \text{SpecLat } L. \end{aligned} \quad (\text{FCQ24})$$

Here $\mathfrak{q}$ is an ordinary prime of A and $\mathfrak{p}$ is a lattice prime. All of these primes are present.

For explicitness, at an arithmetic prime every denominator has nonzero amplitude and top support, so fraction equality is exactly the amplitude fraction equality with each z_ℓ retained; this proves the first localization. At a support prime, e is invertible. As a multiplicative idempotent it becomes the unit; since $e\widehat{a} = e$ for every supported amplitude, all such amplitudes become that unit. The remaining lattice localization is the second displayed stalk. Writing $F = L \setminus \mathfrak{p}$, its elements have the exact equality relation

$$\ell \sim_F m \iff \exists \nu \in F : \nu \wedge \ell = \nu \wedge m. \quad (\text{FCQ25})$$

The relation is just localization equality after clearing one denominator. Finite distributivity proves that it is a lattice congruence. Elements of F become top, and bottom does not become top: a witness to that equality would force a member of F to be bottom. This proves the relevant support stalks without replacing them by Boolean residues.

Let $\mathcal{M}_Y$ be the meromorphic S_A -algebra sheaf obtained by inverting all supported nonzero amplitudes $T = \{\widehat{a} : 0 \neq a \in A\}$. Concretely it is the sheaf of local fractions of $T^{-1}S_A = G_L(K)$ as an S_A -algebra: on each principal-open local chart one also inverts that chart's denominators. These local fractions and their agreement relations define the sheaf, and give

$$(\mathcal{M}_Y)_{Q_q} = G_L(K), \quad (\mathcal{M}_Y)_{P_p} = L[(L \setminus \mathfrak{p})^{-1}]. \quad (\text{FCQ26})$$

The second equality holds because those additional supported denominators are already the unit in the support stalk, by FCQ24.

Define the full quotient on the actual split spectrum by

$$\widetilde{\mathcal{C}}_L(Y) = (\mathcal{M}_Y / \mathcal{O}_Y^\times)^\# . \quad (\text{FCQ27})$$

The orbit and addition relations are sheafified as in FCQ20. At an arithmetic stalk its unit group is $\widehat{A}_q^\times$, and at a support stalk the only multiplicative unit of the lattice semiring is its top: an invertible idempotent equals the unit. Hence its exact two kinds of stalks are

$$\begin{aligned} \widetilde{\mathcal{C}}_L(Y)_{Q_q} &= G_L(K) / A_q^\times, \\ \widetilde{\mathcal{C}}_L(Y)_{P_p} &= L[(L \setminus \mathfrak{p})^{-1}]. \end{aligned} \quad (\text{FCQ28})$$

The quotient–stalk commutation is proved by the same finite-unit-witness argument as FCQ21. In the second line the orbit action is trivial, and addition and multiplication are the single-valued join and meet. Thus both the arithmetic Cartier data and the support open are calculated.

On the closed arithmetic subspace $i : X \cong V_Y(e) \hookrightarrow Y$, restriction of FCQ27 is canonically FCQ20. On a basic open $D_Y(\widehat{a})$ its arithmetic part is $D_X(a)$; the fraction maps restrict to the identical split amplitudes and unit sections there. These charts give the map, and FCQ21 and FCQ28 prove it an isomorphism on every arithmetic stalk. On the open $D_Y(e)$, the restriction of FCQ27 is the ordinary local-fraction sheaf of the lattice semiring L , because inverting e identifies both $\mathcal{O}_Y$ and $\mathcal{M}_Y$ with that sheaf and its unit quotient is trivial.

The generization map from an arithmetic stalk to a support stalk has the exact formula

$$G_L(K) / A_q^\times \longrightarrow L[(L \setminus \mathfrak{p})^{-1}], \quad z_\ell \longmapsto [\ell]_F, \quad [\widehat{x}] \longmapsto [1_L]_F. \quad (\text{FCQ29})$$

It is well defined because every ordinary unit and every nonzero amplitude becomes top there. It sends each multivalued sum to the singleton join of its images, including cancellation, since cancellation has support 1_L . Multiplication also commutes. This is the localized support projection, and is the actual connecting map between the two strata, not an assertion that the support stratum is absent.

Equations FCQ23 and FCQ27–29 supply both requested Cartier bridges. Over the ordinary arithmetic subspace they retain the original unit-orbit Cartier hyperfield together with every z_ℓ . Over the full split spectrum they additionally retain the localized support sheaf and its exact generization maps. The independent additive zero τ and cancellation zero e remain distinct at arithmetic stalks; their behavior on the support open is the explicitly calculated localization $e = 1$. None of these constructions assumes a Frobenius weight, positivity theorem or purity.

9 Exact checks and repairs in the classical circle calculation

This section concerns the original author source `RRZ.tex` of Alain Connes and Caterina Consani, *Riemann–Roch for the ring $\mathbb{Z}$* . The source SHA–256 is

`1975ade5af5d3facf0f954da414a85c07c8adb7a83d1cc08e38f3112c5d072f9`.

The exact locations are the sequence definition at lines 309–311, Lemma `sequenceb` at lines 316–322, and Proposition `caseh1` and its proof at lines 324–338. The sequence formula is verified below. The minimum-distance assertion and the choice of a nearest integer in a fixed interval require the indicated repairs. The resulting classical dimension statement is preserved. Nothing here identifies a first cohomology for the support-sensitive carrier.

9.1 The interval formula, with its actual indices

The author's formula is

$$j(n) = \frac{(-2)^n}{3} - \frac{(-1)^n}{2} + \frac{1}{6}, \quad k(n) = 2\lfloor n/2 \rfloor + 1. \quad (\text{SFC1})$$

In particular $j(1) = 0$, $j(2) = 1$, and $j(3) = -2$. The displayed value list in the source starts at $n = 1$. For $n = 2$, its interval formula is therefore

$$[j(k(2)), j(k(2)) + 2^2 - 1] = [j(3), j(3) + 3] = [-2, 1], \quad (\text{SFC2})$$

exactly the subset sums of $\{1, -2\}$. No change to $k(n)$ is needed.

For all $n \geq 0$ define $G(n) = \{(-2)^r : 0 \leq r < n\}$. The least and greatest subset sums are

$$L_n = -\frac{2(4^{\lfloor n/2 \rfloor} - 1)}{3}, \quad U_n = \frac{4^{\lfloor n/2 \rfloor} - 1}{3}, \quad L_n = j(k(n)), \quad U_n - L_n = 2^n - 1. \quad (\text{SFC3})$$

The first two formulas are the sums of all negative and all positive powers, respectively. Substitution of the odd number $k(n)$ into (SFC1) proves $L_n = j(k(n))$. The two parity cases of n give the last equality. The subset sums are distinct: in the difference of two unequal sums, take the smallest index r with nonzero coefficient. The coefficient is 1 or -1 , so the difference is nonzero modulo 2^{r+1} . There are consequently 2^n distinct integer sums in an interval of exactly that cardinality. They fill $[L_n, U_n]$. The repeated label $\Delta(4)$ on the last listed interval $[-10, 21]$ at source line 320 should be $\Delta(5)$; (SFC3) gives $L_5 = -10$ and $U_5 = 21$ directly.

9.2 The exact circular minimum distance

Use the circle $\mathbb{R}/\mathbb{Z}$ with $d(x, y) = \min_{r \in \mathbb{Z}} |x - y - r|$, and the actual set

$$F(m) = \{(-2)^{-j} \bmod \mathbb{Z} : 1 \leq j \leq m\}. \quad (\text{SFC4})$$

For $m = 0$ it is empty, and for $m = 1$ it is a singleton, so the pairwise-separation requirement is vacuous. For every $m \geq 2$,

$$\min_{x \neq y \in F(m)} d(x, y) = \min \left\{ \frac{1}{4}, 3 \cdot 2^{-m} \right\}. \quad (\text{SFC5})$$

For proof, write $x_j = (-1)^j 2^{-j}$. The distance between $x_1 = -1/2$ and x_j is $1/2 - 2^{-j}$ for every $j \geq 2$: if j is even one uses the complementary circular arc. Its minimum is $1/4$, attained at $j = 2$. For $2 \leq i < j \leq m$, the absolute difference is at most $3/8 < 1/2$, so it already equals the circular distance. It is

$$2^{-j}(2^{j-i} - 1) \quad \text{when } i, j \text{ have equal parity,} \quad 2^{-j}(2^{j-i} + 1) \quad \text{otherwise.}$$

Equal parity implies $j - i \geq 2$; different parity implies $j - i \geq 1$. Both expressions are at least $3 \cdot 2^{-j} \geq 3 \cdot 2^{-m}$. For $m \geq 3$, the adjacent pair $i = m - 1, j = m$ attains this value. For $m = 2$, only the first pair exists. This proves (SFC5).

In particular, at $m = 2$ the two points are $-1/2$ and $1/4$, whose distance is $1/4$, not $3/4$. At $m = 3$ the minimum is still $1/4$, not $3/8$. These are the small cases missed by the minimum-distance expression in source line 331. For $m \geq 4$, its expression $3 \cdot 2^{-m}$ does equal (SFC5).

9.3 The periodic grid proves the covering step

The subset sums of (SFC4) are exactly the dyadic grid

$$\left\{ \sum_{j=1}^m \alpha_j (-2)^{-j} \bmod \mathbb{Z} : \alpha_j \in \{0, 1\} \right\} = \left\{ \frac{r}{2^m} \bmod \mathbb{Z} : 0 \leq r < 2^m \right\}. \quad (\text{SFC6})$$

To verify this without an endpoint assumption, multiply two proposed equal subset sums by $(-2)^m$. Their difference is an integer combination of $1, -2, \dots, (-2)^{m-1}$ with coefficients in $\{-1, 0, 1\}$, divisible by 2^m . A smallest nonzero coefficient would give valuation less than m , by the argument after (SFC3), which is impossible. The 2^m subset sums therefore give distinct elements of the group of 2^m grid points, proving equality.

For any $x \in \mathbb{R}$, choose an integer r nearest to $2^m x$, so $|2^m x - r| \leq 1/2$. Reducing r modulo 2^m does not alter the circle class of $r/2^m$. Hence

$$\max_{x \in \mathbb{R}/\mathbb{Z}} \min_{y \text{ a subset sum of } F(m)} d(x, y) = 2^{-m-1}. \quad (\text{SFC7})$$

The upper bound follows from the nearest-integer construction; the midpoints of adjacent grid points attain it.

The fixed nonperiodic nearest-integer step at source lines 336–337 cannot replace this reduction. For $m = 2$, its integer interval is $\Delta(2) = [-2, 1]$. The point $x = 7/16$ belongs to the corresponding half-open real interval $[-1/2, 1/2)$, but

$$\min_{q \in \{-2, -1, 0, 1\}} |4x - q| = \frac{3}{4} > \frac{1}{2}. \quad (\text{SFC8})$$

The nearest integer is 2, outside that chosen interval. Modulo four it is represented by -2 , and the associated circle point is $-1/2 \bmod \mathbb{Z}$, at distance $1/16$ from x . Thus the periodic-grid argument gives the required covering even at this endpoint case.

9.4 The classical dimension conclusion is retained

For the original tolerant circle object in Connes–Consani’s Proposition **caseh1**, all group subset sums are admissible, and the degree-one tolerance is $d(x, y) \leq \lambda$. When $\lambda \geq 1/2$, the empty set generates because every circle point is within tolerance of zero. When

$$2^{-m-1} \leq \lambda < 2^{-m}, \quad m \geq 1, \quad (\text{SFC9})$$

the m -element set $F(m)$ generates by (SFC7). Its separation condition is also valid: it is vacuous for $m = 1$; for $m = 2$, (SFC5) gives $1/4 > \lambda$; and for $m \geq 3$ both entries in the minimum in (SFC5) exceed $\lambda < 2^{-m}$.

Conversely, a set of r generators has at most 2^r subset sums. For $\lambda < 1/2$, each closed circular tolerance ball has measure 2λ , in the length-one Haar measure. Covering the circle therefore requires $2\lambda 2^r \geq 1$. If $r \leq m - 1$ in (SFC9), this left side is strictly less than 1. Thus $r \geq m$, and the original formula is proved with the repaired steps:

$$\dim_{\mathbb{S}} U(1)_{\lambda} = \begin{cases} 0, & \lambda \geq 1/2, \\ m, & 2^{-m-1} \leq \lambda < 2^{-m}, \quad m \geq 1. \end{cases} \quad (\text{SFC10})$$

This calculation supplies corrected finite-circle distances and a periodic covering proof for the existing classical object. It makes no claim that the support fibres calculated in (SRR4)–(SRR5) have disappeared in a new first cohomology construction.

The source is Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* , arXiv:2609.00299v1, specifically the original author file `CC.tex`. Its complete 1,228-line source was read for this calculation. The source labels used below are

```
opens; eq:restriction_maps;
def:alpha_symmetry; thm:twistor_space_scheme;
prop:twistor_frobenius.
```

Their local signs and generators are retained. The source file's SHA-256 is

```
57a10dcef5cd758e6b2f1c54b1c0a11b0fb0
93638f74fb1be515ec3bc7080ba4.
```

This note evaluates the source's *pointed multiplicative chart monoids*, glues the resulting chart-point sets, and takes their explicit α -orbits. These constructions are defined below. No assertion that this calculation classifies all geometric morphisms into a quotient topos, or all torsors over a general target, is needed or made. This specification is essential because the target is not a ring. No Riemann-hypothesis conclusion is asserted.

10 The target and its exact morphism to the complex numbers

Let L be a bounded distributive lattice with bottom 0_L and top 1_L . The target is the given Split-Zero semiring

$$\begin{aligned} S = G_L(\mathbb{C}) &= \{(0, \lambda) : \lambda \in L\} \cup \{(a, 1_L) : a \in \mathbb{C}\}, \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu). \end{aligned} \tag{CSP1}$$

Pairs in the displayed union have their literal equality; thus the two occurrences of $(0, 1_L)$ are one element. Put

$$z_\lambda = (0, \lambda), \quad \tau = z_{0_L}, \quad e = z_{1_L}, \quad \widehat{a} = (a, 1_L), \quad 1 = \widehat{1}, \quad \epsilon_S = -\widehat{1}. \tag{CSP2}$$

The multiplicative monoid is pointed by its absorbing element τ . In particular e is not its distinguished absorbing element when $0_L \neq 1_L$. Addition and multiplication in (CSP1) are closed: a nonzero output amplitude forces every factor with a nonzero amplitude to have top support. The lattice laws and complex arithmetic prove the semiring laws, with distributivity using $\lambda \wedge (\mu \vee \nu) = (\lambda \wedge \mu) \vee (\lambda \wedge \nu)$.

Lemma 33. *The units of S and the solutions of the imaginary-generator equation are*

$$S^\times = \{\widehat{a} : a \in \mathbb{C}^\times\}, \quad \widehat{a}^{-1} = \widehat{a^{-1}}, \quad \{j \in S : j^2 = \epsilon_S\} = \{\widehat{i}, \widehat{-i}\}. \tag{CSP3}$$

For every $\lambda \in L$ and positive integer n ,

$$z_\lambda^n = z_\lambda, \quad \epsilon_S z_\lambda = z_\lambda, \quad z_\lambda z_\mu = z_{\lambda \wedge \mu}. \tag{CSP4}$$

Proof. An inverse equation $xy = 1$ forces the product of the amplitudes to be 1. Both amplitudes are nonzero, so both supports equal 1_L and the inverse has the displayed value. Conversely these values multiply to 1. If $j^2 = -\widehat{1}$, its amplitude is nonzero and squares to -1 , so it equals i or $-i$ and its support is 1_L . Formula (CSP4) follows directly from lattice idempotence and (CSP1). $\square$

Proposition 34. *The amplitude map and a pointed multiplicative section are*

$$q : S \longrightarrow \mathbb{C}, \quad q(a, \lambda) = a, \quad b : \mathbb{C} \longrightarrow S, \quad b(0) = \tau, \quad b(a) = \widehat{a} \ (a \neq 0). \tag{CSP5}$$

Here q is a unital semiring homomorphism; b is the unique pointed unital multiplicative section of q and preserves -1 . For $0_L \neq 1_L$, b is not additive, and no unital semiring homomorphism from a nonzero ring to S exists.

Proof. The formula for q preserves each operation in (CSP1), together with zero and one. Multiplicativity of b holds when both factors are nonzero because $\mathbb{C}$ is a field; when at least one is zero it follows from absorption by τ . Any section must take each nonzero a to the only point with amplitude a , namely $\widehat{a}$, and pointedness forces its value at zero. However $b(1) + b(-1) = e \neq \tau = b(0)$. More generally, a unital semiring homomorphism $f : R \rightarrow S$ from a ring would give $1 + f(-1) = f(0) = \tau$. The support of the left side is $1_L \vee \text{supp}(f(-1)) = 1_L$, contradicting the bottom support of τ when $0_L \neq 1_L$. $\square$

Complex conjugation is the semiring automorphism $c_S(a, \lambda) = (\bar{a}, \lambda)$. It fixes every z_λ , preserves ϵ_S , and exchanges $\widehat{i}$ with $\widehat{-i}$.

11 All chart points and all overlaps

Let $B = \{0, 1, \epsilon\}$ with $\epsilon^2 = 1$ be the source's signed pointed base monoid. Its target structure sends 0 to τ , 1 to 1, and ϵ to ϵ_S . The chart monoids, with the source's generators unchanged, are

$$M_+ = B[T, J_+]/(J_+^2 = \epsilon), \quad M_- = B[S, J_-]/(J_-^2 = \epsilon), \quad M_0 = B[T, T^{-1}, J]/(J^2 = \epsilon). \quad (\text{CSP6})$$

Each nonzero element of M_+ is uniquely $J_+^a T^m$ with $a \in \mathbb{Z}/4\mathbb{Z}$ and $m \in \mathbb{N}$; M_- has the corresponding form in J_- , S , and M_0 has $m \in \mathbb{Z}$. Thus these presentations have no additive relation identifying $1 + \epsilon$ with the distinguished zero. All homomorphisms in this section are pointed unital multiplicative B -homomorphisms.

The restrictions are exactly the source's restrictions:

$$\rho_+(T) = T, \quad \rho_+(J_+) = J, \quad \rho_-(S) = T^{-1}, \quad \rho_-(J_-) = \epsilon J. \quad (\text{CSP7})$$

Theorem 35. *The complete point sets and overlap maps are*

$$U_\pm(S) = \text{Hom}_B(M_\pm, S) = S \times \{\widehat{i}, \widehat{-i}\}, \quad U_0(S) = S^\times \times \{\widehat{i}, \widehat{-i}\}, \quad (\text{CSP8})$$

$$\rho_+^*(t, j) = (t, j)_+, \quad \rho_-^*(t, j) = (t^{-1}, \epsilon_S j)_-. \quad (\text{CSP9})$$

No z_λ belongs to the overlap, including the supported zero e . The glued chart-point set

$$X_L = U_+(S) \coprod_{U_0(S)} U_-(S) \quad (\text{CSP10})$$

is the disjoint union of two copies of

$$P_L = \mathbb{C}^\times \sqcup \{0_\lambda : \lambda \in L\} \sqcup \{\infty_\lambda : \lambda \in L\}. \quad (\text{CSP11})$$

The symbols 0_λ and ∞_μ are distinct for all λ, μ ; equality within either family is precisely equality of the lattice labels.

Proof. A map on M_+ is determined by $t = \phi(T)$ and $j = \phi(J_+)$. The relation forces $j^2 = \epsilon_S$, whose solutions are (CSP3), and imposes no condition on t . Conversely, for any such pair, the formula $\phi(J_+^a T^m) = j^a t^m$ and $\phi(0) = \tau$ defines a unique map. The same proof applies to M_- . On M_0 , the equation $TT^{-1} = 1$ forces t to be a unit, and (CSP3) gives all possibilities. Precomposition with (CSP7) yields (CSP9).

Both overlap maps are injective. Their gluing relation pairs exactly $(t, j)_+$ with $(t^{-1}, -j)_-$ for $t \in S^\times$. The first component consists of U_+ with $j = \widehat{i}$ and U_- with $j = \widehat{-i}$. In it, a nonzero complex a is represented by $(\widehat{a}, \widehat{i})_+$, equivalently $(\widehat{a^{-1}}, \widehat{-i})_-$. Its two families of additional points are represented by $(z_\lambda, \widehat{i})_+$ and $(z_\lambda, \widehat{-i})_-$, which define respectively 0_λ and ∞_λ . None is in an overlap, so none can be identified by a chain of gluing relations. Reversing the two imaginary signs gives the second component and proves the assertion. $\square$

This proof constructs P_L as the two-chart gluing $S \coprod_{S^\times} S$ with reciprocal transition. It neither declares a labelled zero invertible nor replaces the pointed zero by e .

12 The geometric quotient and its amplitude fibres

The source's symmetry is given on chart points by

$$\alpha(t, j)_+ = (\epsilon_S t, j)_-, \quad \alpha(s, j)_- = (\epsilon_S s, j)_+, \quad (\text{CSP12})$$

and on the overlap by

$$\alpha(t, j) = (\epsilon_S t^{-1}, \epsilon_S j). \quad (\text{CSP13})$$

These formulas are evaluations of $\alpha_+^*(S) = \epsilon T$, $\alpha_+^*(J_-) = J_+$ and their minus-chart counterparts, exactly as in the source.

Theorem 36. *Formulas (CSP12)–(CSP13) define an involution of X_L exchanging its two components. Consequently the orbit set $C_L = X_L / \langle \alpha \rangle$ is in bijection with P_L , by the unique representative in the first component. Amplitude gives the surjection*

$$\pi_L : C_L \cong P_L \longrightarrow \mathbb{P}^1(\mathbb{C}), \quad a \longmapsto a \ (a \neq 0), \quad 0_\lambda \longmapsto 0, \quad \infty_\lambda \longmapsto \infty. \quad (\text{CSP14})$$

Its fibre is a singleton over every $a \in \mathbb{C}^\times$ and an exact copy of L over each of 0 and ∞ .

Proof. To check the only gluing issue, start with the equivalent points $(t, j)_+$ and $(t^{-1}, -j)_-$. Their images are $(-t, j)_-$ and $(-t^{-1}, -j)_+$. These are equivalent because the inverse of $-t^{-1}$ is $-t$ and the imaginary sign changes in the transition. Applying (CSP12) twice multiplies the spatial coordinate by $\epsilon_S^2 = 1$ and leaves j unchanged. A first-component plus point has $j = i$ and maps to a minus point with $j = i$, which lies in the second component; the other three chart/sign combinations give the same exchange. Every orbit therefore meets the first component exactly once. The map q in (CSP5) preserves all chart relations, the reciprocal transition, and the symmetry. Its resulting quotient map is exactly (CSP14); Theorem 35 supplies all its fibres. $\square$

$$\begin{array}{ccc} \{0_\lambda : \lambda \in L\} & \mathbb{C}^\times & \{\infty_\lambda : \lambda \in L\} \\ \pi_L \downarrow & \parallel & \downarrow \pi_L \\ \{0\} & \mathbb{C}^\times & \{\infty\} \end{array}$$

Figure 4: Exact disjoint decomposition of the amplitude map, proved in Theorems 35 and 36. Every support label survives in the upper row; only the map to the lower row identifies them. The real structure exchanges 0_λ and ∞_λ with the same label (Theorem 37). The chart and sign conventions come from Connes–Consani [1], formulas `opens` and `eq:restriction_maps`.

For every $\lambda \in L$, the formula

$$s_\lambda : \mathbb{P}^1(\mathbb{C}) \rightarrow P_L, \quad s_\lambda(a) = a \ (a \neq 0), \quad s_\lambda(0) = 0_\lambda, \quad s_\lambda(\infty) = \infty_\lambda \quad (\text{CSP15})$$

is an injective set-theoretic section of π_L . Only s_{0_L} is induced by a pointed multiplicative section of q : it is induced by b in (CSP5). Indeed a pointed map must send zero to τ . Thus using s_{1_L} to retain supported zeros is a well-defined section of the point map but is not an unmentioned pointed-monoid section. All of these distinctions disappear when L has one element, in which case $S = \mathbb{C}$ and $P_L = \mathbb{P}^1(\mathbb{C})$.

If $f : L \rightarrow L'$ is a bounded lattice homomorphism, the map

$$G_f : S \rightarrow G_{L'}(\mathbb{C}), \quad z_\lambda \mapsto z_{f(\lambda)}, \quad \hat{a} \mapsto \hat{a} \ (a \neq 0) \quad (\text{CSP16})$$

preserves addition, multiplication, zero, one, and ϵ . For addition of two nonzero amplitudes with a zero sum, the image is $z_{f(1_L)} = z_{1_{L'}}$, so cancellation is also checked. All other cases follow directly from meet and join preservation. It therefore induces $P_L \rightarrow P_{L'}$ taking each pole label λ to $f(\lambda)$ and fixing $\mathbb{C}^\times$. This is the exact change-of- support map, including every zero.

13 Real structure and every odd Frobenius

Define two bijections of P_L :

$$\begin{aligned} \beta(a) &= -a^{-1}, & \beta(0_\lambda) &= \infty_\lambda, & \beta(\infty_\lambda) &= 0_\lambda, \\ \sigma(a) &= -\bar{a}^{-1}, & \sigma(0_\lambda) &= \infty_\lambda, & \sigma(\infty_\lambda) &= 0_\lambda. \end{aligned} \tag{CSP17}$$

These are formulas on disjoint subsets; no inverse of z_λ is present.

Theorem 37. *Postcomposition of chart points with c_S commutes with α , and the induced real structure on $C_L = P_L$ is σ of (CSP17). It has no fixed points, retains every support label, and obeys*

$$\pi_L \sigma(a) = -1/\overline{\pi_L(a)}, \quad \sigma s_\lambda = s_\lambda \sigma_{\mathbb{C}}. \tag{CSP18}$$

The first formula uses the usual projective exchange of 0 and ∞ on its right side.

Proof. Complex conjugation preserves multiplication, inverse of units, and ϵ_S , hence commutes with (CSP12)–(CSP13) and respects (CSP9). It exchanges the two components. On a first-component plus point $(t, i)_+$, apply conjugation and then α to return to the first component: $(t, i)_+ \mapsto (\bar{t}, -i)_+ \mapsto (-\bar{t}, -i)_-$. If $t = \hat{a}$ is a unit, the first-component global coordinate is $-1/\bar{a}$; if $t = z_\lambda$, the point is ∞_λ . The minus-chart calculation similarly takes ∞_λ to 0_λ . For a unit, a fixed point would satisfy $a\bar{a} = -1$, which is impossible in $\mathbb{C}$. The two labelled pole families are disjoint, so there are no fixed points there either. Equation (CSP18) follows on all three disjoint subsets. $\square$

For a positive integer n let P_n on P_L raise $a \in \mathbb{C}^\times$ to a^n and fix every labelled pole. Let $d(n) \in \{0, 1\}$ be the residue of $(n-1)/2$ modulo 2 for odd n .

Theorem 38. *Raising every element of the source chart monoids to its n th power is a pointed B -endomorphism precisely when n is odd. These maps commute with restriction and α . Their induced action on P_L is*

$$F_n = \beta^{d(n)} P_n, \quad F_n(a) = \begin{cases} a^n & n \equiv 1 \pmod{4}, \\ -a^{-n} & n \equiv 3 \pmod{4}, \end{cases} \tag{CSP19}$$

	$F_n(0_\lambda)$	$F_n(\infty_\lambda)$	
$n \equiv 1 \pmod{4}$	0_λ	∞_λ	.
$n \equiv 3 \pmod{4}$	∞_λ	0_λ	

(CSP20)

They satisfy, without any removal of labelled poles,

$$F_m F_n = F_{mn}, \quad F_n \sigma = \sigma F_n, \quad \pi_L F_n = \text{Fr}_n \pi_L, \quad F_n s_\lambda = s_\lambda \text{Fr}_n. \tag{CSP21}$$

Here Fr_n is the source's complex twistor action, from the proposition labelled `prop:twistor_frobenius`.

Proof. Power maps are monoid endomorphisms because the chart monoids are commutative. Their restriction to B fixes ϵ precisely when $\epsilon^n = \epsilon$, namely for odd n . For odd n the relation $(J^n)^2 = \epsilon^n = \epsilon$ is preserved. On the overlap, the two orders of restriction send J_- to $(\epsilon J)^n = \epsilon J^n$ and send S to T^{-n} , so they agree. Power maps commute with every monoid homomorphism, in particular the maps defining α .

On chart points they take (t, j) to (t^n, j^n) . Since $j^n = (-1)^{(n-1)/2} j$, for $n \equiv 1 \pmod{4}$ the first component is preserved, while for $n \equiv 3 \pmod{4}$ it is exchanged with the second. In the latter case applying α returns the image to the first component and changes a unit coordinate to $-a^{-n}$. For $t = z_\lambda$, (CSP4) gives $t^n = z_\lambda$ and multiplication by ϵ_S does not change it. The chart exchange gives exactly (CSP20), including $\lambda = 0_L, 1_L$.

For odd n , $\beta P_n = P_n \beta$: on units this is the identity $(-a^{-1})^n = -a^{-n}$, and both sides have the same pole exchange. Since $P_m P_n = P_{mn}$ and $d(mn) = d(m) + d(n)$ modulo 2, (CSP19) proves the first identity in (CSP21). The map σ is β followed by amplitude conjugation; the latter commutes with every P_n and β . This proves the second identity. The other two follow from (CSP14), (CSP15), (CSP19), and (CSP20) on every part of P_L . $\square$

Proposition 39. *For odd n , every unit point of P_L has exactly n distinct F_n -preimages, and each labelled pole has exactly one preimage. For $n > 1$ the fixed-point sets are*

$$\text{Fix}(F_n) = \begin{cases} \{a \in \mathbb{C}^\times : a^{n-1} = 1\} \sqcup \{0_\lambda, \infty_\lambda : \lambda \in L\}, & n \equiv 1 \pmod{4}, \\ \{a \in \mathbb{C}^\times : a^{n+1} = -1\}, & n \equiv 3 \pmod{4}. \end{cases} \quad (\text{CSP22})$$

In particular

$$\bigcap_{n \text{ odd}} \text{Fix}(F_n) = \emptyset. \quad (\text{CSP23})$$

Proof. For a nonzero target w , the preimage equation is $a^n = w$ or $a^n = -1/w$. Its right side is nonzero and its n roots in $\mathbb{C}$ are distinct, since a repeated root would also annihilate na^{n-1} . No labelled pole maps to a unit. Formula (CSP20) gives the single preimage of each pole, with its unchanged support label. Setting the expressions in (CSP19) equal to a gives (CSP22); (CSP20) settles the poles. A simultaneous fixed point cannot be a pole because F_3 exchanges the two pole families. A unit fixed by F_3 satisfies $a^4 = -1$, whereas one fixed by F_5 satisfies $a^4 = 1$. Their intersection is empty, proving (CSP23). $\square$

Statement (CSP23) concerns *points*. It does not contradict the source's Frobenius-invariant *sections* calculation: those are different explicitly defined functors. On the source monoid M_+ the simultaneous fixed elements are exactly B : a nonzero element $J^a T^m$ fixed by its third power has $3m = m$, whence $m = 0$, and $3a = a$ modulo 4, whence $a = 0$ or 2. Zero is also fixed. On evaluated points, however, the image of J must be one of the two square roots of -1 ; neither is fixed by its third power. This calculates the difference directly.

14 The precise Hopf limitation and the surviving torsor law

The spatial core $B[T, T^{-1}]$ has the source's coproduct $T \mapsto T \otimes T$, with the base sign fixed. Its S -points form exactly $S^\times = \mathbb{C}^\times$. The action on the enriched overlap obtained by multiplying its spatial coordinate is

$$D_a(t, j) = (\widehat{at}, j), \quad a \in \mathbb{C}^\times. \quad (\text{CSP24})$$

It extends to X_L by multiplying the plus coordinate by a and the minus coordinate by a^{-1} , preserving each component. It obeys $\alpha D_a = D_{a^{-1}} \alpha$. If $a^2 \neq 1$, it does not induce an action on C_L : for a generic point $x = (t, i)$ the two points $D_a x$ and $D_a \alpha x$ would have to be α -related, but their second coordinates force that relation and their first coordinates then force $-a/t = -1/(at)$, or $a^2 = 1$.

There is nevertheless an exact lift of the usual quotient scaling. Write $\chi(\widehat{i}) = 1$, $\chi(\widehat{-i}) = -1$. Define on each affine chart, using its own local imaginary generator,

$$A_a(t, j)_+ = (\widehat{a^{\chi(j)} t}, j)_+, \quad A_a(s, j)_- = (\widehat{a^{\chi(j)} s}, j)_-. \quad (\text{CSP24a})$$

On the overlap the sign of j changes in the transition and t is inverted, so the two formulas agree. Formula (CSP12) preserves the local j while exchanging charts and multiplying by -1 ; therefore A_a commutes with α . Also $A_a A_b = A_{ab}$. On the first component, (CSP24a) scales the global unit coordinate by a and fixes every labelled pole, hence descends to that precise action $a \cdot x$ on P_L . This is a lift that uses the imaginary branch; it is not the unmodified action (CSP24). It obeys

$$\sigma(a \cdot x) = \bar{a}^{-1} \cdot \sigma(x) \quad (a \in \mathbb{C}^\times, x \in P_L). \quad (\text{CSP25})$$

For unit x this is complex arithmetic, and for a pole it follows because both scaling maps fix the poles and σ exchanges their two families.

Proposition 40. *The prescription $J \mapsto J \otimes J$ does not extend the spatial diagonal to a B -coproduct on M_0 . On its full enriched point set $U_0(S) = \mathbb{C}^\times \times \{\widehat{i}, \widehat{-i}\}$ the choice-free operation that survives is the ternary law*

$$[(t_1, j_1), (t_2, j_2), (t_3, j_3)] = (t_1 t_2^{-1} t_3, j_1 j_2^{-1} j_3). \quad (\text{CSP26})$$

It satisfies

$$[x, y, y] = x = [y, y, x], \quad [[x, y, z], u, v] = [x, y, [z, u, v]]. \quad (\text{CSP27})$$

*Both α and complex conjugation preserve it, as does every odd power map. Selecting any one point o gives the group law $x *_o y = [x, o, y]$; selecting $o = (1, \widehat{i})$ gives the explicit law*

$$(t_1, j_1) *_o (t_2, j_2) = (t_1 t_2, j_1 \widehat{-i} j_2). \quad (\text{CSP28})$$

Proof. In the balanced monoid tensor product over B , $\epsilon \otimes 1 = 1 \otimes \epsilon$. Therefore $(J \otimes J)^2 = \epsilon \otimes \epsilon = 1$, while a B -map must send $J^2 = \epsilon$ to $\epsilon \otimes 1$. These two elements are different: evaluate both tensor factors at $T = 1, J = \widehat{i}$ in S . The balanced tensor evaluation sends 1 to 1 and $\epsilon \otimes 1$ to $\widehat{-1} \neq 1$. This is an obstruction to this exact diagonal on the *enriched* monoid; it is no obstruction to the coproduct on the spatial core used in (CSP24).

In (CSP26), every displayed inverse exists, and $(j_1 j_2^{-1} j_3)^2 = (-1)(-1)^{-1}(-1) = -1$, so the law lands in the enriched point set. The cancellation and association identities (CSP27) follow by multiplying the coordinates; for example both association expressions have first coordinate $t_x t_y^{-1} t_z t_u^{-1} t_v$ and the corresponding second coordinate. Equivalently its source ternary co-operation is $T \mapsto T \otimes T^{-1} \otimes T$ and $J \mapsto J \otimes J^{-1} \otimes J$: the square of the latter is $\epsilon \otimes \epsilon^{-1} \otimes \epsilon = \epsilon$, so the source relation is respected. The image of ϵ is the base element, as required.

On the overlap $\alpha(t, j) = (-t^{-1}, j^{-1})$. Substitution into (CSP26) shows the first coordinate becomes $-t_1^{-1} t_2 t_3^{-1}$ in either order, and the second becomes $j_1^{-1} j_2 j_3^{-1}$ in either order. Conjugation preserves products and inverses, and power maps do so in this commutative group; for odd powers they also preserve the equation $j^2 = -1$. Hence they preserve (CSP26). Formula (CSP27) gives identity o , associativity, and inverse $[o, x, o]$ for $*_o$, proving it is a group. Formula (CSP28) uses $(\widehat{i})^{-1} = \widehat{-i}$. $\square$

The last proposition supplies the exact enriched replacement when the group-like imaginary-generator prescription fails. A binary law requires the stated choice of o ; the ternary law and all chart, support, real-structure, and Frobenius maps above require no such choice. The calculation retains the source's distinguished choice of i only when identifying the quotient with its first component.

References

- [1] Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* , arXiv:2609.00299v1 (2026). Original author source `CC.tex`; source formulas and labels are specified at the beginning and at their mathematical uses.

Source and scope. The source read is Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* , arXiv:2609.00299v1, original author source `CC.tex`, 1228 lines, SHA-256 57a10dcef5cd758e6b2f1c54b1c0a11b0fb093638f74fb1be515ec3bc7080ba4. Reading coverage is the complete file, including its bibliography. The signed base, imaginary chart, geometric involution, and odd maps are Connes–Consani's constructions, not new programme constructions. All assertions below are proved in the specified category of pointed commutative monoids, or in ordinary algebraic $K_0(\mathbb{P}_{\mathbb{C}}^1)$. No claim about a more general category of spherical algebras is inferred merely from this monoid calculation.

15 The actual base and tensor product

Let

$$B = \{0, 1, \epsilon\}, \quad \epsilon^2 = 1, \quad \epsilon \neq 1, \quad D = B[J]/(J^2 = \epsilon). \quad (\text{HA1})$$

Here 0 is absorbing and each displayed presentation is a presentation of a pointed commutative monoid. In particular there is no additive relation $1 + \epsilon = 0$. A B -monoid is a pointed commutative monoid with a specified map from B , and a B -morphism fixes that map. Put

$$C_+ = B[T], \quad C_\times = B[T, T^{-1}], \quad A_+ = D[T], \quad A_\times = D[T, T^{-1}]. \quad (\text{HA2})$$

These are the spatial chart and overlap, and their imaginary enrichments. The relation in (HA1) is the one in [1, original `CC.tex`, lines 576–600].

The tensor product $P \otimes_B Q$ means the smash product of pointed monoids modulo $(bp) \otimes q = p \otimes (bq)$ for $b \in B$. Its multiplication is $(p \otimes q)(p' \otimes q') = pp' \otimes qq'$. In particular

$$\epsilon \otimes 1 = 1 \otimes \epsilon = \epsilon, \quad \epsilon \otimes \epsilon = 1. \quad (\text{HA3})$$

These are exactly the balancing relations specified in [1, original `CC.tex`, lines 437–465].

Lemma 41. *Every nonzero element of A_+ has a unique expression $J^a T^b$ with $a \in \mathbb{Z}/4\mathbb{Z}$ and $b \in \mathbb{N}$. For $A_\times$, replace $\mathbb{N}$ by $\mathbb{Z}$. The units of $A_+ \otimes_B A_+$ have the presentation*

$$\langle j_1, j_2, \epsilon \mid j_1^2 = j_2^2 = \epsilon, \epsilon^2 = 1, [j_1, j_2] = 1 \rangle \cong (\mathbb{Z}/4\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z}).$$

In this tensor product $\epsilon \neq 1$.

Proof. The pointed monoid obtained by adjoining an absorbing zero to $(\mathbb{Z}/4\mathbb{Z}) \times \mathbb{N}$, with $J = (1, 0)$, $T = (0, 1)$ and $\epsilon = (2, 0)$, satisfies (HA1)–(HA2). Conversely those presentations reduce uniquely to these coordinates. The same construction works with $\mathbb{Z}$ in the second coordinate. In the tensor product use $j_1 = J \otimes 1$, $j_2 = 1 \otimes J$. The only additional relation between the two factors is $j_1^2 = j_2^2$. Setting $h = j_2 j_1^{-1}$ gives $h^2 = 1$; the inverse change of generators is $j_2 = j_1 h$. Thus the unit group is the asserted direct product. Its element $\epsilon = j_1^2$ has order two. Alternatively, the compatible pair of maps to the multiplicative monoid of $\mathbb{C}$, sending both J 's to i and ϵ to -1 , induces a map on the balanced tensor product which separates ϵ and 1. $\square$

Proposition 42 (The imaginary diagonal does not descend). *The assignments*

$$\Delta(T) = T \otimes T, \quad \Delta(J) = J \otimes J, \quad \Delta(\epsilon) = \epsilon \quad (\text{HA4})$$

do not define a B -morphism on A_+ or $A_\times$. Moreover, neither of those two B -monoids admits a counit to B .

Proof. A B -morphism must preserve $J^2 = \epsilon$. But (HA3) gives

$$(J \otimes J)^2 = \epsilon \otimes \epsilon = 1 \neq \epsilon = \Delta(J^2). \quad (\text{HA5})$$

For the second assertion, the image of the unit J under a unital monoid map is a unit. The only units of B are 1 and ϵ , both with square 1. Neither squares to ϵ . Consequently there is no B -morphism from either enriched chart to B at all. A counital comonoid structure, and hence a Hopf structure over B , would include such a morphism. $\square$

The scope matters: the coproduct in the author's definition is on the *spatial* monoid M of $B[M]$ ([1, original `CC.tex`, lines 462–465]), not an assertion that a diagonal on an additional generator survives every signed quotient relation. Equation (HA5) identifies the exact relation which prevents that extension; it does not erase the spatial coproduct.

16 The surviving coaction and the stronger torsor correspondence

Proposition 43 (Spatial coaction). *There is a counital and coassociative B -morphism*

$$\rho : A_+ \longrightarrow C_+ \otimes_B A_+, \quad T \longmapsto T \otimes T, \quad J \longmapsto 1 \otimes J. \quad (\text{HA6})$$

It extends to $A_\times \rightarrow C_\times \otimes_B A_\times$. The spatial bialgebra has $\Delta_C(T) = T \otimes T$ and counit $\eta_C(T) = 1$. On $C_\times$ the antipode is $T \mapsto T^{-1}$.

Proof. The image of J in (HA6) has square $1 \otimes \epsilon = \epsilon$, as required. There is no other relation to check beyond invertibility of T on the overlap, where its image is invertible. Both coassociativity composites send T to $T \otimes T \otimes T$ and J to $1 \otimes 1 \otimes J$. Applying the spatial counit sends them back to T and J . The assertions about C follow by the same checks on T and the base. $\square$

For a commutative B -monoid R let

$$X(R) = \text{Hom}_B(A_\times, R) = \{(t, j) \in R^\times \times R^\times : j^2 = \epsilon_R\}.$$

The spatial coaction is the actual map

$$R^\times \times X(R) \longrightarrow X(R), \quad (u, (t, j)) \longmapsto (ut, j). \quad (\text{HA7})$$

This includes, on complex points, $\mathbb{C}^\times$ acting without changing the chosen root of -1 . It proves the precise action on the unfolded chart.

There is a stronger intrinsic relation. Define

$$H = B[u, u^{-1}, h]/(h^2 = 1), \quad G(R) = R^\times \times \{h \in R^\times : h^2 = 1\}. \quad (\text{HA8})$$

The letter h in (HA8) is an independent generator; it is not the base element ϵ . The Hopf structure on H sends u and h to their tensor diagonals, with counit $u, h \mapsto 1$, and antipode $u \mapsto u^{-1}$, $h \mapsto h$.

Theorem 44 (Exact torsor identities and ternary product). *There is a coaction and a canonical isomorphism*

$$\rho_H : A_\times \longrightarrow H \otimes_B A_\times, \quad T \longmapsto u \otimes T, \quad J \longmapsto h \otimes J, \quad (\text{HA9})$$

$$\text{can} : A_\times \otimes_B A_\times \xrightarrow{\sim} H \otimes_B A_\times, \quad (\text{HA10})$$

where can sends

$$T_1 \mapsto uT, \quad T_2 \mapsto T, \quad J_1 \mapsto hJ, \quad J_2 \mapsto J.$$

When $X(R)$ is nonempty, the action $(u, h)(t, j) = (ut, hj)$ makes it a principal homogeneous set for $G(R)$. The ternary operation below requires no choice of a root:

$$[x, y, z] = (t_x t_y^{-1} t_z, j_x j_y^{-1} j_z), \quad X(R)^3 \longrightarrow X(R). \quad (\text{HA11})$$

This operation satisfies $[x, y, y] = x = [y, y, x]$ and $[[x, y, z], u, v] = [x, y, [z, u, v]]$.

Proof. For (HA9), $(h \otimes J)^2 = 1 \otimes \epsilon = \epsilon$; coassociativity and the counit follow on each generator. The inverse to (HA10) is the explicitly defined B -morphism

$$u \longmapsto T_1 T_2^{-1}, \quad h \longmapsto J_1 J_2^{-1}, \quad T \longmapsto T_2, \quad J \longmapsto J_2. \quad (\text{HA12})$$

Indeed $(J_1 J_2^{-1})^2 = \epsilon \epsilon^{-1} = 1$, so (HA12) respects the defining equation of h ; each composite fixes the displayed generators. Given $(t, j), (t', j') \in X(R)$, the unique group element carrying the second to the first is (tt'^{-1}, jj'^{-1}) , whose second coordinate has square 1. The square of the second coordinate in (HA11) is $\epsilon_R \epsilon_R^{-1} \epsilon_R = \epsilon_R$. The stated identities follow by cancellation in the group of units of R . In particular they are identities of the actual functor X , not only of its complex points. The coordinate morphism for (HA11) is $T \mapsto T_1 T_2^{-1} T_3$, $J \mapsto J_1 J_2^{-1} J_3$. $\square$

Here “torsor identities” means the proved coaction and canonical-map isomorphism (HA9)–(HA12); no unspecified Grothendieck topology or descent theorem is being used.

Proposition 45 (The precise effect of choosing an imaginary root). *After the explicit base extension $B \rightarrow D$, let j_0 denote the root in the base copy of D . There is an isomorphism of D -monoids*

$$D \otimes_B A_\times \cong D[T, T^{-1}, k]/(k^2 = 1), \quad k = Jj_0^{-1}. \quad (\text{HA13})$$

Transporting the usual Hopf structure on the right gives

$$\Delta_D(J) = j_0^{-1}(J \otimes J), \quad \eta_D(J) = j_0, \quad \mathcal{S}_D(J) = J, \quad \Delta_D(T) = T \otimes T. \quad (\text{HA14})$$

Replacing j_0 by ϵj_0 changes the displayed coproduct on J by the factor ϵ and changes its counit by the same factor.

Proof. One has $k^2 = J^2 j_0^{-2} = \epsilon \epsilon^{-1} = 1$. The inverse to (HA13) sends J to $j_0 k$. The Hopf formula $\Delta(k) = k \otimes k$ gives $\Delta(J) = j_0 k \otimes k = j_0^{-1} J \otimes J$; the counit gives $\eta(J) = j_0$. Since $k^{-1} = k$, the antipode fixes $J = j_0 k$. The last assertion follows from $\epsilon^{-1} = \epsilon$. This calculation also shows explicitly what root choice the missing B -counit would have had to make. $\square$

17 The quotient symmetry and the spatial action

On the unfolded complex overlap the source defines

$$\alpha(z, j) = (-1/z, -j), \quad j \in \{i, -i\}, \quad (\text{HA15})$$

by $\alpha^*(T) = \epsilon T^{-1}$, $\alpha^*(J) = \epsilon J$ ([1, original `CC.tex`, lines 773–806]). If $r_\lambda(z, j) = (\lambda z, j)$, then

$$\alpha r_\lambda = r_{\lambda^{-1}} \alpha. \quad (\text{HA16})$$

This is verified by evaluating both sides at (z, j) : both give $(-\lambda^{-1}/z, -j)$. Thus the spatial action and involution have an exact semidirect-product relation.

The same uniform map r_λ does not descend to the orbit quotient for arbitrary λ . Indeed two representatives (z, j) and $(-1/z, -j)$ are sent to $(\lambda z, j)$ and $(-\lambda/z, -j)$. The latter is in the orbit of the former, for all $z \neq 0, \infty$, exactly when $-\lambda/z = -1/(\lambda z)$, that is, $\lambda^2 = 1$. After choosing the representative component $j = i$, the action which *does* lift the usual quotient scaling is

$$\tilde{r}_\lambda(z, i) = (\lambda z, i), \quad \tilde{r}_\lambda(z, -i) = (\lambda^{-1} z, -i). \quad (\text{HA17})$$

Its commutation with α follows from (HA16), and its multiplication law follows component by component. This supplies the exact additional branch rule needed to transport scaling through the geometric fold. It does not replace the underlying spatial-core coaction by a diagonal on J .

18 Odd Frobenius maps on the projective line

For a positive odd integer n , define

$$\chi(n) = (-1)^{(n-1)/2}, \quad e(n) = \frac{1 - \chi(n)}{2}, \quad a(z) = -\frac{1}{z}, \quad p_n(z) = z^n, \quad F_n = a^{e(n)} p_n. \quad (\text{HA18})$$

Thus $F_n(z) = z^n$ for $n \equiv 1 \pmod{4}$ and $F_n(z) = -1/z^n$ for $n \equiv 3 \pmod{4}$, as in Connes–Consani’s Proposition `prop:twistor_frobenius` ([1, original `CC.tex`, lines 1159–1191]). In homogeneous coordinates $z = X/Y$, these are everywhere-defined maps

$$F_n[X : Y] = \begin{cases} [X^n : Y^n], & \chi(n) = 1, \\ [-Y^n : X^n], & \chi(n) = -1. \end{cases} \quad (\text{HA19})$$

The two coordinate polynomials never vanish simultaneously on $\mathbb{P}^1$, so each map has degree n .

Theorem 46 (Composition, real structure, and the exact equivariance). *For all positive odd m, n ,*

$$F_m F_n = F_{mn}, \quad F_1 = \text{id}. \quad (\text{HA20})$$

For $\sigma(z) = -1/\bar{z}$ one has

$$F_n \sigma = \sigma F_n. \quad (\text{HA21})$$

For the usual scaling action of $\mathbb{C}^\times$ the exact formula is

$$F_n(\lambda z) = \theta_n(\lambda) F_n(z), \quad \theta_n(\lambda) = \lambda^{n\chi(n)}. \quad (\text{HA22})$$

Proof. For odd n , $p_n a(z) = (-1/z)^n = -1/z^n = a p_n(z)$ on the nonzero affine coordinate, hence on $\mathbb{P}^1$ by (HA19). Also $a^2 = \text{id}$ and $p_m p_n = p_{mn}$. The identity $\chi(mn) = \chi(m)\chi(n)$ follows by checking the two invertible residue classes modulo four. Consequently $e(mn) \equiv e(m) + e(n) \pmod{2}$, proving (HA20). Complex conjugation c commutes with a and each p_n , which have real coefficients; $\sigma = ac$ gives (HA21). Finally (HA22) follows by substitution in the two cases of (HA18). The homogeneous formulas prove all these equalities at 0 and ∞ as well. $\square$

19 A complete computation of the ordinary K_0 action

Here $K_0(\mathbb{P}_{\mathbb{C}}^1)$ is the Grothendieck ring of algebraic vector bundles, with addition from exact sequences and multiplication from tensor product. Let $L = [\mathcal{O}(1)]$ and $\eta = L - 1$.

Lemma 47.

$$K_0(\mathbb{P}_{\mathbb{C}}^1) = \mathbb{Z}[\eta]/(\eta^2), \quad [\mathcal{O}(d)] = 1 + d\eta \quad (d \in \mathbb{Z}). \quad (\text{HA23})$$

Every vector bundle E has

$$[E] = r + q\eta, \quad r = \text{rank } E, \quad q = \chi(E) - r. \quad (\text{HA24})$$

Proof. The Euler sequence on $\mathbb{P}^1$ is $0 \rightarrow \mathcal{O}(-1) \rightarrow \mathcal{O}^{\oplus 2} \rightarrow \mathcal{O}(1) \rightarrow 0$; its maps in homogeneous coordinates are $(-Y, X)$ and (X, Y) . The composition is zero, and checking the charts $X \neq 0$ and $Y \neq 0$ shows that its kernel and cokernel have the asserted form. Hence $L + L^{-1} = 2$ and $(L - 1)^2 = 0$.

For completeness, generation of K_0 by $1, L^{-1}$ follows without assuming that every bundle has already been split. On $\mathbb{P}^1 \times \mathbb{P}^1$, with projections q_1, q_2 , the diagonal is cut out by $X_1 Y_2 - Y_1 X_2$, giving the exact sequence

$$0 \longrightarrow \mathcal{O}(-1, -1) \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}_{\Delta} \longrightarrow 0.$$

Tensor it by $q_1^* E$ and apply the alternating direct image under q_2 . The diagonal term is E . The other two terms have direct images $H^j(E) \otimes \mathcal{O}$ and $H^j(E(-1)) \otimes \mathcal{O}(-1)$ respectively. These identities can be checked by the two-chart Čech complex on the first factor, tensored with either indicated line bundle on the second; there are no terms in degrees greater than one. The alternating identity of the resulting long exact sequence gives

$$[E] = \chi(E)[\mathcal{O}] - \chi(E(-1))[\mathcal{O}(-1)]. \quad (\text{HA25})$$

Thus $1, L^{-1}$ generate K_0 , and so do $1, \eta$. The same two-chart Čech calculation gives $H^0(\mathcal{O}) = \mathbb{C}$, $H^1(\mathcal{O}) = 0$, and $H^0(\mathcal{O}(-1)) = H^1(\mathcal{O}(-1)) = 0$. Rank and Euler characteristic therefore map $[\mathcal{O}], [\mathcal{O}(-1)]$ to $(1, 1), (1, 0)$ in $\mathbb{Z}^2$. Their determinant is -1 , so there are no additional additive relations. This proves the ring presentation. For $d \geq 0$, $(1 + \eta)^d = 1 + d\eta$; for negative d use $(1 + \eta)^{-1} = 1 - \eta$. Finally rank sends $1, \eta$ to $1, 0$, while Euler characteristic sends them to $1, 1$ (the Euler sequence gives $\chi(\mathcal{O}(1)) = 2$). Formula (HA24) follows. $\square$

Theorem 48 (Exact ordinary Adams equality). *For every positive odd n ,*

$$F_n^*(1 + q\eta) = 1 + nq\eta, \quad F_n^*(r + q\eta) = r + nq\eta, \quad F_n^* = \psi^n \quad \text{on } K_0(\mathbb{P}_{\mathbb{C}}^1). \quad (\text{HA26})$$

Proof. The two homogeneous coordinates of (HA19) are sections of $\mathcal{O}(n)$ with no common zero. The universal tautological quotient defining the map therefore gives $F_n^*\mathcal{O}(1) \simeq \mathcal{O}(n)$. This can equally be checked by pulling back the transition ratio X/Y on the inverse images of the two standard charts, with their interchange in the second case of (HA19). Thus $F_n^*\eta = L^n - 1 = n\eta$, proving the two explicit formulas.

Define the Adams operations from the usual exterior-power operations by the formal identity

$$\frac{d}{dt} \log \lambda_t(x) = \sum_{n \geq 1} (-1)^{n-1} \psi^n(x) t^{n-1}, \quad \lambda_t(x+y) = \lambda_t(x) \lambda_t(y).$$

It shows additivity of each ψ^n . For a line-bundle class M , $\lambda_t(M) = 1 + Mt$, so $\psi^n(M) = M^n$. The additive generators $1, L^{-1}$ from (HA25) therefore have identical images under F_n^* and ψ^n , proving the equality on every class. In particular the holomorphic involution a acts trivially on this K_0 , because $a^*\mathcal{O}(1) \simeq \mathcal{O}(1)$. $\square$

This is the exact ordinary- K_0 statement justified by the maps. It proves a comparison on the receiving group in (HA23), rather than an identity of all additional structures which a bundle may carry.

20 What happens to equivariant weights and real structures

Let E be a $\mathbb{C}^\times$ -equivariant bundle for the usual scaling action on $\mathbb{P}^1$. Pullback by F_n acquires a specified equivariant structure using the group homomorphism θ_n in (HA22):

$$\lambda \cdot (z, v) = (\lambda z, \theta_n(\lambda) \cdot v), \quad v \in E_{F_n(z)}. \quad (\text{HA27})$$

Indeed the right-hand side belongs to $E_{F_n(\lambda z)}$ by (HA22), and associativity follows from multiplicativity of θ_n . If the character weights at 0 and ∞ are the multisets W_0, W_∞ , the exact transformed multisets are

$$(W'_0, W'_\infty) = \begin{cases} (nW_0, nW_\infty), & n \equiv 1 \pmod{4}, \\ (-nW_\infty, -nW_0), & n \equiv 3 \pmod{4}. \end{cases} \quad (\text{HA28})$$

The fibre identification is $(F_n^*E)_z = E_{F_n(z)}$; F_n fixes the poles in the first case and interchanges them in the second. A target character λ^m becomes $(\lambda^{n\chi(n)})^m$, which proves (HA28).

There is an explicit test showing that (HA26) cannot simply be promoted to the same assertion about this equivariant pullback. Give the trivial line bundle the fibre action of character λ . Under (HA27) its character becomes $\lambda^{n\chi(n)}$. Its equivariant Adams image has character λ^n . For $n \equiv 3 \pmod{4}$ these are different: restricting to the fixed point 0 detects the distinct Laurent monomials t^{-n} and t^n in the representation ring $R(\mathbb{C}^\times) = \mathbb{Z}[t, t^{-1}]$. Both forget to the identical class 1 in the ordinary K_0 used in (HA26).

Suppose now that a bundle is supplied with an actual antilinear lift $\tau : E_z \rightarrow E_{\sigma(z)}$ of the antipodal real structure. The pullback has the exact antilinear lift

$$\tau_n(z, v) = (\sigma(z), \tau v). \quad (\text{HA29})$$

Its target fibre is correct because $\sigma F_n(z) = F_n \sigma(z)$ by (HA21). If the supplied lift squares to $+\text{id}$ or $-\text{id}$, its pullback squares to the same operator. This is a construction from an existing lift; it does not conjure one from the class of the underlying vector bundle. For example an actual isomorphism $E \simeq \mathcal{O}(w)^{\oplus r}$ pulls back to $\mathcal{O}(nw)^{\oplus r}$, while the rank and degree calculation

alone does not specify a filtration, a polarization, a connection, or the equivariant fibre weights. Equations (HA27)–(HA29) specify precisely the further structures transported here.

The source’s discussion of real Hodge structures and Adams operations appears in [1, original CC.tex, lines 416–417] and in its section *Frobenius and Twistor Equivariance*. The calculation here supplies the exact ordinary- K_0 map, the character correction for the second residue class, and the transport of any supplied real lift. No equality with a programme cohomology group, no positivity theorem for a trace pairing, and no Riemann-hypothesis conclusion is asserted by these identities.

References

- [1] Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* , arXiv:2609.00299v1 (2026), original author LaTeX source CC.tex. Versioned paper identity; original source archive. The exact author-source line locations used above refer to the SHA-256 edition identified at the beginning of this derivation.

21 The odd twistor dynamics on the original annulus object

We apply the concrete dynamics in Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* , Proposition prop:twistor_frobenius, to the exact companion and annulus space of 033, SBT1–23. This section proves the receiving maps and their complete metric defects. Neither a spherical-algebra point functor nor an arithmetic cohomology comparison is silently substituted for these maps.

21.1 The actual circle map and its linear pullback

For every positive odd integer d put

$$\chi_d = (-1)^{(d-1)/2}, \quad m_d = \chi_d d, \quad f_d(z) = \chi_d z^{m_d} \quad (z \in \mathbb{C}^\times). \quad (\text{TOB1})$$

At 0 and ∞ use the unique continuous extension to $\mathbb{P}^1(\mathbb{C})$. Thus $f_d = z^d$ when $d \equiv 1 \pmod{4}$, and $f_d = -z^{-d}$ when $d \equiv 3 \pmod{4}$, exactly as in the source. For odd c, d , the equality $\chi_{cd} = \chi_c \chi_d$ and the oddness of m_c give

$$f_c(f_d(z)) = \chi_c \chi_d^{m_c} z^{m_c m_d} = \chi_{cd} z^{m_{cd}} = f_{cd}(z).$$

The formula extends at both poles. The real structure $\sigma(z) = -1/\bar{z}$ satisfies $f_d \sigma = \sigma f_d$: substituting either sign of χ_d proves this equality on $\mathbb{C}^\times$, and continuity proves it at the poles.

Give S^1 its probability Haar measure $dt/(2\pi)$, and write $e_j(z) = z^j$, $j \in \mathbb{Z}$. Define the pullback

$$V_d : L^2(S^1) \longrightarrow L^2(S^1), \quad (V_d \varphi)(z) = \varphi(f_d(z)), \quad V_d e_j = \chi_d^j e_{m_d j}. \quad (\text{TOB2})$$

It is an isometry: substitution in the integral of a trigonometric polynomial, or orthogonality of the displayed distinct basis vectors, proves the norm equality on their span; that span is dense, so the operator extends uniquely to all of L^2 . Pullback by the measurable degree- d circle covering is the same extension, since that covering preserves Haar measure. The adjoint and range projection are exactly

$$V_d^* e_j = \begin{cases} \chi_d^{j/m_d} e_{j/m_d}, & m_d \mid j, \\ 0, & m_d \nmid j, \end{cases} \quad V_d V_d^* e_j = \mathbf{1}_{d \mid j} e_j. \quad (\text{TOB3})$$

For $S e_j = e_{j+1}$, direct substitution yields

$$V_d S = \chi_d S^{m_d} V_d, \quad V_c V_d = V_{cd}. \quad (\text{TOB4})$$

These identities include all signs for the 3 (mod 4) case.

We use the same coefficient operator on E -valued distributions $E^{\mathbb{Z}}$, paired with finitely supported covectors:

$$(V_dx)_j = \begin{cases} \chi_d^{j/m_d} x_{j/m_d}, & m_d \mid j, \\ 0, & m_d \nmid j. \end{cases} \quad (\text{TOB5})$$

Every pairing is a finite sum; no convergence hypothesis is needed. Equation (TOB4) continues to hold by its coordinate calculation. On $\ell^2(\mathbb{Z}; E, G_N)$ the same map is an isometry.

21.2 An exact finite receiver with the original metric

Keep the original parameters and operators:

$$\begin{aligned} k \geq 17, \quad k \equiv 1 \pmod{4}, \quad q = (k+1)^2, \quad \delta, \gamma > 0, \quad q-1 \leq N \leq 2q, \\ \lambda_{ab} = (2b-k)\gamma - i(2a-k)\delta, \quad Q_k(y) = \prod_{a,b=0}^k (y - \lambda_{ab}), \end{aligned} \quad (\text{TOB6})$$

$$E = \mathbb{C}[y]/(Q_k), \quad M[v] = [yv], \quad F_p = e^{i(\log p)M}, \quad \tilde{F}_p = p^{k/2} F_p, \quad \langle v, w \rangle_N = v^* G_N w.$$

The positive matrix G_N is the full original metric at the stated source cutoff. The four cutoffs $q-1, q, 2q-1, 2q$ are included separately. The new integer d is neither a source cutoff nor the amplification parameter k . Let P_{ab} and ℓ_{ab} be the original Lagrange projectors and vectors. Thus

$$F_p \ell_{ab} = \omega_{ab} \ell_{ab}, \quad \omega_{ab} = p^{(2a-k)\delta} e^{i(2b-k)\gamma \log p}, \quad (\mathcal{J}_p v)_j = F_p^{-j} v. \quad (\text{TOB7})$$

The original embedding satisfies $S\mathcal{J}_p = \mathcal{J}_p F_p$ and $\text{ev}_0 \mathcal{J}_p = I$ by substitution.

Define

$$A_d = \chi_d F_p^{\chi_d}, \quad E_d = E^{\oplus d}, \quad G_{N,d} = \text{diag}(G_N, \dots, G_N), \quad \mathcal{W}_d = V_d \mathcal{J}_p. \quad (\text{TOB8})$$

For every integer ℓ ,

$$(\mathcal{W}_d v)_{d\ell} = A_d^{-\ell} v, \quad (\mathcal{W}_d v)_j = 0 \quad (d \nmid j), \quad S^d \mathcal{W}_d = \mathcal{W}_d A_d. \quad (\text{TOB9})$$

For $\chi_d = 1$ this is (TOB5) directly. For $\chi_d = -1$, $j/m_d = -\ell$, so the coefficient is $(-1)^{-\ell} F_p^{\ell} v = (-F_p^{-1})^{-\ell} v$. This proves both cases without changing any original coordinate.

Let the receiving companion and embedding be

$$\begin{aligned} C_d(v_0, \dots, v_{d-1}) &= (A_d v_{d-1}, v_0, \dots, v_{d-2}), \\ \mathcal{I}_d(v) &= \sum_{r=0}^{d-1} S^r \mathcal{W}_d v_r, \quad (\mathcal{I}_d v)_{d\ell+r} = A_d^{-\ell} v_r \quad (0 \leq r < d). \end{aligned} \quad (\text{TOB10})$$

Then

$$\begin{aligned} \text{ev}_{[0,d-1]} \mathcal{I}_d &= I_{E_d}, \quad S\mathcal{I}_d = \mathcal{I}_d C_d, \quad C_d^d = \text{diag}(A_d, \dots, A_d), \\ \text{im } \mathcal{I}_d &= \{x \in E^{\mathbb{Z}} : x_{j-d} = A_d x_j \text{ for all } j\}. \end{aligned} \quad (\text{TOB11})$$

The first identity follows at the d indicated indices. Shifting the sum in (TOB10), and using its last term (TOB9), proves the second. Following each block through d steps proves the third. In the displayed recurrence, division with remainder $j = d\ell + r$ forces $x_j = A_d^{-\ell} x_r$ for positive and negative ℓ , proving the last equality and its inverse. The image is also the smallest S - and S^{-1} -stable linear subspace containing $\mathcal{W}_d(E)$: its d shifts generate the image, and (TOB9), with invertible A_d , proves stability in both directions. In particular its dimension is exactly dq , with all original q labels retained.

The original metric is transported by the exact first-block evaluation:

$$\langle \mathcal{I}_d v, \mathcal{I}_d w \rangle_{d,N} = \sum_{r=0}^{d-1} v_r^* G_N w_r. \quad (\text{TOB12})$$

This is not the infinite sum of annulus norms. Each original copy has precisely its original norm; no averaging by d is performed. Direct subtraction gives the full defect

$$\begin{aligned} \|C_d v\|_{N,d}^2 - \|v\|_{N,d}^2 &= v_{d-1}^* (A_d^* G_N A_d - G_N) v_{d-1}, \\ \|p^{k/2} C_d v\|_{N,d}^2 - p^k \|v\|_{N,d}^2 &= p^k v_{d-1}^* (A_d^* G_N A_d - G_N) v_{d-1}. \end{aligned} \quad (\text{TOB13})$$

The cancellation concerns the unchanged first $d-1$ summands only; the last block is the complete remaining quadratic form. Equivalently, the singular values of C_d in $G_{N,d}$ consist of $(d-1)q$ copies of 1 and the q singular values of A_d in G_N . Indeed C_d is a unitary block permutation after the block diagonal operator $\text{diag}(I, \dots, I, A_d)$. This proves

$$\|C_d\|_{N,d} = \max\{1, \|A_d\|_N\}, \quad \|C_d^{-1}\|_{N,d} = \max\{1, \|A_d^{-1}\|_N\} \quad (d > 1). \quad (\text{TOB14})$$

For $d=1$, the corresponding norms are just $\|F_p\|_N$ and $\|F_p^{-1}\|_N$. Since $\delta > 0$, these latter norms already exceed 1, by the original eigenvectors of largest and smallest modulus, respectively.

For the last-block vector $v = (0, \dots, 0, \ell_{ab})$, the exact right side in (TOB13) is

$$p^k (p^{2\chi_d(2a-k)\delta} - 1) \|\ell_{ab}\|_N^2. \quad (\text{TOB15})$$

At every odd d , both signs occur as a varies. For either fixed residue class modulo four this expression is independent of d . Thus the original metric defect has not vanished in a growing covering degree.

21.3 All roots, labels, phases, and finite boundaries

Put $b_d = (1 - \chi_d)/2$. The eigenvectors and eigenvalues of C_d are explicitly

$$\begin{aligned} \zeta_{ab,t} &= \exp\left(\frac{\chi_d i(\log p) \lambda_{ab} + i\pi b_d + 2\pi i t}{d}\right), \quad 0 \leq t < d, \\ v_{ab,t} &= (\ell_{ab}, \zeta_{ab,t}^{-1} \ell_{ab}, \dots, \zeta_{ab,t}^{-(d-1)} \ell_{ab}), \\ C_d v_{ab,t} &= \zeta_{ab,t} v_{ab,t}, \quad |\zeta_{ab,t}| = p^{\chi_d(2a-k)\delta/d}. \end{aligned} \quad (\text{TOB16})$$

The first formula specifies the phase explicitly, without a choice of a new logarithm. Its d th power is the eigenvalue $\chi_d \omega_{ab}^{\chi_d}$ of A_d . This proves the second line by substitution in (TOB10). For each fixed (a, b) the d roots are distinct and the corresponding Vandermonde matrix is invertible. The original P_{ab} split E_d into their d -block spaces, so these dq vectors form a basis even if roots from different original labels have equal phases or eigenvalues. Their full Gram entries are

$$\langle v_{ab,t}, v_{ce,u} \rangle_{N,d} = \langle \ell_{ab}, \ell_{ce} \rangle_N \sum_{r=0}^{d-1} (\overline{\zeta_{ab,t}} \zeta_{ce,u})^{-r}. \quad (\text{TOB17})$$

All original off-diagonal metric entries remain in this formula.

The original annulus functions and operator are

$$h_j(\mathbf{x}) = p^{kj/2} (1 - p^{-1})^{-k/2} \mathbf{1}_{(p^j \mathbb{Z}_p^\times)^k}(\mathbf{x}), \quad \mathcal{T}_{p,k} = p^{k/2} \mathcal{U}_p^{\otimes k}, \quad \mathcal{U}_p F(x) = \sqrt{p} F(x/p). \quad (\text{TOB18})$$

Here Haar measure gives $\mu(\mathbb{Z}_p) = 1$. The support in h_j has measure $p^{-kj} (1 - p^{-1})^k$; these functions are orthonormal, and substitution proves $\mathcal{T}_{p,k} h_j = p^{k/2} h_{j+1}$. For an integer $H \geq 0$ define

$$I_{d,H} v = \sum_{j=-H}^H h_j \otimes (\mathcal{I}_d v)_j. \quad (\text{TOB19})$$

Its squared norm is exactly $\sum_{j=-H}^H \|A_d^{-[j/d]} v_{j \bmod d}\|_N^2$, where the remainder lies in $\{0, \dots, d-1\}$ also for negative j . The complete boundary is

$$\begin{aligned} B_{d,H}v &= (\mathcal{I}_{p,k} \otimes I)I_{d,H}v - I_{d,H}(p^{k/2}C_d v) \\ &= p^{k/2}(h_{H+1} \otimes (\mathcal{I}_d v)_H - h_{-H} \otimes (\mathcal{I}_d v)_{-H-1}), \\ \|B_{d,H}v\|^2 &= p^k(\|(\mathcal{I}_d v)_H\|_N^2 + \|(\mathcal{I}_d v)_{-H-1}\|_N^2). \end{aligned} \quad (\text{TOB20})$$

The shift relation (TOB11) cancels every common summation index; the two remaining annuli are disjoint, proving the norm formula. For $d > 2$, this boundary map need not be injective: it observes only the two indicated residue blocks. Its kernel is exactly the subspace whose entries at residues $H \bmod d$ and $(-H-1) \bmod d$ vanish. This retains a new boundary kernel that cannot be removed by citing the injective $d = 1$ boundary.

For the eigenvector $v_{ab,t}$, (TOB10) gives the full sequence $(\mathcal{I}_d v_{ab,t})_j = \zeta_{ab,t}^{-j} \ell_{ab}$. With $r = |\zeta_{ab,t}|$, equation (TOB20) becomes

$$\frac{\|B_{d,H}v_{ab,t}\|^2}{\|I_{d,H}v_{ab,t}\|^2} = p^k \frac{r^{-2H} + r^{2H+2}}{\sum_{j=-H}^H r^{-2j}} \rightarrow p^k |r^2 - 1| = p^k \left| p^{2\chi_d(2a-k)\delta/d} - 1 \right|. \quad (\text{TOB21})$$

For $r > 1$ divide numerator and denominator by r^{2H} ; for $r < 1$ divide by r^{-2H} . The resulting geometric series prove the displayed limits. They are positive for every finite d , since k is odd. As d tends to infinity they tend to zero, but the exact d -step return (TOB11) and original defect (TOB15) remain. These statements concern different, explicitly related operators; the root estimate has not been substituted for its unchanged return.

For a nonzero $v \in E_d$ define

$$R_{p,d}(v) = \frac{\delta \log p}{d} \max\{|2a-k| : P_{ab}v_r \neq 0 \text{ for some } r\}. \quad (\text{TOB22})$$

Then

$$\lim_{H \rightarrow \infty} \frac{1}{H} \log \max_{|j| \leq H} \|(\mathcal{I}_d v)_j\|_N = R_{p,d}(v) > 0. \quad (\text{TOB23})$$

For the upper bound expand each of the d blocks with the original P_{ab} and use $\lfloor |j/d| \rfloor \leq H/d + 1$. The finite sum is bounded by a constant times $\exp(HR_{p,d})$. For the lower bound choose a nonzero projected block attaining the maximum. Choose $j = d\ell + r$ within $[-H, H]$ with the sign of ℓ that makes $A_d^{-\ell}$ grow on that block and with $|\ell| \geq H/d - 2$. Projection bounds its norm from below by the growing projected norm divided by $\|P_{ab}\|_N$. The constants are independent of H and their logarithms disappear after division by H . This proves (TOB23), retaining the original metric. Consequently the image contains no nonzero polynomial-growth sequence for any fixed finite d .

21.4 Exact observation-kernel and support transport

Retain the original isometric section $J : B \rightarrow E$, $P_B = JJ^\dagger$, $Q = I - P_B$, and $K = \ker \Lambda = QE$. Let $J_d = \text{diag}(J, \dots, J)$, and define $\widehat{C}_d = J_d^\dagger C_d J_d$. It is (TOB10) with the last corner A_d replaced by $J^\dagger A_d J$. Therefore

$$C_d J_d - J_d \widehat{C}_d : (b_0, \dots, b_{d-1}) \mapsto (Q A_d J b_{d-1}, 0, \dots, 0). \quad (\text{TOB24})$$

This follows by multiplying each block, using $J^\dagger J = I$. Applying the injective $\mathcal{I}_d$ gives the full corresponding defect for the distribution comparison. More explicitly,

$$p^{k/2} S \mathcal{I}_d J_d - \mathcal{I}_d J_d (p^{k/2} \widehat{C}_d) = p^{k/2} \mathcal{I}_d (Q A_d J \text{pr}_{d-1}, 0, \dots, 0). \quad (\text{TOB25})$$

The finite-cutoff version is the sum of this compressed-kernel term and the two annulus terms (TOB20); thus neither contribution is lost by compression.

For the full bounded distributive lattice L , every vector space V here has its synchronized carrier $V^\uparrow = \{(0, \lambda) : \lambda \in L\} \cup \{(v, 1_L) : v \in V\}$. A linear map T lifts as $(v, \lambda) \mapsto (Tv, \lambda)$. Substitution in the addition and scalar multiplication laws proves linearity over $G_L(\mathbb{C})$. Applying this to (TOB10)–(TOB11) proves that the first-block evaluation recovers every $(0, \lambda)$ and every nonzero original vector. In (TOB24)–(TOB25), zero amplitude in the exact defect is still a labelled zero. All these identities therefore have exact full- L lifts, including the observation-kernel inclusion and the growing finite receiver; no free-semimodule basis change is assumed.

21.5 The complete boundary window restores the original exponent

A single boundary in (TOB20) reads at most two residue blocks. The exact joint map through a complete window reads every block. For an integer $h \geq 0$ put

$$\mathcal{B}_{d,h}v = (B_{d,dh}v, B_{d,dh+1}v, \dots, B_{d,dh+d-1}v). \quad (\text{TOB26})$$

Give its target the direct-sum Hilbert norm. For $H = dh + r$, equation (TOB10) gives $x_H = A_d^{-h}v_r$ and $x_{-H-1} = A_d^{h+1}v_{d-1-r}$. Summing the exact two-annulus identity proves

$$\|\mathcal{B}_{d,h}v\|^2 = p^k \sum_{r=0}^{d-1} (\|A_d^{-h}v_r\|_N^2 + \|A_d^{h+1}v_r\|_N^2). \quad (\text{TOB27})$$

In particular this joint map is injective for every h : every summand is nonnegative, and invertibility of A_d implies that a zero sum forces each $v_r = 0$.

For comparison with exactly the same complete blocks, define

$$L_{d,h}v = \sum_{\ell=-h}^h \sum_{r=0}^{d-1} h_{d\ell+r} \otimes A_d^{-\ell}v_r, \quad \|L_{d,h}v\|^2 = \sum_{\ell=-h}^h \sum_{r=0}^{d-1} \|A_d^{-\ell}v_r\|_N^2. \quad (\text{TOB28})$$

This is the explicitly specified annulus interval $[-dh, dh + d - 1]$, not a redefinition of the symmetric truncation (TOB19). Its norm formula follows from orthonormality of all the different annuli in (TOB18). For any nonzero v with each block in the same original one-dimensional subspace $\mathbb{C}\ell_{ab}$, put $R = |\omega_{ab}|^{\chi_d} = p^{\chi_d(2a-k)\delta}$. Equations (TOB27)–(TOB28) yield

$$\frac{\|\mathcal{B}_{d,h}v\|^2}{\|L_{d,h}v\|^2} = p^k \frac{R^{-2h} + R^{2h+2}}{\sum_{\ell=-h}^h R^{-2\ell}} \longrightarrow p^k |R^2 - 1| = p^k \left| p^{2\chi_d(2a-k)\delta} - 1 \right| > 0. \quad (\text{TOB29})$$

The common factor $\sum_r \|v_r\|_N^2$ cancels from this ratio; no metric was changed. The geometric-series proof after (TOB21) applies to R . The formula includes every $v_{ab,t}$ of (TOB16). It is independent of the covering degree within each residue class modulo four. Therefore the full boundary window, with all residue classes restored, retains the original nonzero radial defect exactly, while the one-step eigenvalue radius in (TOB16) has its explicitly proved $1/d$ exponent. Equations (TOB10), (TOB26) and (TOB28) are the connecting maps between these two quantities.

22 Exact prime traces and determinants of the odd twistor receiver

The input dynamics is due to Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* , Proposition `prop:twistor_frobenius`; its original author file `CC.tex`, lines 1159–1191, gives the two odd residue classes and their compatibility with the real structure. The source’s spatial coproduct restriction is at lines 462–465. The maps used here are the explicit receiving maps (TOB1)–(TOB11), constructed from the original companion in 033, SBT1–SBT23. The calculation concerns one fixed prime p . No product over primes is introduced.

22.1 The complete cyclic receiver and its labelled traces

Retain $k, \delta, \gamma, q, N, E, G_N$ and all original Lagrange labels (a, b) from (TOB6)–(TOB7). In particular

$$\omega_{ab} = p^{(2a-k)\delta} e^{i(2b-k)\gamma \log p}, \quad 0 \leq a, b \leq k, \quad q = (k+1)^2. \quad (\text{TPT1})$$

For a positive odd integer d , put $\chi = \chi_d = (-1)^{(d-1)/2}$, $A = \chi F_p^\chi$, and

$$C = C_d : E^{\oplus d} \longrightarrow E^{\oplus d}, \quad C(v_0, \dots, v_{d-1}) = (Av_{d-1}, v_0, \dots, v_{d-2}), \quad \tilde{C} = p^{k/2} C. \quad (\text{TPT2})$$

The metric is the unchanged direct sum of d copies of G_N . Trace and determinant below are the ordinary trace and determinant of these actual finite-dimensional complex-linear operators; their definitions do not replace this metric or its defects. Let $P_{ab}^{(d)} = \text{diag}(P_{ab}, \dots, P_{ab})$.

For every integer n , including negative integers, the exact formula is

$$\text{tr}(C^n) = \begin{cases} 0, & d \nmid n, \\ d\chi^m \sum_{a,b=0}^k \omega_{ab}^{\chi^m}, & n = dm. \end{cases} \quad (\text{TPT3})$$

For $n = 0$ the second line means dq . The stronger labelled identity is

$$\text{tr}(P_{ab}^{(d)} C^n) = \begin{cases} 0, & d \nmid n, \\ d\chi^m \omega_{ab}^{\chi^m}, & n = dm. \end{cases} \quad \text{tr}(P_{ab}^{(d)} \tilde{C}^n) = p^{kn/2} \text{tr}(P_{ab}^{(d)} C^n). \quad (\text{TPT4})$$

To prove these statements, let $\iota_r : E \rightarrow E^{\oplus d}$ be the inclusion into block r and let π_r be its projection. Iteration of (TPT2) sends every block to the block whose index is increased by n modulo d . If $d \nmid n$, no diagonal block is present, so the trace is zero. If $n = dm$, the exact identity $C^d = \text{diag}(A, \dots, A)$ gives A^m in every diagonal block. This also holds for negative m , because A and C are invertible. Finally $P_{ab} A^m = \chi^m \omega_{ab}^{\chi^m} P_{ab}$ and $\text{tr} P_{ab} = 1$. This proves (TPT3)–(TPT4) without an orthogonality assumption on the Lagrange basis.

For later comparison define the original, fully labelled power sum

$$t_p(m) = \text{tr}(F_p^m) = \sum_{a=0}^k \sum_{b=0}^k p^{m(2a-k)\delta} e^{im(2b-k)\gamma \log p} \quad (m \in \mathbb{Z}). \quad (\text{TPT5})$$

Thus the surviving trace at time $n = dm$ samples original exponent $\chi m = \chi n/d$, not exponent n . Its multiplicity is exactly d . For the weighted receiver the same trace is

$$\text{tr}(\tilde{C}^{dm}) = d\chi^m p^{kdm/2} t_p(\chi m) = d\chi^m p^{k(d-\chi)m/2} \text{tr}(\tilde{F}_p^{\chi m}), \quad \tilde{F}_p = p^{k/2} F_p. \quad (\text{TPT6})$$

All signs, original phases, and both powers of p are retained.

22.2 Characteristic polynomial and the two local determinants

The characteristic polynomials and local determinant polynomials are

$$\det(zI - C) = \det(z^d I - A) = \prod_{a,b=0}^k (z^d - \chi \omega_{ab}^\chi), \quad (\text{TPT7})$$

$$\det(zI - \tilde{C}) = \prod_{a,b=0}^k (z^d - \chi p^{kd/2} \omega_{ab}^\chi), \quad (\text{TPT8})$$

$$D_{p,d}(u) := \det(I - uC) = \prod_{a,b=0}^k (1 - \chi \omega_{ab}^\chi u^d), \quad (\text{TPT9})$$

$$\tilde{D}_{p,d}(u) := \det(I - u\tilde{C}) = \prod_{a,b=0}^k (1 - \chi p^{kd/2} \omega_{ab}^\chi u^d). \quad (\text{TPT10})$$

Indeed $P_{ab}^{(d)} E^{\oplus d}$ is an invariant d -dimensional space, and C on it is the cyclic companion with scalar corner $\chi \omega_{ab}^\chi$. Its eigenvalues are all d roots of that nonzero scalar, with the labelled eigenvectors of (TOB16). The roots within this block are distinct. The original projectors decompose the full space as the direct sum of these invariant blocks, so the products above include the correct multiplicities even when roots from different original labels coincide. Multiplication by $p^{k/2}$ gives (TPT8); putting the eigenvalues in $\det(I - uC)$ gives (TPT9)–(TPT10).

Write, with the indicated original operator retained,

$$D_p^\pm(v) = \det(I - vF_p^{\pm 1}), \quad \tilde{D}_p^\pm(v) = \det(I - v\tilde{F}_p^{\pm 1}). \quad (\text{TPT11})$$

Here D_p^+ is the original $\det(I - vF_p)$, and D_p^- is its inverse-operator comparison. Equations (TPT9)–(TPT10) give the exact substitutions

$$\begin{aligned} D_{p,a}(u) &= D_p^\chi(\chi u^d), \\ \tilde{D}_{p,a}(u) &= D_p^\chi(\chi p^{kd/2} u^d) = \tilde{D}_p^\chi(\chi p^{k(d-\chi)/2} u^d). \end{aligned} \quad (\text{TPT12})$$

For $d \equiv 1 \pmod{4}$, the argument of $\tilde{D}_p^+$ is $p^{k(d-1)/2} u^d$. For $d \equiv 3 \pmod{4}$, the argument of $\tilde{D}_p^-$ is $-p^{k(d+1)/2} u^d$.

There is also an exact relation between the two original labelled spectra. Define the involution $R : E \rightarrow E$ on the original basis by

$$R\ell_{ab} = \ell_{k-a, k-b}, \quad R^2 = I, \quad \omega_{k-a, k-b} = \omega_{ab}^{-1}, \quad RF_p R = F_p^{-1}. \quad (\text{TPT13})$$

The scalar identity follows directly from (TPT1). Applying both operators in the last identity to each ℓ_{ab} proves it. Consequently $D_p^- = D_p^+$ and $t_p(-m) = t_p(m)$ as scalar invariants. These equalities are mediated by the explicit label permutation R ; they do not identify an original labelled eigenline with its reflected one, and they do not assert that R is unitary for G_N . The weighted counterpart is

$$R\tilde{F}_p R = p^k \tilde{F}_p^{-1}, \quad \tilde{D}_p^-(v) = \tilde{D}_p^+(p^{-k}v). \quad (\text{TPT14})$$

This follows by multiplying (TPT13) by $p^{k/2}$; the displayed p^k must not be dropped.

For completeness, the product of all original eigenvalues is 1: the sums of $2a - k$ and $2b - k$ over their full ranges are zero in (TPT1). Since $q = (k+1)^2$ is even and d is odd,

$$\det F_p = 1, \quad \det C = \det A = 1, \quad \det \tilde{C} = p^{kdq/2}, \quad D_p^+(v) = v^q D_p^+(v^{-1}) \quad (v \neq 0). \quad (\text{TPT15})$$

The determinant of a d -block cyclic companion is $(-1)^{(d-1)q} \det A$, which here equals $\det A$. For the final equality, multiply $\prod_{a,b} (1 - v\omega_{ab})$ by writing each factor as $-v\omega_{ab}(1 - v^{-1}\omega_{ab}^{-1})$, and use (TPT13).

22.3 Formal logarithmic series and their precise convergence

In the formal power-series ring $\mathbb{C}[[u]]$, the logarithm of a series with constant coefficient 1 is defined by its zero constant coefficient and the usual formal derivative. Thus no analytic branch

is needed for the following identities:

$$-\log D_{p,d}(u) = \sum_{m \geq 1} \frac{\chi^m t_p(\chi m)}{m} u^{dm}, \quad (\text{TPT16})$$

$$-\log \tilde{D}_{p,d}(u) = \sum_{m \geq 1} \frac{\chi^m p^{kdm/2} t_p(\chi m)}{m} u^{dm}, \quad (\text{TPT17})$$

$$-u \frac{D'_{p,d}(u)}{D_{p,d}(u)} = d \sum_{m \geq 1} \chi^m t_p(\chi m) u^{dm} = \sum_{a,b=0}^k \frac{d \chi \omega_{ab}^\chi u^d}{1 - \chi \omega_{ab}^\chi u^d}, \quad (\text{TPT18})$$

$$-u \frac{\tilde{D}'_{p,d}(u)}{\tilde{D}_{p,d}(u)} = d \sum_{m \geq 1} \chi^m p^{kdm/2} t_p(\chi m) u^{dm} = \sum_{a,b=0}^k \frac{d \chi p^{kd/2} \omega_{ab}^\chi u^d}{1 - \chi p^{kd/2} \omega_{ab}^\chi u^d}. \quad (\text{TPT19})$$

To prove them, expand each factor in (TPT9)–(TPT10) with $-\log(1-x) = \sum_{m \geq 1} x^m/m$, a valid formal identity, and then differentiate term by term. Equivalently, for either receiver $B = C$ or $B = \tilde{C}$,

$$\begin{aligned} -\log \det(I - uB) &= \sum_{n \geq 1} \frac{\text{tr}(B^n)}{n} u^n, \\ -\frac{d}{du} \log \det(I - uB) &= \text{tr}(B(I - uB)^{-1}) = \sum_{n \geq 1} \text{tr}(B^n) u^{n-1}. \end{aligned} \quad (\text{TPT20})$$

Substitution of (TPT3) or (TPT6) into (TPT20) retains both the d -fold trace multiplicity and the factor $1/(dm)$; their product is the exact coefficient $1/m$ in (TPT16)–(TPT17).

The exact radii of these Taylor series about zero are

$$\mathcal{R}_C = p^{-k\delta/d}, \quad \mathcal{R}_{\tilde{C}} = p^{-k/2-k\delta/d}. \quad (\text{TPT21})$$

Both logarithmic series and logarithmic-derivative series converge absolutely on the corresponding open disk. On its boundary the logarithmic series converges conditionally precisely at the points which are not zeros of the determinant; at its zeros it diverges. The logarithmic-derivative Taylor series diverges at every point of that boundary. Neither series converges outside the closed disk.

Here is a proof including boundary behaviour. The eigenvalues of C are those of (TOB16), so their largest modulus is $p^{k\delta/d}$; the labels $a = 0, k$ ensure that this maximum is attained. Multiplication by $p^{k/2}$ gives the second radius. For a fixed receiver B , list its eigenvalues with algebraic multiplicity as β_j . For $|u| < \rho(B)^{-1}$ every $|u\beta_j| < 1$, giving absolute convergence by a finite sum of scalar geometric or logarithmic series. On $|u| = \rho(B)^{-1}$, the eigenvalues of smaller modulus still give absolutely convergent series. Each eigenvalue of maximal modulus gives $\sum_{n \geq 1} (u\beta_j)^n/n$. If $u\beta_j \neq 1$, its partial sums are bounded geometric sums and summation by parts proves convergence. If $u\beta_j = 1$, it contributes the harmonic series with positive multiplicity; all other terms converge, so it cannot be cancelled. This proves the claimed convergence and divergence for the logarithm.

To show that the convergence at the other boundary points is not absolute, group equal maximal phases into distinct unit complex numbers $\alpha_1, \dots, \alpha_s$ with positive integer multiplicities $c_1, \dots, c_s$. Put $b_n = \sum_j c_j \alpha_j^n$. Finite geometric summation gives

$$\lim_{M \rightarrow \infty} \frac{1}{M} \sum_{n=1}^M |b_n|^2 = \sum_{j=1}^s c_j^2 > 0.$$

Since $|b_n| \leq \sum_j c_j$, the average of $|b_n|$ has a positive lower bound. The smaller-modulus contributions tend to zero exponentially and do not change that bound. Summation by parts then shows that the series of absolute values divided by n diverges. The same positive mean square

shows that the terms of the logarithmic-derivative series fail to tend to zero on the boundary. If $|u| > \rho(B)^{-1}$, factor the dominant growth $(|u|\rho(B))^n$ out of its coefficients. The same finite-phase mean-square argument provides a subsequence bounded away from zero; exponentially smaller spectral terms cannot cancel it. Along this subsequence even division by n does not make the terms tend to zero. This proves the asserted exterior divergence. Finally every nearest determinant zero gives a genuine pole of $-D'/D$ whose residue is minus its positive multiplicity. Thus (TPT21) is also the exact analytic Taylor radius, without a possible pole cancellation.

22.4 The exact prime variable and reverse recovery

Only for this fixed prime define

$$Z_{p,d}(s) = \tilde{D}_{p,d}(p^{-s})^{-1}, \quad p^{-s} = e^{-s \log p}. \quad (\text{TPT22})$$

It is a meromorphic function of s . In the precise open half-plane $\text{Re } s > k/2 + k\delta/d$, its branch with the indicated convergent series satisfies

$$\begin{aligned} \log Z_{p,d}(s) &= \sum_{m \geq 1} \frac{\chi^m t_p(\chi m)}{m} p^{-dm(s-k/2)}, \\ -\frac{Z'_{p,d}(s)}{Z_{p,d}(s)} &= d \log p \sum_{m \geq 1} \chi^m t_p(\chi m) p^{-dm(s-k/2)}. \end{aligned} \quad (\text{TPT23})$$

The second identity follows by the chain rule from (TPT19), including the sign $d(p^{-s})/ds = -(\log p)p^{-s}$. The original prime powers which appear are indexed by dm , the retained mass is $p^{kdm/2}$, and the derivative contributes $d \log p$. The support of these prime-power indices is not replaced by all positive integers.

Put $b_d = (1 - \chi)/2$. Every zero of the denominator in (TPT22), with its original label retained, is

$$s_{ab,\ell} = \frac{k}{2} + \frac{\chi(2a-k)\delta}{d} + i \left(\frac{\chi(2b-k)\gamma}{d} + \frac{\pi b_d + 2\pi\ell}{d \log p} \right), \quad \ell \in \mathbb{Z}. \quad (\text{TPT24})$$

Indeed substituting this expression into $\chi \omega_{ab}^x p^{-d(s-k/2)} = 1$ proves the equality; conversely every solution differs by one indicated logarithmic period. If labels produce the same point, their positive multiplicities add. These are poles of the specified finite local function $Z_{p,d}$, not a statement about a global zeta function.

The trace information has an exact reverse map. For every integer m ,

$$t_p(\chi m) = \frac{\chi^m}{d} \text{tr}(C^{dm}) = \frac{\chi^m p^{-kdm/2}}{d} \text{tr}(\tilde{C}^{dm}). \quad (\text{TPT25})$$

For a specified original projector, even its eigenvalue is recovered by

$$\omega_{ab} = \left(\frac{\chi}{d} \text{tr}(P_{ab}^{(d)} C^d) \right)^x = \left(\frac{\chi p^{-kd/2}}{d} \text{tr}(P_{ab}^{(d)} \tilde{C}^d) \right)^x. \quad (\text{TPT26})$$

These follow directly from (TPT3)–(TPT4), using $\chi^2 = 1$ and the nonzero original eigenvalues.

Even without the labelled projectors, the first q nonzero-time trace slots determine the characteristic polynomial of A . Set $a_m = d^{-1} \text{tr}(C^{dm})$ and recursively define

$$e_0 = 1, \quad j e_j = \sum_{m=1}^j (-1)^{m-1} e_{j-m} a_m \quad (1 \leq j \leq q). \quad (\text{TPT27})$$

Then $\det(I - vA) = \sum_{j=0}^q (-1)^j e_j v^j$. For proof, the identity $-vD'_A(v)/D_A(v) = \sum_{m \geq 1} a_m v^m$ follows from its finite eigenvalue product. Multiplying by D_A and comparing coefficients gives

exactly (TPT27), which recursively determines all coefficients. The eigenvalues of $A = \chi F_p^\chi$ then recover the original eigenvalue multiset by $\omega = (\chi\alpha)^\chi$. Trace data alone do not recover an ordering of its labels or the original metric G_N ; (TPT26) uses the actual retained projectors when such labelled recovery is intended.

Finally the exact operator recovery, stronger than spectral recovery, uses the original block inclusions and projections:

$$\begin{aligned} \pi_r C^d_{\iota_r} &= A, & \pi_r \tilde{C}^d_{\iota_r} &= p^{kd/2} A, \\ F_p &= \chi(\pi_r C^d_{\iota_r})^\chi, & \tilde{F}_p &= p^{k/2} \chi(p^{-kd/2} \pi_r \tilde{C}^d_{\iota_r})^\chi \quad (0 \leq r < d). \end{aligned} \tag{TPT28}$$

Substituting $C^d = \text{diag}(A, \dots, A)$ proves every equality. The inclusions remain isometries for the original block metric, and the projections remain their adjoints. Thus the $1/d$ appearing in the labelled radii (TPT24) has exact return maps: it retains the original operator, its inverse twist, all mass factors, all labelled traces, and the metric defects of (TOB13). The finite local determinant calculation therefore does not discard the complete return while retaining only its contracted root radii.

Tensor sectors retained by the original operators

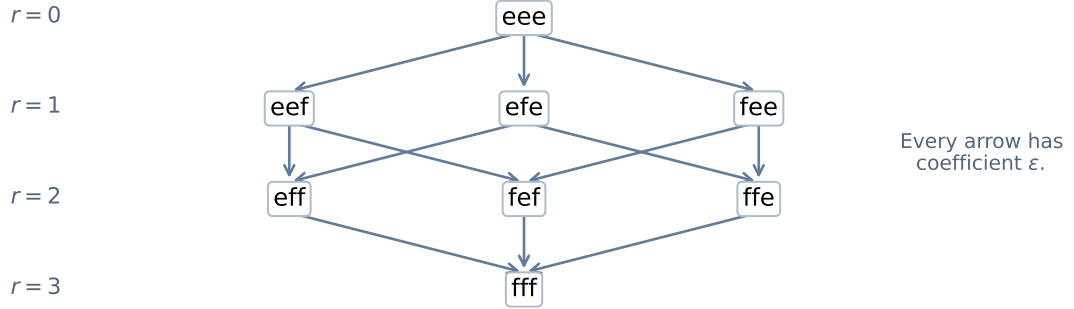
$$A_p \ell_{ab} = \alpha_{ab} \ell_{ab}, \quad \alpha_{ab} = (2a - k)\delta \log p + i(2b - k)\gamma \log p$$

$$\begin{array}{ccc} \ell_{ab} & & \ell_{k-a, k-b} \\ \alpha_{ab} & & -\alpha_{ab} \\ & \searrow \quad \swarrow & \\ & A_p^{[2]}(\ell_{ab} \otimes \ell_{k-a, k-b}) = 0 & \end{array}$$

$q = (k+1)^2$ independent vectors; uncentred exponent $k \log p$ is retained.

Nilpotent tensor block: $Re = \varepsilon f, Rf = 0, \varepsilon > 0$

$$X_3 = R \otimes I \otimes I + I \otimes R \otimes I + I \otimes I \otimes R$$



$$\ker X_3 = \text{span}\{ffe - fef, fef - eff, fff\}$$

Words denote the displayed tensor factors. The original orthonormal e, f and their tensor metric are retained; the three kernel vectors need not be orthogonal.

$$\dim \ker R^{[t]} = \sum_{j=0}^t \binom{t}{j} (q-2)^{t-j} \binom{j}{\lfloor j/2 \rfloor}$$

Complete receiver proofs: OLR23–31. Lattice lifts: OLR16–17.
Source locality: Connes–Consani, arXiv:2606.06604v1, Definition defnloc.
These local tensor sectors do not impose a purity theorem on the original object.

Figure 5: Two tensor mechanisms in the unchanged spectral receiver. Opposite original labels give zero centred exponent; the uncentred mass is still $k \log p$. In the three active nilpotent slots every arrow carries the actual coefficient ε . Its full kernel and the general tensor formula are proved in OLR23–31.

23 Locality of the original finite spectral receiver

Sources and scope of the calculation. Alain Connes and Caterina Consani, *On the Absolute Geometry of Spec $\mathbb{Z}$ and the Fargues–Fontaine curve*, arXiv:2606.06604, original author file `FF.tex`, Definition `defnloc` and the following extension proposition (lines 693–774), define locality by continuity on $H_p = \mathbb{Z}[1/p]$ for its specified place topology. Their archimedean complex points are in `prop:archimedean_points_C` (lines 1465–1500), and their p -adic complex points in `prop:nonarchimedean_points_C` (lines 1643–1659). Those original TeX passages were read in this derivation. ACF23–ACF26 retain their scalar local families on the full support base. The matrix analogues needed below are proved here, including their Jordan terms, rather than inferred from eigenvalue moduli alone.

The original receiver, metric, odd twistor return and complete boundary are TOB6–TOB29 in `TWISTOR_TO_ORIGINAL_BOUNDARY.tex`; its labelled prime traces and reverse maps are TPT1–TPT28 in `TWISTOR_PRIME_TRACE.tex`. Their original programme input is 033, SBT1–SBT23. The rank-one nilpotent and original observation maps below are TDP4, TDP11–TDP13 and TDP23 in the retained `TAU_DELiGNE_PRIME_MEMORY.tex`. These local proof files are programme derivations, not author papers by Connes or Consani.

23.1 The exact matrix extension criterion

Let V be a finite-dimensional complex vector space and $A \in \text{End}_{\mathbb{C}}(V)$. Give $\text{GL}(V)$ its usual complex topology. Fix a rational prime p and put

$$\rho_A(h) = \exp(hA), \quad h \in H_p = \mathbb{Z}[1/p]. \quad (\text{OLR1})$$

This is an additive-group homomorphism into $\text{GL}(V)$: the absolutely convergent exponential series for the commuting matrices hA, tA gives $\rho_A(h+t) = \rho_A(h)\rho_A(t)$, and $\rho_A(-h)$ is its inverse.

The following three statements are equivalent:

$$\begin{aligned} \rho_A \text{ extends continuously to } \mathbb{Q}_p; \quad \exp(p^r A) = I_V \text{ for some integer } r \geq 0; \\ A \text{ is semisimple and every eigenvalue belongs to } 2\pi i\mathbb{Z}[1/p]. \end{aligned} \quad (\text{OLR2})$$

Here semisimple means diagonalizable over $\mathbb{C}$; it is not a claim that the original metric makes the eigenbasis orthogonal. The same statements are equivalent to continuity of OLR1 on H_p for its p -adic topology. The extension is unique.

Here is an elementary proof of necessity that retains nilpotents. For any induced matrix norm, the neighbourhood $\mathcal{U} = \{T : \|T - I_V\| < 1/2\}$ contains no nontrivial subgroup. Indeed, if a subgroup lies in $\mathcal{U}$, all positive and negative integer powers of any one of its elements T lie there. For every eigenvalue λ of T this implies $|\lambda^m - 1| < 1/2$ for all integers m . Positive or negative powers exclude $|\lambda| \neq 1$. For $|\lambda| = 1$, the same bound implies $\Re \lambda^m > 7/8$. If $\lambda \neq 1$, the geometric sum gives $n^{-1} \sum_{m=1}^n \lambda^m \rightarrow 0$, contradicting that inequality. Thus every eigenvalue is one, and $T = I_V + D$ with D nilpotent. If $D \neq 0$, choose the largest j with $D^j \neq 0$. Some entry of $T^m = \sum_{a=0}^j \binom{m}{a} D^a$ is a polynomial of degree j with nonzero leading coefficient, so its powers are unbounded. This again contradicts membership in $\mathcal{U}$. Hence $T = I_V$.

Continuity at zero of an extension provides an integer $r \geq 0$ with $\rho_A(p^r \mathbb{Z}_p) \subset \mathcal{U}$; this image is a subgroup, so it is trivial. In particular $\exp(p^r A) = I_V$. The identical argument applied to the subgroup $p^r \mathbb{Z}$ inside H_p proves this conclusion from continuity on H_p alone.

On a Jordan primary space write $A = \alpha I + D$, $D^s = 0$. The equality $\exp(p^r A) = I$ first implies $\exp(p^r \alpha) = 1$, so $\alpha = 2\pi i a / p^r$ for an integer a . It then requires $\exp(p^r D) = I$. But

$$\exp(p^r D) - I = p^r D \left(\sum_{j=0}^{s-1} \frac{(p^r D)^j}{(j+1)!} \right). \quad (\text{OLR3})$$

The factor in parentheses is I plus a nilpotent and is invertible, with inverse given by a finite geometric sum. Thus $D = 0$. Conversely finitely many eigenvalues in $2\pi i\mathbb{Z}[1/p]$ have a common denominator p^r , and diagonalization proves $\exp(p^r A) = I$.

Sufficiency and uniqueness can be proved directly without assuming completeness of a particular matrix norm on the open set $\mathrm{GL}(V)$. The inclusion induces a group isomorphism

$$H_p/p^r\mathbb{Z} \xrightarrow{\simeq} \mathbb{Q}_p/p^r\mathbb{Z}_p. \quad (\text{OLR4})$$

Its kernel is $H_p \cap p^r\mathbb{Z}_p = p^r\mathbb{Z}$. It is onto because truncating the p -adic digits of any x just before exponent r gives a finite element $h_x \in H_p$ with $x - h_x \in p^r\mathbb{Z}_p$. Define $\tilde{\rho}_A(x) = \exp(h_x A)$. Changing h_x changes it by an integer multiple of p^r , so this definition is independent of the choice. OLR4 proves the group law. It is constant on the open cosets of $p^r\mathbb{Z}_p$, hence continuous. Truncated digit expansions also prove density of H_p in $\mathbb{Q}_p$, and agreement on a dense subset of a Hausdorff target proves uniqueness. This completes OLR2.

Using the original ACF24 convention $\psi_p(x) = \exp(2\pi i\{x\}_p)$, the explicit spectral formula is

$$A = \sum_j 2\pi i b_j P_j \implies \tilde{\rho}_A(x) = \sum_j \psi_p(b_j x) P_j, \quad b_j \in \mathbb{Z}[1/p]. \quad (\text{OLR5})$$

For $h \in H_p$, $b_j h - \{b_j h\}_p$ is an integer, proving agreement with the original exponential. No phases or eigenspace projectors have been changed. For comparison, every A , including every Jordan nilpotent, gives the continuous archimedean representation $t \mapsto \exp(tA)$ on $\mathbb{R}$. Thus OLR2 describes the intersection of that matrix family with the p -adic local family; it does not exclude the archimedean local points admitted in the Connes–Consani source.

23.2 The maximal admissible subspace and its exact defect

For arbitrary A define $D_r(A) = \exp(p^r A) - I$. Then

$$\begin{aligned} V_{p\text{-loc}}(A) &= \bigcup_{r \geq 0} \ker D_r(A) \\ &= \bigoplus_{\alpha \in \mathrm{Spec}(A) \cap 2\pi i H_p} \ker(A - \alpha I). \end{aligned} \quad (\text{OLR6})$$

The kernels increase because $T^p - I = (T - I) \sum_{j=0}^{p-1} T^j$. On an α -primary space, if $e^{p^r \alpha} \neq 1$, then $D_r(A)$ is a nonzero scalar times I plus a nilpotent, so is invertible. If $e^{p^r \alpha} = 1$, OLR3 proves that its kernel is exactly the kernel of the primary nilpotent, the actual eigenspace. This proves OLR6. Finitely many allowable α have a common denominator, so the union is already $\ker D_{r_0}(A)$ for one $r_0 \geq 0$ and remains that kernel for all $r \geq r_0$. The restriction to this subspace satisfies OLR2. Conversely every A -invariant subspace on which the restriction is p -adic local is contained in OLR6 by OLR2. Moreover every linear subspace invariant under $\rho_A(H_p)$ is A -invariant: for its vector v the vectors $p^n(\rho_A(p^{-n}) - I)v$ belong to that subspace and converge to Av by the exponential series; finite-dimensional subspaces are closed. Thus maximality applies to the original group action itself.

The defect is the original quotient, with its exact connecting map:

$$\begin{aligned} 0 \longrightarrow V_{p\text{-loc}}(A) \longrightarrow V \xrightarrow{\pi} \mathfrak{D}_p(A) := V/V_{p\text{-loc}}(A) \longrightarrow 0, \\ \overline{D}_r : \mathfrak{D}_p(A) \xrightarrow{\simeq} \mathrm{im} D_r(A), \quad \overline{D}_r([v]) = D_r(A)v \quad (r \geq r_0). \end{aligned} \quad (\text{OLR7})$$

It is well-defined and injective because its kernel before taking the quotient is exactly OLR6; it is onto its displayed image. All maps intertwine A and $\exp(hA)$ because $D_r(A)$ commutes with them. A good-eigenvalue Jordan block can have a local quotient even though the entire block is not local. OLR7 retains that extension through the original nilpotent; it does not claim a decomposition into a local summand and a complementary invariant summand.

23.3 Application to every label and cutoff of the original receiver

Retain exactly

$$k \geq 17, \quad k \equiv 1 \pmod{4}, \quad q = (k+1)^2, \quad \delta, \gamma > 0, \quad q-1 \leq N \leq 2q,$$

$$\lambda_{ab} = (2b-k)\gamma - i(2a-k)\delta, \quad Q_k(y) = \prod_{a,b=0}^k (y - \lambda_{ab}), \quad (\text{OLR8})$$

$$E = \mathbb{C}[y]/(Q_k), \quad M[v] = [yv], \quad A_p = i(\log p)M, \quad F_p = e^{A_p}, \quad \langle v, w \rangle_N = v^* G_N w.$$

The four cutoffs $q-1, q, 2q-1, 2q$ remain individual members of this range. No replacement of G_N is used. The roots are distinct, since equality forces equality of real and imaginary parts and then of both integer labels. Let ℓ_{ab} be the original Lagrange polynomial with value one at λ_{ab} and zero at the other roots, and let P_{ab} be the projector onto its coefficient. Evaluation proves that these vectors form a basis and $M\ell_{ab} = \lambda_{ab}\ell_{ab}$. Thus the complete exponent and its locality are

$$A_p \ell_{ab} = ((2a-k)\delta \log p + i(2b-k)\gamma \log p) \ell_{ab},$$

$$E_{p\text{-loc}}(A_p) = 0, \quad \mathfrak{D}_p(A_p) = E,$$

$$D_r(A_p) = \sum_{a,b} (\omega_{ab}^{p^r} - 1) P_{ab}, \quad D_r(A_p)^{-1} = \sum_{a,b} (\omega_{ab}^{p^r} - 1)^{-1} P_{ab}, \quad (\text{OLR9})$$

$$\omega_{ab} = p^{(2a-k)\delta} e^{i(2b-k)\gamma \log p}.$$

Indeed k is odd, so $2a-k \neq 0$. Every exponent in OLR9 has nonzero real part, excluding every label from OLR6. Every $\omega_{ab}^{p^r} - 1$ is nonzero for the same reason, proving the displayed inverse. This is the entire original defect, not only an observation of a selected eigenvalue.

For $v = \sum_i c_i \ell_i$ and $w = \sum_j d_j \ell_j$, the complete original Gram form of the defect is

$$\langle D_r v, D_r w \rangle_N = \sum_{i,j} \bar{c}_i d_j \overline{(\omega_i^{p^r} - 1)} (\omega_j^{p^r} - 1) \langle \ell_i, \ell_j \rangle_N. \quad (\text{OLR10})$$

All off-diagonal Gram entries survive. On one original vector its literal squared factor is

$$\|D_r \ell_{ab}\|_N^2 = [p^{2p^r(2a-k)\delta} - 2p^{p^r(2a-k)\delta} \cos(p^r(2b-k)\gamma \log p) + 1] \|\ell_{ab}\|_N^2. \quad (\text{OLR11})$$

The topology failure therefore keeps the full phase as well as the radial exponent. More precisely $D_r v \not\rightarrow 0$ for any nonzero v . If a coefficient with $2a-k > 0$ is nonzero, applying its fixed bounded projector to $D_r v$ proves divergence of that component. Otherwise every occupied label has $2a-k < 0$, and the finite spectral sum gives $D_r v \rightarrow -v \neq 0$.

The uncentred action is also retained:

$$\tilde{A}_p = A_p + \frac{k \log p}{2} I, \quad \tilde{F}_p = e^{\tilde{A}_p} = p^{k/2} F_p. \quad (\text{OLR12})$$

Its admissible subspace is exactly the sum of those original $\mathbb{C}\ell_{ab}$ for which $k/2 + (2a-k)\delta = 0$ and $(2b-k)\gamma \log p / (2\pi) \in H_p$. This follows directly from OLR6 and keeps the mass factor. For the stated general hypothesis $\delta > 0$ one must retain both conditions; discarding $k/2$ would be a different calculation.

23.4 The original nilpotent and observation-kernel return

Retain the programme vectors e, f and the literal coefficient $\epsilon > 0$ with $e^\dagger f = 0$ and $\|e\|_N = \|f\|_N = 1$. For $R = \epsilon f e^\dagger$, $R^2 = 0$, and the complete locality calculation is

$$E_{p\text{-loc}}(R) = \ker R = e^\perp, \quad D_r(R) = p^r \epsilon f e^\dagger, \quad \|D_r(R)v\|_N^2 = p^{2r} \epsilon^2 |e^\dagger v|^2. \quad (\text{OLR13})$$

The exponential is exactly $I + hR$, proving the first two statements by OLR6, and the original unit norm of f proves the third. Thus the nonzero nilpotent is not removed by the fact that its sole eigenvalue is zero. This calculation concerns this specified R ; it does not replace $M + R$ by M when the two fail to commute. For any such sum, OLR3–OLR7 apply to its actual primary spaces.

For the original isometric section $J : B \rightarrow E$, retain $P_B = JJ^\dagger$, $Q = I - P_B$, and $K = QE = \ker \Lambda$. Put $U_r = J^\dagger F_p^{p^r} J$. The complete visible and kernel defect is

$$\begin{aligned} D_r(A_p)Jb &= J(U_r - I)b + QF_p^{p^r}Jb, \\ \|D_r(A_p)Jb\|_N^2 &= \|(U_r - I)b\|_B^2 + \|QF_p^{p^r}Jb\|_N^2. \end{aligned} \quad (\text{OLR14})$$

Insert $I = P_B + Q$ and use $QJ = 0$ for the first identity. The two terms are orthogonal in the original metric for the second. The combined map on the right is injective: its vanishing gives $D_r(A_p)Jb = 0$, then OLR9 gives $Jb = 0$, hence $b = 0$. No such conclusion about the visible term alone is asserted. For arbitrary h, t the original compression has the exact product defect

$$J^\dagger \rho_{A_p}(h + t)J - J^\dagger \rho_{A_p}(h)JJ^\dagger \rho_{A_p}(t)J = J^\dagger \rho_{A_p}(h)Q\rho_{A_p}(t)J. \quad (\text{OLR15})$$

Insertion of $I = P_B + Q$ proves it. Thus treating compression as a group representation would remove precisely the displayed return.

23.5 Full lattice support and the exact lifted correspondence

For the full bounded distributive lattice L , keep $V_L^\uparrow = \{(0, \lambda) : \lambda \in L\} \cup \{(v, 1_L) : v \in V\}$ over $G_L(\mathbb{C})$, and $T^\uparrow(v, \lambda) = (Tv, \lambda)$ for every linear T . The component laws give $(TU)^\uparrow = T^\uparrow U^\uparrow$ and $(T + U)^\uparrow = T^\uparrow + U^\uparrow$ directly. In particular the zero linear map lifts to $\varepsilon_V(v, \lambda) = (0, \lambda)$, not to the map constant at $\tau = (0, 0_L)$. The exact support-retaining version of OLR7 is

$$\begin{aligned} D_r(A)^\uparrow &= \overline{D}_r^\uparrow \pi^\uparrow, & (D_r(A)^\uparrow)^{-1}(Z_L) &= V_{p\text{-loc}}(A)_L^\uparrow \quad (r \geq r_0), \\ Z_L &= \{(0, \lambda) : \lambda \in L\}, & \pi^\uparrow(v, \lambda) &= ([v], \lambda). \end{aligned} \quad (\text{OLR16})$$

Equality is checked on each (v, λ) ; membership in the inverse image is exactly $D_r(A)v = 0$ with its support unchanged. The fibre of $\pi^\uparrow$ over $(0, \lambda)$ for $\lambda < 1_L$ is the singleton $(0, \lambda)$; over $(0, 1_L)$ it is $V_{p\text{-loc}}(A) \times \{1_L\}$. For the original A_p , OLR9 says that the inverse image in OLR16 is precisely Z_L , and the defect lift is an automorphism on the whole synchronized carrier. In particular no support label is erased.

With L discrete and the carrier a subspace of $V \times L$, the lifted representation is continuous whenever its matrix representation is continuous: its action is $(h, (v, \lambda)) \mapsto (\rho_A(h)v, \lambda)$. Every zero state is fixed. This does not turn a zero-valued monoid character $T^h \mapsto z_\lambda$ into a group-valued local point: the fixed vector of a representation and the value of a character are different specified maps, and z_λ is not a unit.

The original free-coordinate support information has its exact correspondence as well for a retained basis of positive dimension. In that basis write a free vector as $(v, \boldsymbol{\mu})$, where $v_i \neq 0$ forces $\mu_i = 1_L$. The idempotent synchronization is

$$P_{\text{syn}}(v, \boldsymbol{\mu}) = (v, \bigvee_i \mu_i), \quad P_{\text{syn}}^{-1}(v, \lambda) = \{(v, \boldsymbol{\mu}) : \bigvee_i \mu_i = \lambda, v_i \neq 0 \Rightarrow \mu_i = 1_L\}. \quad (\text{OLR17})$$

This is the matrix pair $(I, \mathbf{1})$ on free coordinates: its amplitude is I and its support sends every output coordinate to the full finite join. Its image is exactly the synchronized carrier, and its square is itself, proving OLR17. Taking the inverse image of OLR16 retains all masks with $v \in V_{p\text{-loc}}(A)$. In the original case these are all zero-amplitude masks, with arbitrary $\boldsymbol{\mu}$. No group action with the free-module identity has been inferred from an action on this retract. For

dimension zero the free zero-coordinate semimodule has one element, with empty support join 0_L , whereas the synchronized carrier of the zero complex vector space still contains every $(0, \lambda)$, $\lambda \in L$. Thus OLR17 is not a surjection onto that latter carrier in dimension zero unless L has one element. The original receiver and its positive tensor powers have dimensions $q > 0$ and $q^t > 0$, so the stated correspondence applies to them, including their zero-amplitude fibres.

23.6 Frobenius, the complete twistor return, and the boundary

Source Frobenius acts by precomposition $h \mapsto mh$, so its operator parameter is exactly $A \mapsto mA$. For $m = p^j$, $j \in \mathbb{Z}$, OLR6 gives $V_{p\text{-loc}}(p^j A) = V_{p\text{-loc}}(A)$, because multiplication by p^j preserves H_p and the eigenspace kernels. OLR5 gives the corresponding p -adic parameters $b_j \mapsto p^j b_j$. For a general positive integer m the exact condition is $m\alpha \in 2\pi i H_p$; primes dividing m other than p can enlarge the admissible subspace. For $m = 0$ the action is identity on the whole space. These statements retain every source Frobenius.

Keep every positive odd d , $\chi_d = (-1)^{(d-1)/2}$, and the original receiver

$$C_d(v_0, \dots, v_{d-1}) = (\chi_d F_p^{\chi_d} v_{d-1}, v_0, \dots, v_{d-2}), \quad G_{N,d} = \text{diag}(G_N, \dots, G_N). \quad (\text{OLR18})$$

Let ι_s, π_s be the literal block inclusion and projection. The identity $C_d^d = \text{diag}(\chi_d F_p^{\chi_d}, \dots)$ proves the exact recovery of every locality defect:

$$\begin{aligned} D_r(A_p) &= \chi_d^{p^r} \pi_s C_d^{\chi_d p^r} \iota_s - I_E \\ &= \chi_d^{p^r} p^{-k\chi_d p^r/2} \pi_s (p^{k/2} C_d)^{\chi_d p^r} \iota_s - I_E. \end{aligned} \quad (\text{OLR19})$$

Negative powers in the $\chi_d = -1$ case are legitimate because the receiver is invertible. The powers of the sign multiply to one, and the scalar mass in the second equality is explicitly cancelled by its displayed inverse factor; it has not been omitted. Thus the original nonzero defect has an exact return through every odd receiver and every original block.

The explicit logarithm on the full TOB16 eigenbasis has eigenvalues

$$\nu_{ab,t} = \frac{\chi_d^i (\log p) \lambda_{ab} + i\pi(1 - \chi_d)/2 + 2\pi i t}{d}, \quad 0 \leq t < d. \quad (\text{OLR20})$$

Its exponential is C_d . The eigenvectors $(\ell_{ab}, e^{-\nu_{ab,t}} \ell_{ab}, \dots, e^{-(d-1)\nu_{ab,t}} \ell_{ab})$ form a basis: within each label the d distinct roots give an invertible Vandermonde matrix. Every real part is $\chi_d(2a - k)\delta \log p/d \neq 0$. OLR6 therefore gives zero admissible subspace for this entire dq -dimensional exponential family, despite its contracted real parts. The complete Gram entries remain TOB17, not a diagonal substitute for $G_{N,d}$.

The same return is visible in the complete original boundary. Put $T_d = \chi_d F_p^{\chi_d}$ and keep the TOB26–TOB28 maps $\mathcal{B}_{d,H}$ and $L_{d,H}$, for $H \geq 0$. Their exact norms are

$$\begin{aligned} \|\mathcal{B}_{d,H} v\|^2 &= p^k \sum_{s=0}^{d-1} (\|T_d^{-H} v_s\|_N^2 + \|T_d^{H+1} v_s\|_N^2), \\ \|L_{d,H} v\|^2 &= \sum_{j=-H}^H \sum_{s=0}^{d-1} \|T_d^{-j} v_s\|_N^2. \end{aligned} \quad (\text{OLR21})$$

For completeness these maps use the original orthonormal annuli

$$h_j(\mathbf{x}) = p^{kj/2} (1 - p^{-1})^{-k/2} \mathbf{1}_{(p^j \mathbb{Z}_p^\times)^k}(\mathbf{x}).$$

The complete finite receiver has coordinates $x_{d\ell+s} = T_d^{-\ell} v_s$. The boundary at $dH + s$ consists exactly of $p^{k/2} (h_{dH+s+1} \otimes x_{dH+s} - h_{-dH-s} \otimes x_{-dH-s-1})$. Its two annuli are disjoint; summing

over s gives the first line. The interval $[-dH, dH + d - 1]$ gives the second line by the same orthogonality. Invertibility of T_d makes the joint boundary injective. On blocks in $\mathbb{C}\ell_{ab}$, finite geometric summation gives

$$\frac{\|\mathcal{B}_{d,H}v\|^2}{\|L_{d,H}v\|^2} \longrightarrow p^k \left| p^{2\chi_d(2a-k)\delta} - 1 \right| > 0. \quad (\text{OLR22})$$

Indeed the ratio is $p^k(R^{-2H} + R^{2H+2})/\sum_{j=-H}^H R^{-2j}$, $R = p^{\chi_d(2a-k)\delta} \neq 1$; divide by the largest geometric term to obtain the limit. OLR19 and OLR21 are the explicit maps linking this complete boundary with locality's unchanged original prime return.

23.7 All tensor labels and the nonzero local tensor sector

For $t \geq 1$, retain the entire original tensor object $E^{\otimes t}$ with metric $G_N^{\otimes t}$ and generator $A_p^{[t]} = \sum_{j=1}^t I^{\otimes(j-1)} \otimes A_p \otimes I^{\otimes(t-j)}$. The commuting summands give $e^{hA_p^{[t]}} = \rho_{A_p}(h)^{\otimes t}$. On the original vector $\ell_{a_1b_1} \otimes \cdots \otimes \ell_{a_tb_t}$ its exponent is

$$(2\sum_j a_j - tk)\delta \log p + i(2\sum_j b_j - tk)\gamma \log p. \quad (\text{OLR23})$$

The tensor of the original Lagrange bases is a basis, so OLR6 proves that the exact admissible projector is

$$\Pi_{p,t} = \sum_{\substack{\sum a_j = tk/2 \\ (2\sum b_j - tk)\gamma \log p / (2\pi) \in H_p}} P_{a_1b_1} \otimes \cdots \otimes P_{a_tb_t}. \quad (\text{OLR24})$$

An empty index set means the zero projector. These spectral projectors need not be orthogonal for $G_N^{\otimes t}$; that metric remains unchanged, with entries the products of the original Gram entries. The entire complementary defect also has an explicit receiver: $[v] \mapsto (I - \Pi_{p,t})v$ is an isomorphism from $E^{\otimes t}/\text{im } \Pi_{p,t}$ to $\ker \Pi_{p,t}$, with inverse sending a vector to its quotient class. Its kernel before passing to the quotient is exactly the admissible range; $\Pi_{p,t}$ commutes with the tensor generator, so both maps intertwine the complete exponential action. This is a spectral direct sum, not an orthogonal replacement of the metric. OLR7 and OLR16 apply to all its retained components. Since k is odd, the admissible space is zero for odd t . For even t , define $n_{t,j} = [x^j](1+x+\cdots+x^k)^t$. Counting the independent a - and b -tuples gives its exact dimension

$$\dim \text{im } \Pi_{p,t} = n_{t,tk/2} \sum_{j=0}^{tk} n_{t,j} \mathbf{1}_{(2j-tk)\gamma \log p / (2\pi) \in H_p}. \quad (\text{OLR25})$$

Each coefficient counts precisely the tuples with the indicated sum by expansion of the polynomial product, proving the count.

In particular at $t = 2$ the original q independent vectors

$$\ell_{ab} \otimes \ell_{k-a,k-b}, \quad 0 \leq a, b \leq k, \quad (\text{OLR26})$$

have exponent zero and hence the trivial local character. There can be additional labels in OLR24 with nonzero allowed phase, and OLR25 retains them exactly. Thus the local tensor sector is nonzero although the individual receiver has no nonzero local amplitude. This calculation cannot be replaced by tensoring the zero admissible subspace of the individual factor. The involution $R_0\ell_{ab} = \ell_{k-a,k-b}$ satisfies $R_0MR_0 = -M$ by evaluation on each basis vector, so $R_0F_pR_0 = F_p^{-1}$. It identifies explicitly the reverse label in every vector OLR26. Their Gram entry at labels i, j is $\langle \ell_i, \ell_j \rangle_N \langle R_0\ell_i, R_0\ell_j \rangle_N$; no unproved isometry of R_0 in the original metric is assumed.

All mass factors also tensor. For the uncentred generator

$$\tilde{A}_p^{[t]} = A_p^{[t]} + \frac{tk \log p}{2} I, \quad (\text{OLR27})$$

the first condition in OLR24 becomes $tk/2 + (2\sum a_j - tk)\delta = 0$; its phase condition is unchanged. In particular the vectors OLR26 have uncentred exponent $k \log p$, so their centred trivial characters do not make the uncentred characters p -adic local.

For a general primary tensor block, the same calculation retains all nilpotents. If its factor generators are $\alpha_j I + D_j$, the tensor generator is $(\sum_j \alpha_j)I + D_{\text{tot}}$ with $D_{\text{tot}} = \sum_j I \otimes D_j \otimes I$ nilpotent. OLR3 proves that its admissible space is exactly $\ker D_{\text{tot}}$ when $\sum_j \alpha_j \in 2\pi i H_p$, and zero otherwise. For the original rank-one R , its tensor generator $R^{[t]} = \sum_j I \otimes R \otimes I$ satisfies

$$e^{hR^{[t]}} = (I + hR)^{\otimes t}, \quad (R^{[t]})^{t+1} = 0, \quad (R^{[t]})^t = t! R^{\otimes t} \neq 0, \quad E_{p\text{-loc}}(R^{[t]}) = \ker R^{[t]}. \quad (\text{OLR28})$$

Commutativity of the factor operators gives the exponential; every repeated slot in a product vanishes since $R^2 = 0$, and the $t!$ orders using every slot give the last nonzero power. OLR3 gives the final kernel. All these tensor maps admit the support-preserving lifts OLR16–OLR17 on their actual full carriers.

Its kernel dimension can be calculated exactly, with the original rank-two support projector $\Pi = ee^\dagger + ff^\dagger$:

$$\dim \ker R^{[t]} = \sum_{j=0}^t \binom{t}{j} (q-2)^{t-j} \binom{j}{\lfloor j/2 \rfloor}. \quad (\text{OLR29})$$

Here R and ϵ are precisely those in OLR13. To prove the formula, let $W = \text{im } \Pi = \mathbb{C}e \oplus \mathbb{C}f$ and $U = \ker \Pi$, an orthogonal decomposition for the unchanged G_N . The original vectors e, f are orthonormal; $Re = \epsilon f$, $Rf = 0$, and $R|_U = 0$. Tensoring the exact decomposition $E = W \oplus U$ gives a direct orthogonal sum indexed by the set of slots occupied by W . A sector with j such slots has operator X_j on $W^{\otimes j}$, the sum of the original R over these slots, tensored with the identity on the remaining U factors.

Grade $W^{\otimes j}$ by the number r of factors equal to f . Its grade- r subspace V_r has dimension $\binom{j}{r}$. The operator X_j maps V_r to V_{r+1} , with coefficient ϵ on every replacement of one e by f . Its original-metric adjoint $Y_j = X_j^\dagger$ reverses those replacements with the same real coefficient ϵ . Terms in different slots commute. In one slot $(R^\dagger R - RR^\dagger)e = \epsilon^2 e$ and $(R^\dagger R - RR^\dagger)f = -\epsilon^2 f$. Consequently

$$(Y_j X_j - X_j Y_j)|_{V_r} = \epsilon^2 (j - 2r) I_{V_r}, \quad \|X_j v\|^2 - \|Y_j v\|^2 = \epsilon^2 (j - 2r) \|v\|^2. \quad (\text{OLR30})$$

Thus $X_j : V_r \rightarrow V_{r+1}$ is injective when $r < j/2$. The isometry interchanging e and f in each of the j slots interchanges X_j and Y_j and sends grade $r + 1$ to grade $j - r - 1$. For $r \geq j/2$ that latter grade is less than $j/2$, so $Y_j : V_{r+1} \rightarrow V_r$ is injective by the result just proved. Its adjoint $X_j : V_r \rightarrow V_{r+1}$ is then surjective. The total kernel dimension therefore telescopes:

$$\dim \ker X_j = \sum_{r=\lfloor j/2 \rfloor}^j \left(\binom{j}{r} - \binom{j}{r+1} \right) = \binom{j}{\lfloor j/2 \rfloor}. \quad (\text{OLR31})$$

For odd j , the two middle grades have equal dimension and the injective intervening map is an isomorphism, consistent with this sum. For $j = 0$, $W^{\otimes 0} = \mathbb{C}$ and $X_0 = 0$, so the same formula gives one. Each j -slot sector has the further multiplicity $(q-2)^{t-j}$ and there are $\binom{t}{j}$ such sectors. Their direct sum proves OLR29. No coefficient ϵ , tensor slot, or original metric term was removed.

The original single receiver, its nilpotent channels, its complete observation return, its odd twistor receivers, and its tensor local sector have thus been placed in the specified local-point framework by exact maps. The calculation selects no locality condition that Connes–Consani impose on all points: their archimedean points still include every original exponential above. It therefore neither proves RH by imposing p -locality nor produces an RH counterexample by exhibiting a nonlocal receiver.

24 Complete mixed-tensor nilpotent ladders in the original metric

Programme input and scope. This calculation extends OLR13 and OLR28–OLR31 of OPERATOR_LOCALITY_PRIME_RECEIVER.tex, and TDP11, TDP12 and TDP23 of the retained programme proof TAU_DELiGNE_PRIME_MEMORY.tex. Those are the actual input proofs; no theorem from SGA or from Deligne is invoked here. The original source has $k \geq 17$, $k \equiv 1 \pmod{4}$, $q = (k+1)^2$ and each literal cutoff $q-1 \leq N \leq 2q$, with its positive metric G_N on the q -dimensional complex space E . Keep the actual vectors $e, f \in E$ and coefficient $\epsilon > 0$:

$$\begin{aligned} \langle v, w \rangle_N &= v^* G_N w, & \|e\|_N &= \|f\|_N = 1, & e^\dagger f &= 0, \\ R &= \epsilon f e^\dagger, & \Pi &= \epsilon e^\dagger + f f^\dagger, & W &= \text{im } \Pi = \mathbb{C}e \oplus \mathbb{C}f, & U &= \ker \Pi, & E &= W \perp U. \end{aligned} \quad (\text{OLM1})$$

Thus $Re = \epsilon f$, $Rf = 0$, $R|_U = 0$, and $\dim U = q - 2$. Every adjoint below uses this metric or its original tensor product. The grade introduced below counts slots occupied by f . It is not identified with an arithmetic weight or a purity filtration.

24.1 Every active sector and its original adjoint

For a positive integer t let

$$X = R^{[t]} = \sum_{a=1}^t I^{\otimes(a-1)} \otimes R \otimes I^{\otimes(t-a)}, \quad E_t = E^{\otimes t}, \quad G_t = G_N^{\otimes t}. \quad (\text{OLM2})$$

For each $J \subseteq \{1, \dots, t\}$ let E_J be the tensor subspace with factor W at slots in J and factor U elsewhere. Then

$$E_t = \bigoplus_{J \subseteq \{1, \dots, t\}}^\perp E_J, \quad E_J \simeq W^{\otimes j} \otimes U^{\otimes(t-j)}, \quad j = |J|. \quad (\text{OLM3})$$

The isomorphism moves the W factors into increasing order before the U factors, without changing the order within either family. It is an isometry, since the tensor inner product is the product of the factor inner products. Distinct sectors are orthogonal because some slot pairs W with U . Expanding the identity $I_E = \Pi + (I_E - \Pi)$ in every slot proves exhaustiveness. Every sector reduces both X and $X^\dagger$, and the restriction of X is $X_j \otimes I$, where X_j is the sum of the same original R over the j active slots.

For $0 \leq s \leq j$, let $V_{j,s} \subset W^{\otimes j}$ be the span of the original tensor words containing s copies of f and $j-s$ copies of e . These words are orthonormal already by OLM1. Therefore the $V_{j,s}$ are mutually orthogonal and have dimension $\binom{j}{s}$. Put $Y_j = X_j^\dagger$. The operator X_j replaces each occurrence of e in turn by ϵf , and Y_j replaces each f in turn by ϵe . Terms from different slots commute. In one slot $R^\dagger R - R R^\dagger = \epsilon^2(e e^\dagger - f f^\dagger)$, so the complete commutator is

$$[Y_j, X_j]|_{V_{j,s}} = \epsilon^2(j-2s)I_{V_{j,s}}, \quad \|X_j v\|^2 - \|Y_j v\|^2 = \epsilon^2(j-2s)\|v\|^2. \quad (\text{OLM4})$$

The second equality follows by pairing the first with v . In particular $X_j : V_{j,s} \rightarrow V_{j,s+1}$ is injective for $s < j/2$. The isometry interchanging e and f in every slot interchanges X_j and Y_j . It sends $V_{j,s}$ to $V_{j,j-s}$. Thus $Y_j : V_{j,s+1} \rightarrow V_{j,s}$ is injective for $s \geq j/2$, and its adjoint X_j is surjective there. These facts use the original ϵ ; it has not been replaced by one.

24.2 Actual primitive spaces and their orthogonal projections

For $0 \leq r \leq \lfloor j/2 \rfloor$ define

$$\mathcal{P}_{j,r} = \ker(Y_j : V_{j,r} \rightarrow V_{j,r-1}), \quad m_{j,r} = \dim \mathcal{P}_{j,r} = \binom{j}{r} - \binom{j}{r-1}, \quad \binom{j}{-1} = 0. \quad (\text{OLM5})$$

For $r = 0$, interpret $V_{j,-1} = 0$. For $r \geq 1$, the map $X_j : V_{j,r-1} \rightarrow V_{j,r}$ is injective by OLM4. Its adjoint Y_j is consequently onto $V_{j,r-1}$: the orthogonal complement of its image is the kernel of X_j , which is zero. Rank and nullity prove the dimension in OLM5. These are concrete kernels of the actual adjoint, not postulated primitive classes.

The corresponding orthogonal projector is explicit. With all restrictions to the displayed grades understood, put

$$P_{j,0} = I_{V_{j,0}}, \quad P_{j,r} = I_{V_{j,r}} - X_j(Y_j X_j|_{V_{j,r-1}})^{-1} Y_j \quad (r \geq 1). \quad (\text{OLM6})$$

The inverse exists because $\langle v, Y_j X_j v \rangle = \|X_j v\|^2 > 0$ for nonzero $v \in V_{j,r-1}$. The subtracted operator is self-adjoint and idempotent, by multiplication and $Y_j = X_j^\dagger$. It fixes $\text{im}(X_j|_{V_{j,r-1}})$ and vanishes on $\ker Y_j$. These two subspaces are orthogonal complements, so OLM6 is exactly the claimed projector.

24.3 Exact ladder lengths, lowering coefficients and norms

Fix j, r as above and write $d = j - 2r$. For every $u \in \mathcal{P}_{j,r}$ one has

$$Y_j X_j^a u = \epsilon^2 a(d - a + 1) X_j^{a-1} u \quad (a \geq 1). \quad (\text{OLM7})$$

To prove it, expand $[Y_j, X_j^a] = \sum_{b=0}^{a-1} X_j^b [Y_j, X_j] X_j^{a-1-b}$. The vector to which the commutator in summand b is applied has grade $r + a - 1 - b$. OLM4 makes its coefficient $\epsilon^2(j - 2r - 2a + 2 + 2b)$. Summing those a coefficients gives $\epsilon^2 a(d - a + 1)$, and the term $X_j^a Y_j u$ is zero. This proves OLM7 without deleting any power of ϵ .

Define the literal factors

$$s_{d,a} = \epsilon^{2a} a! \frac{d!}{(d-a)!} \quad (0 \leq a \leq d), \quad s_{d,0} = 1.$$

Then the original inner products are

$$\boxed{\langle X_j^a u, X_j^b v \rangle = \mathbf{1}_{a=b} s_{d,a} \langle u, v \rangle, \quad u, v \in \mathcal{P}_{j,r}, \quad 0 \leq a, b \leq d.} \quad (\text{OLM8})$$

Different powers have different grades, hence are orthogonal. At equal powers, move one X_j from the first entry to the second by its actual adjoint and use OLM7. Induction gives $s_{d,a} = \epsilon^2 a(d - a + 1) s_{d,a-1}$, the displayed factorial product. It is strictly positive for $a \leq d$. At the next power OLM7 gives $Y_j X_j^{d+1} u = 0$, and therefore $\|X_j^{d+1} u\|^2 = \langle X_j^d u, Y_j X_j^{d+1} u \rangle = 0$. Thus a nonzero primitive u has the exact ladder

$$u, X_j u, \dots, X_j^d u \neq 0, \quad X_j^{d+1} u = 0, \quad \text{length } d + 1 = j - 2r + 1. \quad (\text{OLM9})$$

For its unaltered power vectors the raising coefficient is one, while its lowering coefficient at level a is exactly $\epsilon^2 a(d - a + 1)$ from OLM7. The metric entries OLM8 record why these are adjoints. No power vector has been divided by its norm, and no source metric has been changed.

Ladders arising from different primitive grades are orthogonal. It suffices to examine equal total grades: let $r < r'$ and $r + a = r' + b$, so $a > b$. Moving X_j^b from the second vector to the first gives $Y_j^b X_j^a u$, a scalar multiple of $X_j^{a-b} u$ by OLM7. Its pairing with $v \in \mathcal{P}_{j,r'}$ is zero because the adjoint expression contains $Y_j^{a-b} v = 0$. Ladders from orthogonal primitive vectors in the same primitive space are orthogonal by OLM8.

They fill every original grade. At grade s the nonzero summands are exactly $X_j^{s-r} \mathcal{P}_{j,r}$ for $0 \leq r \leq \min(s, j - s)$, and their dimensions sum to

$$\sum_{r=0}^{\min(s, j-s)} m_{j,r} = \binom{j}{\min(s, j-s)} = \binom{j}{s} = \dim V_{j,s}. \quad (\text{OLM10})$$

Their orthogonality and the nonzero factors in OLM8 prove independence, so this dimension identity proves equality. Equivalently the map

$$\mathcal{L}_j : \bigoplus_{r=0}^{\lfloor j/2 \rfloor} \bigoplus_{a=0}^{j-2r} \mathcal{P}_{j,r} \longrightarrow W^{\otimes j}, \quad (u_{r,a}) \longmapsto \sum_{r,a} X_j^a u_{r,a} \quad (\text{OLM11})$$

is bijective and satisfies $\|\mathcal{L}_j(u)\|^2 = \sum_{r,a} s_{j-2r,a} \|u_{r,a}\|^2$. This identity specifies the domain metric transported from the original one. It is not an assertion that the unweighted domain direct sum has that metric.

The inverse has an exact formula in the original operators. Let v_s be the orthogonal component in grade s of v . For $a = s - r$ in the indicated range, its primitive coordinate is

$$u_{r,a} = s_{j-2r,a}^{-1} P_{j,r} Y_j^a v_{r+a}. \quad (\text{OLM12})$$

For its own summand, OLM7 iterated gives $Y_j^a X_j^a u = s_{j-2r,a} u$. For a summand of a lower primitive grade, the same iteration leaves a positive power of X_j in grade r , which OLM6 annihilates. For a higher primitive grade it applies more lowerings than available powers and the result vanishes. This proves OLM12 and supplies the connecting inverse, including all factorials and powers of ϵ .

24.4 Full Jordan multiplicities over all original slots

The decomposition in OLM11, followed by the sector maps OLM3, proves that for each pair (j, r) the original $X = R^{[t]}$ has Jordan blocks of length $j - 2r + 1$ with total multiplicity

$$M_{t;j,r} = \binom{t}{j} (q-2)^{t-j} \left(\binom{j}{r} - \binom{j}{r-1} \right), \quad 0 \leq j \leq t, \quad 0 \leq r \leq \lfloor j/2 \rfloor. \quad (\text{OLM13})$$

There are $\binom{t}{j}$ literal choices of active slots. For each, the inactive factor has dimension $(q-2)^{t-j}$, and each primitive vector supplies exactly the one ladder OLM9. The sector decompositions exhaust E_t , so no blocks are missing. For a fixed length $\ell \geq 1$, the multiplicity, with equal lengths combined, is

$$M_{t,\ell} = \sum_{\substack{0 \leq j \leq t \\ j \geq \ell-1, j-\ell+1 \text{ even}}} \binom{t}{j} (q-2)^{t-j} \left[\binom{j}{(j-\ell+1)/2} - \binom{j}{(j-\ell-1)/2} \right]. \quad (\text{OLM14})$$

Binomial coefficients with negative lower index mean zero. The longest block has length $t+1$ and multiplicity one, arising from $j = t, r = 0$. The exact dimensions of every power kernel and image follow:

$$\dim \ker X^a = \sum_{\ell=1}^{t+1} M_{t,\ell} \min(a, \ell), \quad \text{rank } X^a = \sum_{\ell=1}^{t+1} M_{t,\ell} \max(\ell - a, 0) \quad (a \geq 0). \quad (\text{OLM15})$$

Within a length- ℓ ladder, applying X^a shifts each unchanged power vector by a and kills exactly its last $\min(a, \ell)$ positions; independence proves these identities. In particular $\sum_{\ell} \ell M_{t,\ell} = q^t$. At $a = 1$, summing $m_{j,r}$ over r telescopes to $\binom{j}{\lfloor j/2 \rfloor}$ and recovers OLR29 exactly.

24.5 Every exponential entry and its complete Gram energy

Let $h \in \mathbb{C}$. On a ladder with primitive grade r and $d = j - 2r$, the original exponential has the finite exact action

$$e^{hX_j} X_j^a u = \sum_{b=a}^d \frac{h^{b-a}}{(b-a)!} X_j^b u, \quad \langle X_j^b u, e^{hX_j} X_j^a v \rangle = \begin{cases} s_{d,b} \frac{h^{b-a}}{(b-a)!} \langle u, v \rangle, & b \geq a, \\ 0, & b < a. \end{cases} \quad (\text{OLM16})$$

The first identity is the exponential series terminated by OLM9; OLM8 gives the second with the stated complex phase. No additional eigenvalue or unipotent coefficient has been introduced: all powers of ϵ remain in the literal vectors and the factors $s_{d,b}$.

The formula works directly with primitive-space-valued coordinates. For one fixed sector and primitive grade, let $v = \sum_{a=0}^d X_j^a u_a$, where $u_a \in \mathcal{P}_{j,r} \otimes U^{\otimes(t-j)}$ and X_j acts on its active factor. Then

$$\|e^{hX} v\|_{G_t}^2 = \sum_{b=0}^d s_{d,b} \left\| \sum_{a=0}^b \frac{h^{b-a}}{(b-a)!} u_a \right\|^2. \quad (\text{OLM17})$$

This follows by collecting the coefficient of X_j^b in OLM16 and applying OLM8. The norm inside uses the original primitive and inactive-factor metrics. The expression retains every interference term between different positions in the same ladder; such terms must not be discarded merely because the original ladder positions were orthogonal.

More explicitly, define

$$\mathbf{G}_{d;a,c}(h, z) = \sum_{b=\max(a,c)}^d s_{d,b} \frac{\bar{h}^{b-a} z^{b-c}}{(b-a)!(b-c)!}. \quad (\text{OLM18})$$

For $v = \sum_a X_j^a u_a$ and $w = \sum_c X_j^c v_c$ in the same primitive sector, finite expansion gives

$$\langle e^{hX} v, e^{zX} w \rangle_{G_t} = \sum_{a,c=0}^d \mathbf{G}_{d;a,c}(h, z) \langle u_a, v_c \rangle. \quad (\text{OLM19})$$

Indeed the coefficient in the first argument is conjugated and the second is not, which fixes the order of the powers in OLM18. At $h = z = 0$ the matrix is the original diagonal $s_{d,a} \mathbf{1}_{a=c}$. For every h , its diagonal specialization $\mathbf{G}_d(h, h)$ is positive definite: the triangular coefficient map in OLM17 has diagonal entries one and inverse obtained with $-h$, and every $s_{d,b}$ is positive. Its determinant is $\prod_{b=0}^d s_{d,b}$, since the triangular map and its adjoint have determinant one. These statements retain the original Gram matrix and do not impose an arithmetic purity condition.

All sectors and primitive grades are orthogonal and reduce both X and $X^\dagger$. Consequently the norm and pairing on the entire E_t are exactly the sums of OLM17 and OLM19 over the actual $J \subseteq \{1, \dots, t\}$ and all its primitive grades. These formulas give the full energy, including every inactive slot, rather than only its largest Jordan block.

The exact exponential displacement is particularly explicit:

$$\|(e^{hX} - I)v\|_{G_t}^2 = \sum_{b=1}^d s_{d,b} \left\| \sum_{a=0}^{b-1} \frac{h^{b-a}}{(b-a)!} u_a \right\|^2 \quad (\text{OLM20})$$

on each primitive sector, with the same sums over sectors for a general vector. It follows by removing the $a = b$ term in OLM17. For $h \neq 0$, the kernel is precisely the top position $X_j^d \mathcal{P}_{j,r}$ in each ladder: the coefficient at $b = 1$ first forces $u_0 = 0$, then $b = 2$ forces $u_1 = 0$, and induction forces $u_0 = \dots = u_{d-1} = 0$. The coefficient u_d is unrestricted. Thus the displacement has the same kernel as X , exactly as in the locality calculation OLR28.

For a nonzero vector in one ladder sector let a_0 be its smallest occupied position. Along positive real $h \rightarrow \infty$, the exact polynomial OLM17 has leading term

$$\|e^{hX} v\|_{G_t}^2 = \frac{s_{d,d}}{(d-a_0)!^2} \|u_{a_0}\|^2 h^{2(d-a_0)} + O(h^{2(d-a_0)-1}) \quad (a_0 < d). \quad (\text{OLM21})$$

Only $b = d$ and its smallest occupied $a = a_0$ contribute this highest degree; every other term has smaller degree, proving the formula without a possible cancellation of its positive leading coefficient. If $a_0 = d$, the vector is fixed and its energy is exactly $s_{d,d} \|u_d\|^2$. The same leading term holds for OLM20 when $a_0 < d$. Taking $h = p^n$ retains the exact p -adic-to-complex locality failure in every nonfixed ladder position.

24.6 Exact global singular energy as an independent consequence

The tensor exponential identity in TDP23 can be proved directly: the slot operators commute and each has square zero, so

$$e^{hX} = (I_E + hR)^{\otimes t}. \quad (\text{OLM22})$$

On the original orthonormal pair (e, f) , the one-factor matrix of $I + hR$ is $\begin{pmatrix} 1 & 0 \\ h\epsilon & 1 \end{pmatrix}$; on U it is I_U . Writing $a = \epsilon^2|h|^2$, the two eigenvalues of its actual positive Gram operator on W are

$$\lambda_{\pm}(h) = \frac{2+a \pm \sqrt{a^2+4a}}{2}, \quad \lambda_+ \lambda_- = 1. \quad (\text{OLM23})$$

Indeed that Gram matrix is $\begin{pmatrix} 1+a & \epsilon \bar{h} \\ \epsilon h & 1 \end{pmatrix}$, whose trace is $2+a$ and determinant is one. Its characteristic polynomial proves OLM23. The other $q-2$ eigenvalues are one. The squared singular values of the entire tensor exponential therefore consist of

$$\lambda_+^b \lambda_-^c, \quad \text{multiplicity } \frac{t!}{b!c!(t-b-c)!} (q-2)^{t-b-c}, \quad b, c \geq 0, \quad b+c \leq t. \quad (\text{OLM24})$$

To prove this, take orthogonal eigenspaces of the positive one-factor Gram operator, tensor them in the original slots, and count choices of the two one-dimensional factors and the $(q-2)$ -dimensional factor. The tensor Gram operator is the tensor of the original Gram operators by OLM22. If eigenvalues coincide, the displayed positive multiplicities add; the labelling remains valid, also at $h=0$. Thus the complete operator norm and squared Hilbert–Schmidt energy are exactly

$$\|e^{hX}\|_{G_t} = \lambda_+(h)^{t/2}, \quad \|e^{hX}\|_{\text{HS}, G_t}^2 = (q + \epsilon^2|h|^2)^t. \quad (\text{OLM25})$$

The first identity follows from the largest product eigenvalue, and the second follows by summing OLM24, or multiplying the one-factor traces. These are consequences in the same metric as the full ladder Gram formulas; no norm estimate has replaced the exact ladder entries.

24.7 The full support lifts and free-mask correspondence

For the full bounded distributive lattice L keep $V_L^\uparrow = \{(0, \lambda) : \lambda \in L\} \cup \{(v, 1_L) : v \in V\}$ and the original lift $T^\uparrow(v, \lambda) = (Tv, \lambda)$ over $G_L(\mathbb{C})$. The componentwise ring and lattice operations prove directly that $T^\uparrow$ is linear and that composition is preserved. Apply this to the complete isomorphism formed by OLM3 and OLM11 and to its explicit inverse OLM12. They become mutually inverse maps on the full synchronized carriers, recovering every support label as well as every primitive coordinate. The domain of this lift is the synchronized carrier of the entire direct sum in OLM11 and OLM3. It is not a product assigning unrelated support labels to different summands.

All metric identities lift with the same support meet:

$$\langle (v, \lambda), (w, \mu) \rangle_L = \langle (v, w)_{G_t}, \lambda \wedge \mu \rangle, \quad (X^a)^\uparrow (X^b)^\uparrow = (X^{a+b})^\uparrow. \quad (\text{OLM26})$$

The pair on the right of the first formula lies in $G_L(\mathbb{C})$: a nonzero amplitude requires both vectors nonzero and hence both supports top. At a zero amplitude the literal meet is retained. For $a+b \geq t+1$, the second formula is the supported-zero operator $(v, \lambda) \mapsto (0, \lambda)$, not the map constant at τ . All factorial coefficients in OLM7–OLM21 are unchanged on amplitudes.

For $h \neq 0$, the exact locality-defect correspondence is

$$\begin{aligned} ((e^{hX} - I)^\uparrow)^{-1}(Z_L) &= (\ker X)_L^\uparrow, \\ \ker X &= \bigoplus_{J,r}^\perp X_J^{j-2r} \mathcal{P}_{j,r} \otimes U^{\otimes(t-j)}, \quad Z_L = \{(0, \lambda) : \lambda \in L\}. \end{aligned} \quad (\text{OLM27})$$

OLM20 proves the amplitude kernel and its complete ladder description. The lift retains the support of every preimage, which proves the first equality. The quotient map by this kernel, followed by its induced injection into $\text{im}(e^{hX} - I)$, is exactly the lift of OLR7; its formula on a class is $[v] \mapsto (e^{hX} - I)v$. It is well-defined, injective and onto its displayed image because its amplitude kernel is precisely the stated one.

In the original free coordinates of the positive-dimensional space E_t of dimension q^t , write a vector as $(v, \boldsymbol{\mu})$, with $v_i \neq 0 \Rightarrow \mu_i = 1_L$. Keep the original synchronization and its complete fibre:

$$\begin{aligned} P_{\text{syn}}(v, \boldsymbol{\mu}) &= (v, \bigvee_i \mu_i), \\ P_{\text{syn}}^{-1}(v, \lambda) &= \{(v, \boldsymbol{\mu}) : \bigvee_i \mu_i = \lambda, v_i \neq 0 \Rightarrow \mu_i = 1_L\}. \end{aligned} \tag{OLM28}$$

The free operator has amplitude I and all-top support matrix, so it sends every output support coordinate to the displayed join. Repeating that join leaves it unchanged, proving it is an idempotent with precisely the displayed fibres. The free masks over OLM27 are exactly those whose original amplitude belongs to the explicitly computed $\ker X$; all zero-amplitude masks are retained. There is no asserted change of free-semimodule basis to a ladder basis. The exact relation is the correspondence OLM28 followed by the synchronized isomorphism OLM11 and inverse OLM12. This keeps every original mask while using the proved metric ladder coordinates for its amplitude. If an individual zero-dimensional summand is considered on its own, its free zero-coordinate semimodule has one element whereas its synchronized zero-vector carrier still has all of L ; the positive-dimensional assertion just made concerns the full original E_t and does not suppress that distinction.

These constructions give the complete original-metric nilpotent decomposition, every Jordan multiplicity, every exponential coefficient and its joint energy, and the exact full-support maps. Their slot-counting grades and finite ladder commutator have not been identified with an arithmetic weight, monodromy filtration or a Frobenius purity statement.